

ASYMMETRIC PRICE COMPETITION ON HYBRID PLATFORMS: THEORY AND EVIDENCE FROM AMAZON*

Luise Eisfeld[†]

University of Lausanne and Swiss Finance Institute

Jun Yan[‡]

Xiamen University

Li Yu[§]

Liaoning University

This version: August 13, 2026

Abstract

Many e-commerce platforms operate in hybrid mode: they host third-party sellers, earning ad-valorem commissions, while also retailing directly. In an oligopoly model, this creates two platform-specific incentives – a revenue-sharing effect and a commission-saving effect – implying that platform retail presence can exert weaker competitive pressure than third-party entry, and that common cost shocks are passed through asymmetrically across seller types. Using data from Amazon’s U.S. marketplace and repeated within-product availability changes, we find that third-party prices respond modestly more to third-party entry and exit than to Amazon’s. Consistent with the model, we estimate that commodity cost shocks are fully passed on to final prices by third-party sellers, whereas pass-through is significantly lower in the case of Amazon.

Keywords: E-Commerce, Dual Mode Platform, Online Pricing, Pass-Through

JEL Classification: D4, L1, E3

*We thank Philippe Bacchetta, Luca Bennati, Brett Hollenbeck, Joao Montez, Marc Rysman, and Morten Sæthre for helpful conversations and comments, and Eduard Boehm, Clara Jean, and Marc Ivaldi for insightful discussions. We also thank participants at talks given at EARIE (2024), MaCCI (2025), the Paris Conference on Digital Economics (2026), the Bristol Empirical IO Workshop (2026), the Zurich Workshop on Digital Platforms (2026), the Swiss IO Day (2026), the Toulouse School of Economics, the Stockholm School of Economics, NHH Bergen, the Joint Research Centre of the European Commission, and Xiamen University for their questions and comments. We acknowledge generous funding through the HEC Lausanne Research Fund. Any errors are our own.

[†]Email: luise.eisfeld@unil.ch.

[‡]Email: yanjun@xmu.edu.cn.

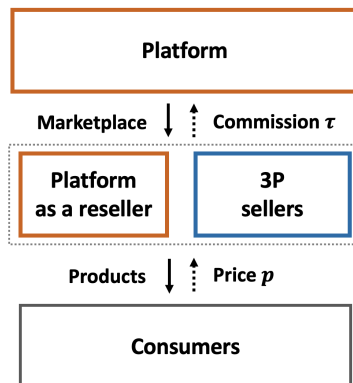
[§]Email: yuli@lnu.edu.cn.

1 Introduction

A growing number of e-commerce platforms – from Amazon and Walmart to Zalando and JD.com – operate in *hybrid* mode: they host third-party sellers, receiving a fraction of their sales revenue through an ad valorem commission, while also selling directly to consumers themselves. Amazon is the leading example. Roughly 60% of sales on the platform are generated by third-party sellers (Federal Trade Commission, 2024), from which Amazon earns ad valorem commissions; yet Amazon also retails many of the same products on its own account. This dual role gives the platform a distinctive stake in downstream competition: it benefits when third-party sellers make sales, but it also competes with them for the transaction itself. As online retail expands¹ and platforms increasingly combine marketplace intermediation with their own retail activity, the competitive implications of hybrid operation have become increasingly important.

In this paper, we study how this distinctive market structure, illustrated in Figure 1, shapes price competition in product markets. Hybrid platforms are not well described by the familiar case of a retailer selling private-label products alongside branded goods. In that setting, the retailer typically controls the retail prices of all products it carries. On a hybrid platform, by contrast, third-party (3P) sellers set their own prices. Nor is a hybrid platform well captured by models of common ownership: through percentage fees, the platform takes a share of sellers’ revenues rather than their profits, and thus does not internalize their costs. These features create pricing incentives that are specific to hybrid platforms and absent from more familiar market structures.

Figure 1: Structure of a hybrid platform



To illustrate how price competition plays out on hybrid platforms, we combine a theoretical model with reduced-form evidence from Amazon. On the theory side, we develop an oligopoly

¹See, e.g., <https://www.census.gov/retail/ecommerce.html> for the United States, https://ec.europa.eu/eurostat/statistics-explained/index.php?title=E-commerce_statistics_for_individuals for the European Union, <https://www.ons.gov.uk/businessindustryandtrade/retailindustry/timeseries/ms6y/drsi> for Great Britain, and <https://dzswgf.mofcom.gov.cn/sjcx.html> for China (accessed 28/08/2024).

pricing model to shed light on the core mechanism governing pricing decisions. A platform mediates price competition among differentiated-product retailers and, under a hybrid mode, also sells on its own account in direct competition with 3P sellers. We identify two platform-specific pricing wedges that arise when the platform sells its own products, relative to an otherwise comparable independent seller. First, because the platform earns an ad-valorem commission on 3P sales, raising its own price diverts demand toward 3P sellers and increases commission revenue. We refer to this as the revenue-sharing effect, which gives the platform an incentive to set a higher retail price. Second, because the platform does not pay the commission on its own sales, it faces a lower effective marginal cost than 3P sellers. We refer to this as the commission-saving effect, which gives the platform an incentive to set a lower retail price.

These two distinctive pricing incentives render the platform’s entry effect different from that of an independent seller, and in general ambiguous. In particular, when the revenue-sharing effect is strong, the platform’s entry may exert little competitive pressure on 3P sellers. As a result, equilibrium prices may decrease only mildly (or not at all), contrary to the common belief that platforms like Amazon inevitably price-squeeze 3P sellers through their own offerings.

However, even if the platform’s entry does not immediately price-squeeze 3P sellers, the commission-saving effect places 3P sellers at a relative competitive disadvantage and makes them particularly vulnerable to cost shocks. We find that, under an industry-wide common cost shock, the platform’s price responds less than that of a 3P seller. This arises because the commission fee inflates 3P sellers’ effective marginal costs, forcing the sellers to adjust their prices more aggressively in response to cost shocks. Our theoretical pass-through results suggest that, in episodes of positive cost shocks, 3P sellers therefore face a trade-off: either absorb the higher costs and compress their margins, or raise prices more sharply than the platform, which causes these sellers a loss in demand and market share. If positive cost shocks occur repeatedly, these temporary disadvantages can accumulate over time, enabling the platform to strengthen its position and expand its market presence from a dynamic perspective. These findings also have an implication for the consumer incidence of cost shocks: the incidence of cost shocks on a market depends on both the commission rate and the composition of first-party and 3P supply.

Guided by the model, we then ask whether the two price implications it isolates are visible in the context of Amazon. For this purpose, we collect a granular panel of prices for tens of thousands of products in narrowly defined product categories sold on [amazon.com](https://www.amazon.com), the U.S.-facing storefront of Amazon.² Our data show that Amazon participation is selective and concentrated in high-turnover products, and that Amazon and 3P sellers enter and exit a given product re-

²In the U.S., Amazon likely has a loyal customer base: 190 million customers are paid subscribers and receive membership benefits; see <https://cirpamazon.substack.com/p/updated-us-amazon-prime-member-estimate> (accessed 11/02/2026).

peatedly in manners that are not straightforward to predict with observables, and possibly due to stockouts, as in [Chen and Tsai \(2024\)](#). Using hundreds of repeated within-product episodes in which sellers temporarily enter or exit as a retailer, we conduct short-run event studies of 3P sellers’ pricing response upon entry (exit) by Amazon, and by other 3P sellers, respectively. Under the assumption that the timing of these entry and exit episodes is conditionally exogenous – an assumption for which we provide supporting pre-trend and hazard diagnostics – Amazon entry is followed by a 6% decrease in the prices of 3P sellers selling the same product, while Amazon exit is followed by a 3% increase. Comparing the pricing response of 3P sellers upon 3P entry (exit) to Amazon entry (exit), we find that the 3P seller prices tend to move modestly more strongly upon entry (exit) of other 3P sellers, as opposed to Amazon.

The second implication of our theory relates to how both seller types’ prices react to common cost shocks. We estimate the impact of recent shocks to commodity costs, in particular, to cocoa and coffee beans, on consumer prices, using distributed lag regressions, without imposing structural assumptions. We find that Amazon passes on common cost shocks to a smaller degree than 3P sellers: while pass-through to 3P sellers’ prices is complete in long horizons, Amazon displays significantly lower pass-through that is indistinguishable from zero. These empirical patterns hold when conditioning on product-weeks in which both seller types are jointly active, and are robust across different subsamples. Overall, these results line up with the empirical prediction that our model suggests.

All in all, our theory and empirical patterns show that hybrid platform organization shapes downstream price competition and cost pass-through. This matters for platform regulation: the downstream price effect of restricting the platform’s own retail activity depends on whether its offer disappears or is replaced by a 3P offer. Further, our paper shows that the ad valorem commission is not merely a transfer between the platform and sellers: it raises 3P sellers’ effective marginal costs and simultaneously gives the platform a stake in demand diverted toward 3P sellers.

Related Literature

A hybrid platform creates a competitive environment distinct from conventional business models in the retail industry, as classified by [Johnson \(2017\)](#). First, a marketplace platform differs from the wholesale model commonly adopted by traditional retailers (e.g., supermarkets), because it neither owns nor sets the prices of 3P sellers’ products ([Hagiu and Wright, 2015](#)). Therefore, pricing incentives on marketplace platforms, whether pure or hybrid, differ from those under

the wholesale model.³ Second, although the pure marketplace mode in our model resembles the so-called agency model studied by [Johnson \(2017\)](#), [Foros, Kind and Shaffer \(2017\)](#), and [Hu, Zheng and Pan \(2022\)](#), that literature does not consider the possibility of a hybrid mode, in which the platform also competes directly with 3P sellers as a retailer.⁴

A growing literature studies this novel market structure. The theoretical literature can be organized around three broad questions: the endogenous emergence of the hybrid mode ([Etro, 2021](#); [Gautier, Hu and Watanabe, 2023](#); [Hervas-Drane and Shelegia, 2025](#)), its competitive effects ([Etro, 2021](#); [Hagiu, Teh and Wright, 2022](#); [Etro, 2023](#); [Anderson and Bedre Defolie, 2024](#); [Hervas-Drane and Shelegia, 2022](#)), and the antitrust implications of this business model, including self-preferencing and imitation (e.g., [Hagiu et al., 2022](#); [Choe, Dubus, Matsushima and Shekhar, 2025](#); [Madsen and Vellodi, 2025](#)). Our focus is specifically on how hybrid mode shapes price competition. Several papers are particularly relevant. [Hagiu et al. \(2022\)](#) show that the platform’s own product “price-squeezes” a superior 3P seller, intensifying competition and thus increasing total transactions on the marketplace. In contrast, [Anderson and Bedre Defolie \(2024\)](#) find adverse competitive effects of hybrid operation: the platform optimally raises its commission fee to steer demand toward its own offerings, reducing 3P participation and product variety and ultimately raising final prices. Our paper differs from those in two key aspects. First, these papers endogenize both the platform’s choice of business model and its commission fee. In contrast, we view the commission fee as a longer-term choice and therefore treat it as fixed, while allowing the platform to flexibly enter or exit competition with 3P sellers at the product level. This is consistent with the empirical fact in our data that Amazon frequently begins and ceases selling individual products without any corresponding change in the category-level referral fee. Second, by adopting a general oligopoly model of price competition, we can systematically investigate how the interplay between the revenue-sharing channel and the double-markup channel shapes price competition. The homogeneous-preferences and fixed-fee framework of [Hagiu et al. \(2022\)](#) and the monopolistic-competition setting of [Anderson and Bedre Defolie \(2024\)](#) do not generate such strategic pricing interactions. We discuss these differences in detail in Section 3.3.1. Last, [Hervas-Drane and Shelegia \(2022\)](#) is also closely related to our paper in terms of research question. Like us, the authors study how a platform’s stake in rival sellers affects competitive interactions in the marketplace. However, as the authors build on [Varian \(1980\)](#)’s model of sales, their setting and mechanism differ substantially from ours.

³This distinction also implies that a platform’s pricing incentives when selling its own products differ from those of a supermarket selling private-label products. When setting the price of its own product, a hybrid platform internalizes the commission revenue generated by competing 3P sellers, whereas a supermarket internalizes the retail profits from the competing brands that it resells.

⁴More broadly, a related literature in contract theory and operations management studies revenue-sharing contracts (e.g., [Cachon and Lariviere, 2005](#)), but focuses primarily on vertical coordination within the supply chain rather than on the implications for competition.

Existing empirical literature on hybrid platforms has investigated self-preferencing and steering on hybrid platforms (Lee and Musolff, 2025; Chen and Tsai, 2024; Dendorfer and Seibel, 2026; Farronato, Fradkin and MacKay, 2023, 2025), Amazon’s entry strategy (Zhu and Liu, 2018), and the effects of restricting Amazon’s retail activity (Gutiérrez, 2022; Bennati, 2026; Bedre Defolie and Sokullu, 2026). Evidence on how hybrid platform operation shapes within-platform price competition remains limited. Crawford, Courthoud, Seibel and Zuzek (2022) and Korganbekova and Korganbekov (2026) may be most closely related to our paper, as these also study the effects of Amazon entry (as reseller or under private label brand) on 3P merchants. However, they focus primarily on whether Amazon selectively enters successful products pioneered by 3P sellers and study the effects of Amazon entry relative to no entry, whereas we ask whether Amazon’s dual role makes the competitive effect of its entry differ from that of an otherwise identical 3P seller. We further study the implications of hybrid operation for asymmetric cost pass-through, which has not been examined in this literature.

A further contribution of our paper is hence to uncover a novel implication of hybrid platforms for cost shock transmission. This connects us to the literature on pass-through and incidence under imperfect competition (e.g., Weyl and Fabinger, 2013; Miklos-Thal and Shaffer, 2021). Studying the pass-through of coffee-bean prices, Nakamura and Zerom (2010), Bettendorf and Verboven (2000) and Sangani (2026) are most closely related. In contrast of estimating average pass-through, we are, to the best of our knowledge, the first to identify asymmetric cost pass-through as a consequence of hybrid-platform organization, and document empirical patterns that are in line with our theory.

Outline. This paper is structured as follows. Section 2 describes the main institutional features of e-commerce platforms like Amazon, introduces the data we have collected, and provides stylized facts on seller participation and prices. Section 3 builds an oligopoly model of price competition on a hybrid platform, and proposes empirical implications. Sections 4 and 5 present empirical results on the prevailing price level and pass-through on Amazon respectively. Section 6 concludes.

2 Institutional Setting and Stylized Facts

2.1 Hybrid Selling on Amazon Marketplace

Amazon – the world’s largest e-commerce platform – has operated in dual mode since 2000. 3P sellers account for over 60% of sales on Amazon, and 560,000 sellers – predominantly small and medium-sized firms – were active on the U.S. marketplace in 2021 (Federal Trade Commission,

2024).⁵ Amazon levies an *ad valorem* commission (called “referral fee”) on each mediated 3P transaction. The referral-fee schedule is organized primarily by product category and revised infrequently;⁶ rates are typically between 5 and 20% of the final price. Amazon’s first-party retail activity includes private-label offerings (e.g., Amazon Basics), but a major fraction of Amazon’s sales in fact concerns non-private-label products that Amazon resells.

Amazon identifies a product by an Amazon Standard Identification Number (ASIN), and aggregates all offers for a given product onto a single detail page. Sellers of the same ASIN compete primarily for the “Buy Box” – the prominent section on the top-right of the page – which captures the vast majority of sales and the allocation of which is governed by a proprietary algorithm (Chen, Mislove and Wilson, 2016; Lee and Musolff, 2025; Breslin, 2025). Offers from non-winning sellers are relegated to a less visible list (“Other Sellers on Amazon”) below the Buy Box. Besides the ad-valorem referral fee, sellers can choose to outsource storage and shipping to Amazon (“Fulfilled by Amazon”), with an additional per-unit fulfillment fee, or avoid these fees by managing the logistics themselves (“Fulfilled by Merchant”).⁷ All in all, prior research suggests that seller identity, delivery mode, and Buy Box status make same-ASIN offers imperfect substitutes (e.g., Musolff, 2026), which motivates the use of an oligopoly model to capture price competition.

2.2 Data and Measurement

Data source. We collect data on prices of products sold on Amazon from Keepa (keepa.com), a company that provides data on prices, reviews, sales rank, and further both time-varying as well as cross-sectional attributes of products sold on Amazon by repeatedly scraping its product pages. Keepa updates its information multiple times per day, and its data have been used in previous academic literature (e.g., He, Reimers and Shiller (2022); Gutiérrez (2022); Cabral and Xu (2021); He, Hollenbeck and Proserpio (2022); Bennati (2026)). Our sample period runs from January 2021 to March 2025, and thereby deliberately excludes the later period in which U.S. tariff changes and geopolitical turmoil may have influenced prices and supply chains.

Sample. We focus on subcategories on Amazon that experienced sizable, well-identified upstream cost shocks in recent years and where demand fundamentals are relatively stable, in

⁵Amazon sellers made an average yearly revenue of \$250,000 in 2023; see <https://sell.amazon.com/blog/amazon-stats> (accessed 27/09/2024).

⁶See <https://sellercentral.amazon.com/help/hub/reference/external/GTG4BAWSY39Z98Z3> (accessed 27/11/2025); the most recent referral fee change on amazon.com we are aware of occurred in January 2024.

⁷FBA fee depends on product size and weight of the product, not on its price, and hence does not generate the effects that ad valorem fees generate. See https://sellercentral.amazon.com/revcalpublic?ref=sdus_pri_xscus_rc2&initialSessionID=135-2662106-4270236&ld=NSGoogle_SD_PRI_RC2&pageName=US%3ASD%3AFBA-main&ldStackingCodes=NSGoogle (accessed 19/06/2026).

particular the ground coffee and cocoa categories (see Section 5).⁸ We also collect data for other food (wheat pasta) and non-food subcategories (TVs, washers, T-shirts, USB hubs, and SD cards). Within each product category, we treat each ASIN as the relevant product and restrict attention to the 10,000 products with the highest turnover, as proxied by both the current and the 180-day average sales rank⁹.

Whenever a product is being offered on Amazon, we observe the lowest price among 3P offers under each of the delivery modes (FBA and FBM), the lowest price across all offers (including Amazon retail), Amazon’s own price, the total number of active sellers, and the seller ID and price of the offer displayed in the Buy Box. Additionally, we observe each product’s sales rank. We use the latter to construct variables that identify relatively high-turnover products. For food categories, we recover package sizes (weight in grams) from product titles using the OpenAI API.¹⁰ We then construct a product-day panel using prices recorded at midnight, and aggregate these data to the product-week level by averaging all time-variant variables.

Table 1: Summary of sample by category (ASIN-week panel; 222 weeks)

Category	Products (ASINs)	Avg. seller count (per ASIN-week)	ASIN-weeks with ≥ 2 sellers	With reviews (across ASINs)	Amazon private label (across ASINs)
Cocoa	1,432	2.3	32.4%	67.0%	0.2%
Coffee	14,532	2.5	35.8%	90.6%	0.3%
Pasta	7,709	2.5	42.0%	72.9%	0.4%
Washers	2,167	1.8	21.7%	22.8%	0.0%
TV	6,875	2.3	32.5%	64.8%	0.0%
Shirts	22,825	1.5	16.0%	92.4%	0.0%
USB hubs	7,269	2.2	20.0%	43.3%	0.4%
SD cards	12,851	1.3	14.3%	11.4%	0.0%

Notes: We track sampled ASINs daily from January 1, 2021 to March 31, 2025 and collapse the data to the ASIN-week level. We only include ASINs that have at least one week where Amazon or 3P has a valid price. The average seller count corresponds to the average number of sellers listing an ASIN in a week, computed over ASIN-weeks with at least one observed seller, and includes Amazon retail when present. “With reviews” and “Amazon private label” are product-level shares. The review indicator is measured from the latest observed product page and is therefore time-invariant within the sample. Because we query the data repeatedly over time, the final number of products in a category can substantially exceed 10,000.

Table 1 reports summary statistics on sample coverage and seller participation for each product category. Product variety is substantial in every category: for example, the coffee category

⁸Grocery products now account for a substantial share of Amazon’s revenue, suggesting that these categories capture mainstream products. In 2024, Amazon CFO Brian Olsavsky noted “strong growth in items like everyday essentials. This includes items like health, beauty, and personal care, as well as nonperishable grocery,” and stated that “the strength in everyday essential’s revenue is a positive indicator that customers are turning to us for more of their daily needs.” See <https://www.fool.com/earnings/call-transcripts/2024/10/31/amazoncom-amzn-q3-2024-earnings-call-transcript/> (last accessed 01/02/2025).

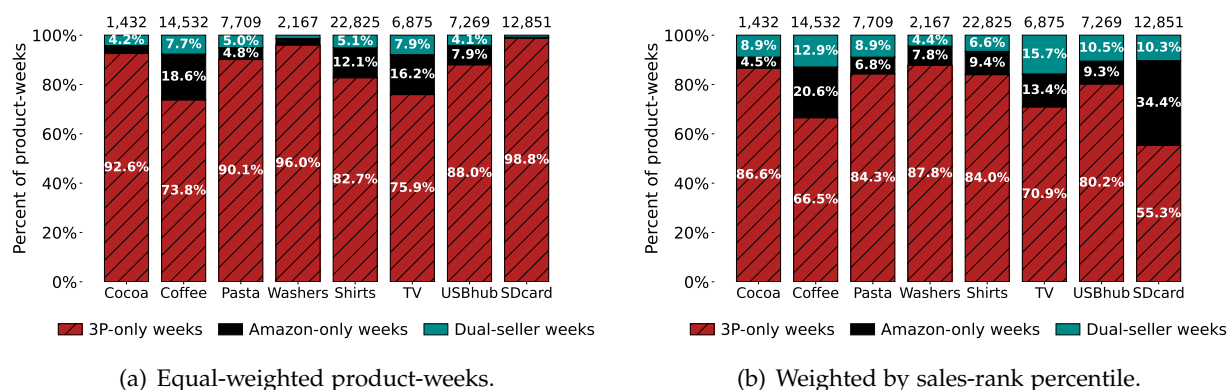
⁹This “Best Sellers Rank” aggregates recent and historical sales relative to the category peer group. See <https://sell.amazon.com/blog/amazon-best-sellers-rank> for further information (accessed 02/09/2024).

¹⁰Manual checks confirm a very high accuracy. In the Online Appendix, we provide details on how the data was built and also shows that food products are sold predominantly in standard household sizes rather than bulk formats

alone contains almost 40,000 distinct ASINs, 14,532 of which are in our dataset. Competitive conditions also vary meaningfully across categories. The average number of sellers per ASIN-week ranges from 1.3 for SD cards to 2.5 for coffee, and the share of ASIN-weeks with at least two observed sellers ranges from 14.3% to 42.0%. Review prevalence also differs markedly across categories. By contrast, Amazon private-label products account for only a very small share of products in every category, never exceeding 0.4% of the products scraped.

2.3 Stylized Facts

Figure 2: Extent of hybrid selling across categories



Notes: Each bar shows the share of ASIN-week observations that are 3P-only, Amazon-only, or dual-seller for a given category. Panel (a) weights each ASIN-week equally. Panel (b) weights ASIN-weeks by within-category sales-rank percentile, giving more weight to higher-turnover products. The sample covers ASINs observed weekly from January 2021 through March 2025. A product-week is classified as 3P-only (Amazon-only) if at least one 3P (Amazon) offer is recorded and the other side is absent; it is classified as dual-seller if both offer the products in the same week. The number of ASINs for each category is displayed above each bar.

Fact 1: Amazon retailing is selective and concentrated in high-demand ASINs. Figure 2 displays, for each category, the share of ASIN-week observations that are supplied exclusively by 3P sellers (3P-only), exclusively by Amazon (Amazon-only), or by both (dual-seller). Panel (a) weights each ASIN-week equally and shows that dual-seller states account for between about 0.8% and 7.9% of product-weeks across categories. 3P sellers are the exclusive suppliers in at least 73.8% of product-weeks in every category, while Amazon is the exclusive seller in between 0.4% and 18.6% of product-weeks. Overall, Amazon’s retail participation varies markedly across product categories: Amazon is active (either alone or alongside 3P sellers) in roughly 26% of ASIN-weeks in *Coffee* and about 24% in *TVs*, but in less than 2% of ASIN-weeks in *SD Cards*.

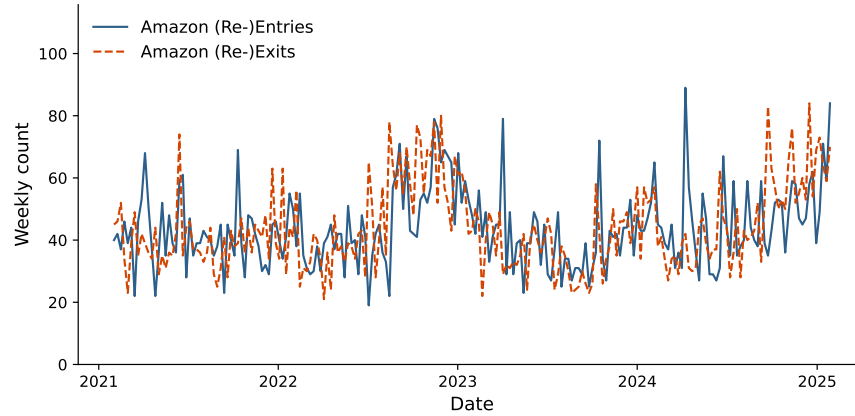
Panel (b) of Figure 2 weights ASIN-weeks by their within-category sales-rank percentile, giving more weight to higher-turnover products. Comparing the equal-weighted shares in panel (a) and the sales-rank-weighted shares in panel (b) shows that the prevalence of Amazon-only and

dual-seller ASIN-weeks increases for all categories except for *Shirts*. For instance, in both *TVs* and *Coffee*, the share of ASIN-weeks in which Amazon is active (Amazon-only or dual-seller) rises from about 24-26% under equal weighting to around 30-34% under sales-rank percentile weighting. For *SD Cards*, Amazon-only and dual-seller states account for around 34% and 10% of sales-rank-weighted ASIN-weeks, respectively, which is markedly larger than before. Taken together, these patterns indicate that Amazon’s direct retail presence is relatively more common among higher-turnover ASINs, while 3P sellers appear to account for most of the low-turnover “long tail” within categories.

Fact 2: Amazon’s retail availability is intermittent and recurrent within ASINs. A central feature of Amazon’s first-party participation is that its availability changes repeatedly within the same ASIN. Among ASINs for which Amazon is observed as a retailer before the start of our sample, Amazon frequently moves between periods of presence and absence¹¹. Figure 3 plots the weekly numbers of Amazon (re-)entries and (re-)exits for ASINs in the coffee category. These entry and exit events occur throughout the sample period and are not concentrated on a small number of calendar dates. Among ASINs that switch Amazon-presence status, the median ASIN experiences three Amazon-present spells and four Amazon-absent spells, with median spell durations of three and nine days, respectively (see Appendix C.1). Similar patterns arise in the other categories. These events therefore mark transitions between recurring spells of Amazon’s presence and absence as a reseller, representing intermittent retail availability rather than permanent product-market entry or exit. They are consistent with stockouts or replenishment delays, as discussed by Chen and Tsai (2024), although our data do not identify the operational cause of each transition.

¹¹We distinguish these recurrent ASIN-level transitions from a small number of coordinated brand-level assortment changes, in which Amazon begins or ceases to offer several ASINs from the same brand at approximately the same time; see Figure C.1 in the Appendix for example. These episodes are more consistent with deliberate assortment decisions and conceptually distinct from the recurrent within-ASIN variation emphasized here.

Figure 3: Recurring Amazon entries and exits over time

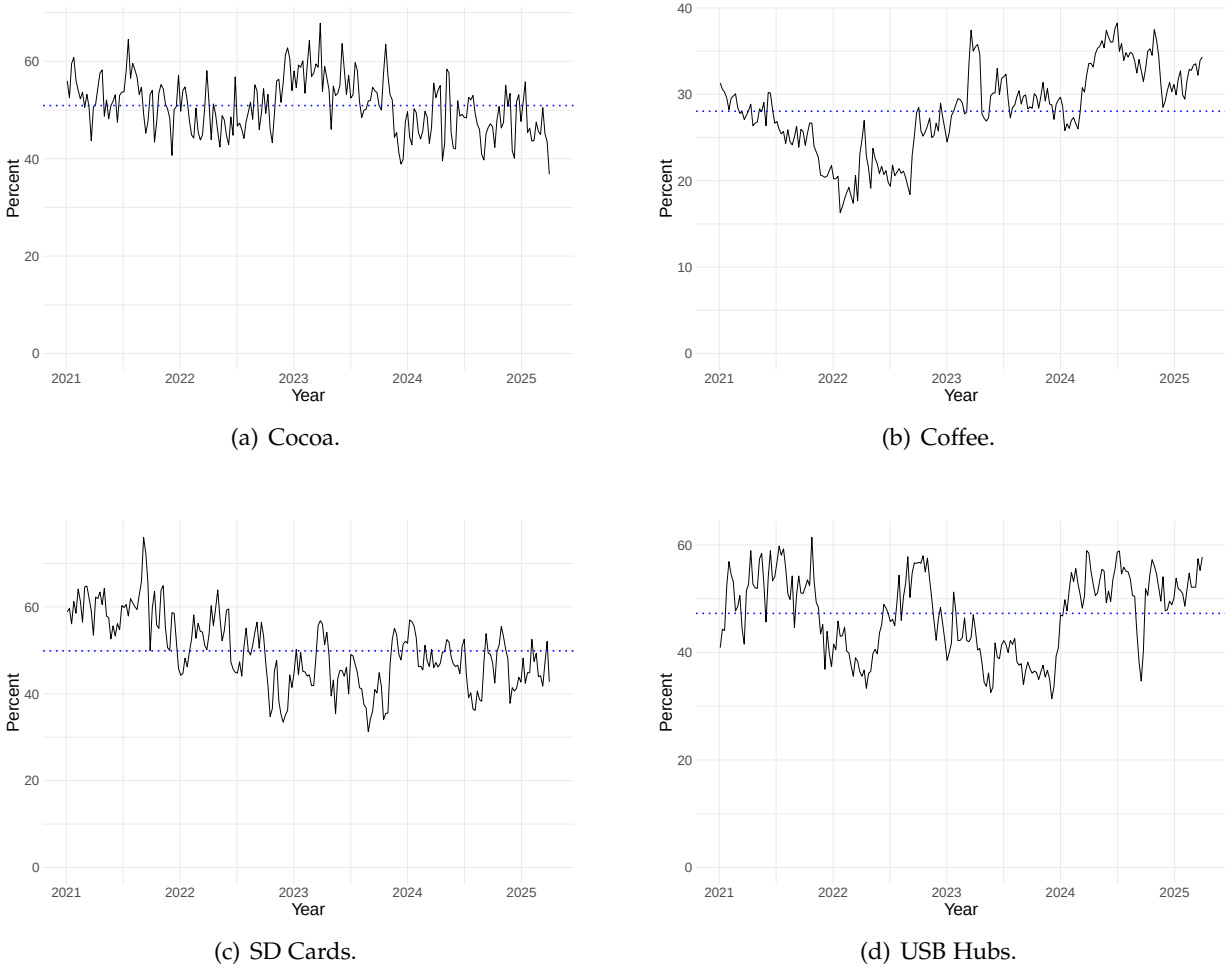


Notes: The Figure shows the weekly number of re-exits and re-entries by Amazon over time for the Coffee category. We exclude brands that account for more than 20% of aggregate Amazon entry or exit flows in the ten highest-activity weeks (which applies to one brand in the case of coffee), as those spikes would reflect systematic brand-level assortment changes rather than product-level entry or exit decisions (see Figure C.1). We only count a re-entry (re-exit) as such if they are followed by a spell of presence (absence) of 5 days or more (the lines look similarly stable when using other cutoffs).

Fact 3: Conditional on same-ASIN overlap, Amazon does not systematically price above the lowest 3P offer. As Amazon sells a different set of products than 3P sellers, cross-ASIN price comparisons would confound pricing behavior with product selection. We therefore compare prices only within ASIN-weeks in which Amazon and at least one 3P seller simultaneously sell the same product. In these dual-seller observations, Figure 4 shows that Amazon’s price is at or below the lowest observed 3P offer in roughly one-half to two-thirds of overlapping ASIN-weeks across the displayed categories. The same pattern obtains in regressions with ASIN and year-week fixed effects (Tables C.3 to C.6): Amazon’s price is generally statistically indistinguishable from, or below, the lowest 3P or FBA offer, with the clearest Amazon discount in coffee.¹² Thus, conditional on same-ASIN overlap, Amazon does not appear to occupy a systematically high-price position.

¹²Relative to FBM offers, Amazon is lower throughout, but this comparison is less direct because FBM prices include shipping costs.

Figure 4: Percent of products offered by both 3P and Amazon for which Amazon’s price is higher than the lowest 3P’s price



Notes: Plots use product-week observations in which both a 3P seller’s, as well as Amazon’s price is available (i.e., both seller types are active). Blue dotted line indicates median.

Taken together, these facts show that Amazon’s retail arm is a selective competitor: it is more prevalent on high-turnover ASINs, frequently alternates between presence and absence within the same ASIN, and, conditional on direct overlap, prices at or below comparable 3P offers. However, similar price levels need not imply similar pricing incentives. In the next section, we develop a theoretical framework characterizing the pricing incentives of Amazon and 3P sellers.

3 Theory

In this section, we develop a model of price competition on a hybrid platform that charges an ad valorem commission to 3P sellers. The platform’s dual role as both retailer and intermediary sets it apart from 3P sellers. In this setting, the ad valorem commission shapes price competition

through two distinct channels. We then examine how the interplay between these two channels generates asymmetric pricing behavior between the platform and 3P sellers.¹³

3.1 General Framework

Suppose that a monopoly e-commerce platform P operates a marketplace in a given product market. There are n retailers, each selling a differentiated product on the marketplace. The demand faced by retailer $i \in \{1, \dots, n\}$, denoted $D_i(\mathbf{p}) = D_i(p_i, \mathbf{p}_{-i}) = D_i(p_1, \dots, p_n)$, depends on its own price p_i and the prices of its rivals $\mathbf{p}_{-i}$. We assume a symmetric demand system. The marginal cost of retailer $i \in \{1, \dots, n\}$ is c_i . The platform charges a commission fee τ on the revenue generated by each 3P seller on the platform. Consistent with the institutional setting described in Section 2, in which referral-fee schedules are revised infrequently while Amazon can repeatedly start and stop retailing a given product, we take the commission fee as given and assume that it does not vary with the platform's mode of operation.

We consider two possible modes of platform operation in this product market: the pure marketplace mode and the hybrid mode. In the pure marketplace mode, the platform hosts only 3P products, so all n competing retailers are 3P sellers. In the hybrid mode, the platform not only runs the marketplace but also sells its own product and competes directly with 3P sellers. With n competing retailers, we let the platform be retailer 1 and denote the price and marginal cost of its product by p_P and c_P , respectively, where $p_P = p_1$ and $c_P = c_1$. The remaining retailers $i = 2, \dots, n$ operate as 3P sellers.

Timing: Under each mode, all retailers simultaneously choose their prices $p_i, i \in \{1, 2, \dots, n\}$.

3.2 Commission-Induced Pricing Forces

We start by characterizing two distinctive channels through which the ad valorem commission affects price competition: the *double-markup channel* in 3P sellers' pricing and the *revenue-sharing channel* in the platform's pricing of its own products. These two channels generate the key pricing forces that shape both equilibrium prices and cost pass-through.

In the pure marketplace mode, each retailer $i \in \{1, 2, \dots, n\}$ chooses its price p_i to solve

$$\begin{aligned} & \max_{p_i} [(1 - \tau)p_i - c_i] D_i(\mathbf{p}) \\ \Leftrightarrow & \max_{p_i} (1 - \tau) \left(p_i - \frac{c_i}{1 - \tau} \right) D_i(\mathbf{p}), \end{aligned}$$

where each retailer retains only a fraction $1 - \tau$ of its sales revenue while bearing the full marginal

¹³All omitted proofs are collected in Appendix A.1.

cost of the product. Rearranging the profit function, the commission is equivalent to scaling up retailer i 's marginal cost by a factor of $1/(1 - \tau)$. Following Johnson (2017), we call $\frac{c_i}{1-\tau}$ the retailer's "perceived marginal cost." We use the term *double-markup channel* to refer to the pricing effects generated by τ through inflating 3P sellers' perceived marginal costs.. As the first-order condition implies,

$$p_i - \frac{c_i}{1 - \tau} = -\frac{D_i(\mathbf{p})}{\partial D_i(\mathbf{p})/\partial p_i}, \quad i \in \{1, \dots, n\} \quad (1)$$

a 3P seller with market power adds another layer of markup over the perceived marginal cost $c_i/(1 - \tau)$, so that its retail price effectively reflects two layers of markup over the original marginal cost c_i .

In the hybrid mode, 3P sellers $i = 2, \dots, n$ solve the same profit-maximization problem as in the pure marketplace mode. The platform chooses p_P to maximize the sum of its retail profit and commission revenue from 3P sales:

$$\max_{p_P} (p_P - c_P)D_P(\mathbf{p}) + \tau \sum_{i=2}^n p_i D_i(\mathbf{p}).$$

The first-order condition is

$$p_P - \left(c_P + \tau \sum_{i=2}^n p_i \delta_{Pi}(\mathbf{p}) \right) = -\frac{D_P(\mathbf{p})}{\partial D_P(\mathbf{p})/\partial p_P}, \quad (2)$$

where

$$\delta_{ij} \equiv -\frac{\partial D_j(\hat{\mathbf{p}}^*)/\partial p_i}{\partial D_i(\hat{\mathbf{p}}^*)/\partial p_i}, \quad i, j \in \{1, \dots, n\}, \quad i \neq j,$$

is the diversion ratio from product i to product j . It measures the fraction of the marginal demand lost by product i following an increase in its price that is captured by product j . Therefore, the additional term $\tau \sum_{i=2}^n p_i D_i(\mathbf{p})$ in the platform's profit function captures the second channel through which τ affects price competition, which we call the *revenue-sharing channel*. As Equation (1) shows, through its share of 3P sellers' revenue, the platform internalizes the demand diversion toward 3P products caused by increasing its own price and, consequently, its positive effect on commission revenue.

Taken together, these two channels create an inherent asymmetry between the pricing incentives of the platform and those of 3P sellers. This asymmetry arises from the platform's dual role as both retailer and intermediary under an ad valorem commission structure and does not depend on a particular functional form for demand, provided that the products are substitutes.

3.3 Pricing Asymmetries

We now analyze how the interplay between the two pricing forces shapes price competition along two dimensions: equilibrium price levels and retailers' price adjustments in response to cost shocks.

3.3.1 Equilibrium Price Comparisons

Comparing the first-order conditions of 3P sellers and the platform in Equations (1) and (2), we see that the two channels generate opposing effects on the platform's retail price relative to 3P sellers' prices. The revenue-sharing channel creates upward pressure on the platform's price, as a higher own-product price diverts demand toward 3P products and thereby increases commission revenue. In contrast, because the platform's retail and intermediation activities are vertically integrated, it avoids the double-markup distortion faced by 3P sellers, creating downward pressure on its price relative to theirs. The equilibrium price ranking therefore depends on the relative strength of these two forces.

For analytical tractability, we assume in what follows that all retailers have the same marginal cost, $c_i = c, \forall i \in \{1, \dots, n\}$, regardless of whether retailer i is the platform or a 3P seller. We further impose the following linear specification on the symmetric demand system:

$$D_i(\mathbf{p}) = \frac{1 - \gamma - (1 + (n - 2)\gamma) p_i + \gamma \sum_{j \neq i} p_j}{(1 - \gamma)(1 + (n - 1)\gamma)}, \quad \forall i \in \{1, \dots, n\}.$$

Therefore, under the pure marketplace mode, we focus on the symmetric equilibrium, characterized by Equation (1) and denote the common equilibrium price with n competing retailers by $p_M^*(n)$. Under the hybrid mode, we consider the semi-symmetric equilibrium in which all 3P sellers charge the same price. This equilibrium is characterized by Equation (1) for $i = 2, \dots, n$ and Equation (2) for the platform. We denote the platform's equilibrium price by $\hat{p}_P^*(n)$ and the common equilibrium price of 3P sellers by $\hat{p}_S^*(n)$. Throughout the analysis, parameters are restricted to configurations under which the relevant equilibria in both modes are interior, so that all sellers are active in equilibrium.

Proposition 1. *When $0 < \tau \leq \max \left\{ 0, 1 - \frac{2c(1+(n-2)\gamma)}{(n-1)\gamma} \right\}$, the upward force from the revenue-sharing channel dominates the downward force from avoiding the double-markup distortion. Consequently:*

- (i) *the platform charges a higher equilibrium price than 3P sellers: $\hat{p}_P^*(n) > \hat{p}_S^*(n)$;*
- (ii) *the platform's retail entry exerts less competitive pressure on 3P sellers' prices than entry by an otherwise identical 3P seller: $\hat{p}_S^*(n) - p_M^*(n - 1) > p_M^*(n) - p_M^*(n - 1)$.*

For commission rates above this threshold, the downward force from avoiding the double-markup distortion dominates, and both inequalities are reversed.

This result may appear counterintuitive: the revenue-sharing effect dominates when the commission rate τ is relatively low, even though a lower commission rate mechanically reduces the platform's commission revenue per dollar of 3P sales. The intuition is that 3P sellers' perceived marginal cost, $\frac{c}{1-\tau}$, increases convexly with the commission τ . Consequently, as the commission rate rises, the downward pricing force associated with eliminating the double-markup distortion becomes increasingly strong. Only when the commission rate is sufficiently high does this force become strong enough to outweigh the revenue-sharing effect, leading the platform to charge a lower price than 3P sellers. The same condition under which the revenue-sharing effect dominates and the platform charges a higher price than 3P sellers under hybrid mode also implies that platform entry exerts less competitive pressure on incumbent 3P prices than entry by an otherwise identical 3P seller.

In sum, the efficiency advantage from eliminating the double-markup distortion does not unambiguously lead the platform to price more aggressively. Which of these opposing forces dominates depends on the underlying market conditions. Greater product substitutability, a larger number of competitors, and a lower marginal cost all expand the range of commission rates over which the revenue-sharing effect dominates. Appendix A.2 provides numerical illustrations of how the relative importance of the two pricing forces varies across parameter configurations.

Proposition 2. *The platform's entry as a retailer strictly lowers the equilibrium prices charged by 3P sellers, $\hat{p}_S^*(n) < p_M^*(n-1)$.*

In our framework of price competition with differentiated products, the effect of platform entry on 3P sellers' prices can be decomposed into two components:

$$\underbrace{\hat{p}_S^*(n) - p_M^*(n-1)}_{\text{total effect of platform entry}} = \underbrace{p_M^*(n) - p_M^*(n-1)}_{\text{standard 3P entry effect}} + \underbrace{\hat{p}_S^*(n) - p_M^*(n)}_{\text{platform-specific pricing effect}}.$$

The first term on the right-hand side captures the standard entry effect associated with an increase in the number of competitors. The second term isolates the platform-specific pricing effect by holding the number of competitors fixed and comparing 3P sellers' equilibrium prices across the two modes. This difference is therefore driven by the platform's two distinctive pricing incentives relative to those of an otherwise identical 3P seller. Under the linear demand system, the standard entry effect is sufficiently negative to offset any upward platform-specific pricing effect. Consequently, the platform's retail entry strictly lowers the equilibrium prices charged by incumbent 3P sellers.

To conclude the analysis on equilibrium prices, we compare our results with those of [Hagiu et al. \(2022\)](#) and [Anderson and Bedre Defolie \(2024\)](#). [Hagiu et al. \(2022\)](#) consider asymmetric Bertrand competition among the platform, a superior seller, and several fringe sellers. In their model, the platform’s entry as a retailer “price-squeezes” the superior seller, forcing it to charge a lower price, because the platform does not incur the fixed transaction fee paid by 3P sellers and may also have a product-quality advantage over the fringe sellers. Similarly, Proposition 2 shows that the platform’s retail entry also always lowers 3P sellers’ equilibrium prices in our model. However, this result does not by itself imply that the platform’s own product “price-squeezes” 3P sellers. Such a price squeeze arises only when the platform’s advantage from avoiding the commission-induced double-markup distortion dominates the revenue-sharing effect, so that the platform-specific pricing effect is also negative. In this case, the platform charges a lower equilibrium price than 3P sellers and exerts additional downward pressure on their prices relative to entry by an otherwise identical 3P seller. Our analysis also identifies cases in which the revenue-sharing effect dominates, so that the platform does not exert such a price squeeze and instead charges a higher price than 3P sellers in the hybrid equilibrium. This possibility does not arise in [Hagiu et al. \(2022\)](#). Our analysis also differs from [Anderson and Bedre Defolie \(2024\)](#). Their monopolistic-competition framework shuts down the effect of entry on incumbent sellers’ market power. Consequently, the effect of the platform’s entry on 3P sellers’ prices operates entirely through the platform’s adjustment of the commission rate following its switch from the pure marketplace mode to the hybrid mode. Their framework therefore abstracts from the pricing interactions between the platform and 3P sellers that are central to our analysis.

3.3.2 Asymmetric Cost Pass-Through

We now investigate how the two pricing forces identified above generate asymmetries in the pass-through of cost shocks to the platform’s and 3P sellers’ prices under hybrid mode. We allow retailers to have heterogeneous marginal costs c_i , while maintaining the linear demand specification. Under this specification, the diversion ratio is identical for every pair of distinct products, so we write $\delta \equiv \delta_{ij}$ for all $i, j \in \{1, \dots, n\}, i \neq j$.

We first define the pass-through matrix for seller-specific cost shocks. Let $\hat{\mathbf{p}}^* = (\hat{p}_1^*, \dots, \hat{p}_n^*)^\top$ denote the vector of equilibrium prices under the hybrid mode, and let $\mathbf{c} = (c_1, \dots, c_n)^\top$ denote the vector of marginal costs. Then, $\rho \equiv \frac{\partial \hat{\mathbf{p}}^*}{\partial \mathbf{c}^\top}$ is the pass-through matrix, where each element $\rho_{ij} \equiv \frac{\partial \hat{p}_i^*}{\partial c_j}$ measures the pass-through of a shock to retailer j ’s marginal cost into retailer i ’s price. We refer to the diagonal elements ρ_{ii} as own-cost pass-through rates and the off-diagonal elements $\rho_{ij}, i \neq j$, as cross-cost pass-through rates.

To distinguish the two channels through which τ affects pass-through, we write τ as τ_D in

the 3P sellers' first-order condition in Equation (1) to capture the double-markup channel, and as τ_R in the platform's first-order condition in Equation (2) to capture the revenue-sharing channel. Substituting the linear demand system into these equilibrium conditions yields the following result.

Proposition 3 (Pass-Through Matrix). *The pass-through matrix under the hybrid mode is given by*

$$\rho \equiv \frac{\partial \hat{\mathbf{p}}^*}{\partial \mathbf{c}^\top} = M(\tau_R)^{-1} C(\tau_D),$$

where

$$M(\tau_R) \equiv \begin{bmatrix} 2 & -\delta(1 + \tau_R)\mathbf{1}_{n-1}^\top \\ -\delta\mathbf{1}_{n-1} & (2 + \delta)I_{n-1} - \delta J_{n-1} \end{bmatrix}, \quad C(\tau_D) \equiv \begin{bmatrix} 1 & \mathbf{0}_{n-1}^\top \\ \mathbf{0}_{n-1} & \frac{1}{1 - \tau_D} I_{n-1} \end{bmatrix}, \quad \text{and } \tau_R = \tau_D = \tau.^{14}$$

This decomposition highlights the distinct roles the two channels play in determining pass-through. First, through the double-markup channel, τ_D directly amplifies the cost shocks to 3P sellers by a factor of $1/(1 - \tau_D)$, as captured by the lower-right block of $C(\tau_D)$. In contrast, the platform's cost shock is not amplified. Then, through the revenue-sharing channel, τ_R strengthens the platform's strategic response to 3P sellers' prices, thereby making its price more responsive to cost shocks affecting 3P sellers. This can be seen from the matrix $M(\tau_R)$, which governs the strategic interactions among retailers' prices. Letting $BR_i(\mathbf{p}_{-i})$ denote retailer i 's best-response function implied by its first-order condition, we have $\frac{\partial BR_i}{\partial p_j} = \frac{\delta(1 + \tau_R)}{2} > \frac{\delta}{2} = \frac{\partial BR_i}{\partial p_j}$, $i, j \in \{2, \dots, n\}$, $i \neq j$. The inequality arises because an increase in 3P seller j 's price affects the platform's pricing incentive not only through standard strategic complementarity but also by increasing the opportunity cost $\tau \sum_{i=2}^n \hat{p}_i^* \delta$ of selling its own product. Through the revenue-sharing channel, the platform therefore has an additional incentive to raise its own price in response to an increase in p_j . Taken together, the two channels determine equilibrium pass-through through the decomposition $\rho = M(\tau_R)^{-1} C(\tau_D)$: $C(\tau_D)$ determines the effective cost shocks, while $M(\tau_R)^{-1}$ maps these shocks into equilibrium price responses by accounting for strategic interactions among prices.

We next compare pass-through between the platform and 3P sellers for both seller-specific cost shocks and an industry-wide cost shock. To define the latter, let ε denote a common shock that raises the marginal costs of all retailers one-for-one, so that $\frac{d\mathbf{c}}{d\varepsilon} = \mathbf{1}_n$. The pass-through of ε

¹⁴Here, $\mathbf{1}_k$ and $\mathbf{0}_k$ denote the k -dimensional column vectors of ones and zeros, respectively, I_k denotes the $k \times k$ identity matrix, and $J_k \equiv \mathbf{1}_k \mathbf{1}_k^\top$ denotes the $k \times k$ matrix of ones.

to retailer i 's equilibrium price is then given by

$$\rho_i \equiv \frac{d\hat{p}_i^*}{d\varepsilon} = \sum_{j=1}^n \frac{\partial \hat{p}_i^*}{\partial c_j} \frac{dc_j}{d\varepsilon} = \sum_{j=1}^n \rho_{ij}, \quad \forall i \in \{1, \dots, n\}.$$

Proposition 4. PASS-THROUGH ASYMMETRIES. *Under $\tau_R = \tau_D = \tau \in (0, 1)$, the pass-through rates exhibit the following asymmetries:*

- (1) *The platform's own-cost pass-through rate is lower than that of any 3P seller: $\rho_{PP} < \rho_{SS}$.*
- (2) *If $n \geq 3$, then, for any two distinct 3P sellers $S, S' \in \{2, \dots, n\}$, the cross-cost pass-through rates satisfy $\rho_{SP} < \rho_{SS'} < \rho_{PS}$: a 3P seller responds more strongly to another 3P seller's cost shock than to the platform's cost shock, whereas the platform responds more strongly to a 3P seller's cost shock than does another 3P seller.*
- (3) *The pass-through rate of an industry-wide cost shock to the platform's price is lower than that to any 3P seller's price: $\rho_P < \rho_S$.*

The intuition is simple. Before accounting for strategic feedback, a one-unit increase in the platform's marginal cost raises its price by $\frac{1}{2}$, whereas the same increase in a 3P seller's marginal cost raises its price by $\frac{1}{2(1-\tau_D)} > \frac{1}{2}$. Thus, the double-markup channel associated with τ_D tends to make a 3P seller's own-cost pass-through exceed the platform's. By contrast, the revenue-sharing channel tends to increase the platform's own-cost pass-through because it strengthens the platform's strategic response to 3P sellers' prices. When both channels are governed by the same commission rate, the cost-shock amplification effect dominates because it operates directly on the 3P seller's pricing condition, whereas the revenue-sharing effect operates only indirectly through strategic price feedback, with the effect of the original cost shock attenuated at each stage of the feedback process. Hence, $\rho_{PP} < \rho_{SS}$.

Regarding the cross-cost pass-through asymmetries, consider first the responses of the platform and an unshocked 3P seller to the same 3P seller's cost shock. Both responses are subject to the same cost-shock amplification effect, but the platform responds more strongly because revenue sharing makes its best response more sensitive to the shocked seller's price. Hence, $\rho_{PS} > \rho_{SS'}$. Next, a 3P seller responds more strongly to another 3P seller's cost shock than to the platform's cost shock because the former is amplified by $\frac{1}{1-\tau_D}$ before entering the strategic price-feedback process, whereas the latter enters this process without such amplification. Hence, $\rho_{SS'} > \rho_{SP}$.

Finally, the seller-specific pass-through results allow us to interpret the common-shock pass-through asymmetry. An industry-wide cost shock combines each retailer's own-cost response with its responses to all rivals' cost shocks. The higher own-cost pass-through of 3P sellers tends

to make their prices respond more strongly to the common shock than the platform’s price does, whereas the cross-cost asymmetry works in the opposite direction. The key distinction is that an own-cost shock enters a retailer’s pricing decision directly, whereas a rival’s cost shock must first affect the rival’s price and then pass through the retailer’s strategic response to that price. The strategic responses to all rival prices, $\sum_{j \neq i} \partial BR_i / \partial p_j$, sum to $(n - 1)\delta/2 < 1$ for any 3P seller and to $(n - 1)\delta(1 + \tau_R)/2 < 1$ for the platform. Thus, even in aggregate, rivals’ cost shocks are attenuated by this additional transmission layer. Consequently, when the cost-amplification and revenue-sharing channels are governed by the same commission rate, the own-cost pass-through asymmetry dominates, implying $\rho_P < \rho_S$.

3.4 Empirical Implications

The theoretical analysis yields two key findings that guide our empirical analysis:

- **Finding 1 (Differential Competitive Effects of Entry):** Although platform entry intensifies competition relative to no entry, the platform does not necessarily compete as aggressively as an otherwise identical 3P entrant.
- **Finding 2 (Asymmetric Pass-Through):** The platform’s retail price is less responsive to industry-wide cost shocks than 3P sellers’ prices.

In the next sections, we examine whether these patterns are present in the Amazon marketplace data. To study the differential competitive effects of entry, in Section 4, we exploit repeated within-ASIN changes in Amazon’s and 3P sellers’ retail availability and estimate event-time patterns in incumbent 3P prices around these changes. Then, to examine the asymmetric pass-through implication, in Section 5, we use shocks to commodity prices to compare the responsiveness of Amazon’s and 3P sellers’ prices to exogenous input cost changes.

4 Empirical Evidence: Price Levels

4.1 Empirical Approach

Our objective is to compare changes in the observed 3P price around transitions in Amazon’s retail availability and around comparable transitions in 3P seller availability. Estimating the effect of seller participation on market outcomes confronts a classic identification challenge: seller entry and exit are endogenous (Bresnahan and Reiss, 1991). In particular, a cross-sectional comparison of ASINs with and without an observed Amazon retail offer would be difficult to interpret because Amazon’s retail presence is selective across products (see Fact 1 in Section 2.3) and

likely correlated with product-specific demand, inventory, and 3P competitive conditions. Similar concerns apply to the entry and exit of 3P sellers.

To avoid relying on cross-ASIN differences in seller participation, we exploit the repeated within-ASIN changes in seller availability documented as Fact 2. Our empirical design relates to reduced-form studies of how entry affects prices (e.g., [Arcidiacono, Ellickson, Mela and Singleton, 2020](#); [Crawford et al., 2022](#); [Fischer, Martin and Schmidt-Dengler, 2025](#)), but the source of identifying variation differs from the canonical entry setting. The relevant changes on Amazon are frequent, short-lived, and reversible: a given seller type may appear or disappear on the same ASIN multiple times, and may reflect stockouts or replenishment frictions. We therefore employ data at the daily level and interpret “entry” and “exit” as changes in observed seller availability, not as permanent product-market entry or exit.¹⁵ As treatment status is reversible rather than absorbing, and products can undergo treatment repeatedly, we estimate short-window event-time patterns around well-defined transitions in seller composition.

We implement the event-study analysis in two steps. First, we study Amazon entry and exit events alone to characterize how incumbent 3P prices evolve around changes in Amazon’s retail availability. Second, we estimate stacked event studies that include both Amazon and 3P entry or exit events in the same regression, allowing us to directly compare the corresponding price responses. We next describe the sample construction, event definitions, and estimation details for these two exercises.

Sample selection and outcome variable. We implement the main analysis in the coffee category, where Amazon retail participation is frequent and where most products have reviews, indicating high turnover (see Table 1).¹⁶ For all analyses in this subsection, we restrict attention to ASINs that exhibit meaningful variation in seller composition (specifically, we require that, at any point in our sample, Amazon and 3P sellers are simultaneously active for at least 10 consecutive days, and only 3P sellers are active for at least 10 consecutive days). We also exclude the brand that displays a brand-level assortment change in our period of observation (as discussed Section 2.3).

As Keepa does not report the full vector of 3P prices, we use the lowest observed 3P offer under each fulfillment mode (FBA and FBM) as the empirical counterpart to the model’s equilibrium 3P price. Under the assumption that this frontier price is a sufficient empirical proxy for the relevant competitive 3P price, movements in this frontier provide reduced-form evidence on

¹⁵We hence use “entry” and “exit” as shorthand for changes in observed seller availability within an ASIN, rather than permanent product-market entry or exit. We restrict the analysis to ASINs for which Amazon had already been observed as a retailer before the start of our sample. This restriction allows us to focus on recurrent changes in Amazon availability rather than first observed Amazon entry during the sample period.

¹⁶As opposed to apparel or electronics, grocery categories like coffee are also not exposed to contemporaneous entry by low-cost import platforms such as Temu or Shein.

the price-level margin emphasized by the model.

Event definition. We define a clean “Amazon entry” event as a sequence in which (i) no Amazon offer is observed and the relevant 3P price is observed for at least K consecutive days before the event, and (ii) both an Amazon offer and the relevant 3P price are observed for at least K consecutive days afterward. We define “Amazon exit” analogously. Because we do not observe the identities of all active 3P sellers in the data, we define “3P entry” as a one-unit change in the 3P seller count: an increase of exactly one for entry, and a decrease of exactly one for exit. For 3P events, Amazon remains absent and the relevant 3P price remains observed throughout the pre- and post-event periods. This implies that the window length K defines our “qualifying” entry and exit events; in the baseline specification, $K = 10$, while alternative specifications are shown in the Appendix.

Econometric specification. We perform two types of “sharp” event studies (Kleven, Landaís and Søgaaard, 2019) around symmetric windows of these events. For each event e , let $i(e)$ denote the ASIN associated with the event, and let T_e denote the event date. Define relative event time as

$$r = t - T_e$$

Our first specification focuses only on Amazon entry and exit to examine whether either event is followed by a response in the lowest 3P price:

$$y_{er}^m = \sum_{\substack{j=-10 \\ j \neq -1}}^{10} \beta_j^m \mathbf{1}\{r = j\} + \alpha_e + \varepsilon_{er}^m \quad (3)$$

In our baseline analysis, the outcome variable y_{er}^m is the logarithm of the lowest observed 3P price under fulfillment mode m , $p_{it}^{3P,m}$ for $m \in \{\text{FBA}, \text{FBM}\}$. α_e is an event fixed effect¹⁷, and the coefficient β_j^m measures the change in the log lowest 3P price at event day j , relative to day -1 , within the same event.

Second, we stack Amazon and 3P events in a common regression, estimated separately for entry and exit. To rule out simultaneous changes in seller composition, we restrict the stacked sample to event windows in which the total number of active sellers (including Amazon when present) is constant within the pre- and post-event periods and changes by exactly one at the event date. Let A_e be an indicator equal to one for Amazon events and zero for 3P seller-count

¹⁷This fixed effect is more granular than ASIN fixed effects would be, given that ASINs may undergo multiple relevant entry or exit events. Using calendar-date and ASIN fixed effects instead yields very similar results.

events. We estimate the following specification separately for entry and exit:

$$y_{er}^m = \sum_{\substack{j=-10 \\ j \neq -1}}^{10} \left[\beta_{j,A}^m \mathbf{1}\{r = j\} \mathbf{1}\{A_e = 1\} + \beta_{j,3P}^m \mathbf{1}\{r = j\} \mathbf{1}\{A_e = 0\} \right] + \alpha_e + \varepsilon_{er}^m \quad (4)$$

The primary outcome variable y_{er}^m is once again the logarithm of $p_{it}^{3P,m}$ for $m \in \{\text{FBA}, \text{FBM}\}$. The coefficients $\beta_{j,A}^m$ and $\beta_{j,3P}^m$ trace the Amazon and 3P event-time paths, respectively. The formal Amazon-vs.-3P difference at event time j is $\beta_{j,A}^m - \beta_{j,3P}^m$.¹⁸

Identification and diagnostics. A causal interpretation requires that, among qualifying events, the transition date is unrelated to contemporaneous or anticipated shocks to the counterfactual 3P price frontier, that 3P sellers do not adjust their behavior in anticipation, and that, absent the transition, the price frontier would remain at its pre-event path over the short event window.

As shown in Fact 2 (Figure 3, see also Appendix C.1), Amazon’s recurrent presence and absence spells have heterogeneous durations, which reduce concerns that permanent assortment changes or a few platform-wide shocks drive the variation. We further assess whether qualifying changes in Amazon’s availability are predictable from observable pre-event price and demand dynamics using hazard models for the probability that Amazon enters or exits an ASIN on the following day, conditional on the product’s current seller regime. Our estimates show little relation between Amazon transitions and recent changes in prices or sales rank, although the hazards vary with some lagged levels, weekdays, and marketplace activity (Tables C.1 and C.2). Finally, the pre-event coefficients reported below provide complementary diagnostics for anticipation and pre-existing trends and, in the stacked design, for differential pre-trends.

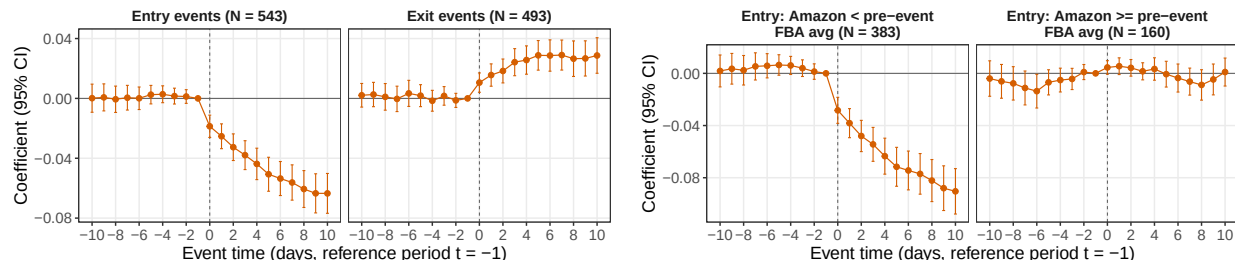
4.2 Effect of Amazon Entry and Exit on 3P Prices

We first estimate event-study specifications for Amazon entry and exit (Equation 3). Our baseline specification uses the lowest 3P seller price under FBA as an outcome variable, as sellers operating under FBA are arguably closer substitutes to Amazon. Panel (a) in Figure 5 plots the estimated event-study coefficients, while the corresponding coefficient estimates and fit statistics are reported in Table D.3. The pre-event coefficients are small and jointly statistically indistinguishable from zero across the entry and exit specifications (Table D.1). For entry events, we document a strong decline in the lowest 3P price by almost 6% (3% for FBM prices). For exit

¹⁸As noted in Appendix B, prior to 2026, Keepa updated FBA and FBM prices at somewhat lower frequency than other price types. Event-time dynamics therefore cannot speak to the day-by-day speed of adjustment, but note that Musolf (2026) finds high re-pricing frequencies, suggesting that price adjustments occur faster than our figures suggest.

events, we find the opposite pattern: mostly flat pre-trends are followed by an increase in the price by up to 3% (similar for FBM prices).

Figure 5: Event study of entry and exit by Amazon as a seller of a given ASIN; outcome variable: $\log(p^{3P,FBA})$



(a) Lowest 3P price under FBA over event window.

(b) Split by Amazon's price at entry.

Notes: Figures show coefficient estimates stemming from Equation 3, as well as 95% confidence intervals computed using two-way clustered standard errors (ASIN and calendar-date). Panel (a) displays specifications (1) and (3) of Table D.3. Panel (b) shows a sample split of specification (1) using, first, observations in which Amazon enters with a price that is at least as high as the pre-event average lowest FBA price, and second using observations in which Amazon enters with a price that is lower. All regressions are estimated using “qualifying” events in the daily coffee data, as defined earlier.

One possibility is that the 3P price response to Amazon entry reflects Amazon's platform identity rather than the competitive position of its offer. To assess whether the response varies with Amazon's entry price, Panel (b) splits entry events into two groups: Amazon enters with a relatively high price (defined as a price above the pre-event average of 3P FBA prices; 160 out of 543 events) or with a low price (383 out of 543 events). We can see that the responses are strikingly different: price declines upon entry are significant only if Amazon enters with a price that is lower than the average FBA price in the pre-window period¹⁹. The substantially stronger price response when Amazon enters at a relatively low price is consistent with 3P sellers responding to the competitive position of Amazon's offer and, more generally, with the price-mediated competitive channel illustrated by the theoretical model.²⁰

We supplement these results with two additional checks. First, on the event date, total seller count rises by about one at Amazon entry and falls by about one at exit, consistent with no additional same-day change in 3P participation alongside Amazon entry or exit (see panel (b) of Figure D.2). Second, demand shocks are potential confounders that could cause price changes. To gauge this possibility, we examine the patterns of sales ranks, which are informative about demand dynamics around events. A lower sales rank indicates stronger sales performance. The

¹⁹The analogous finding arises in the case of exit events (see Figure D.3 in the Appendix).

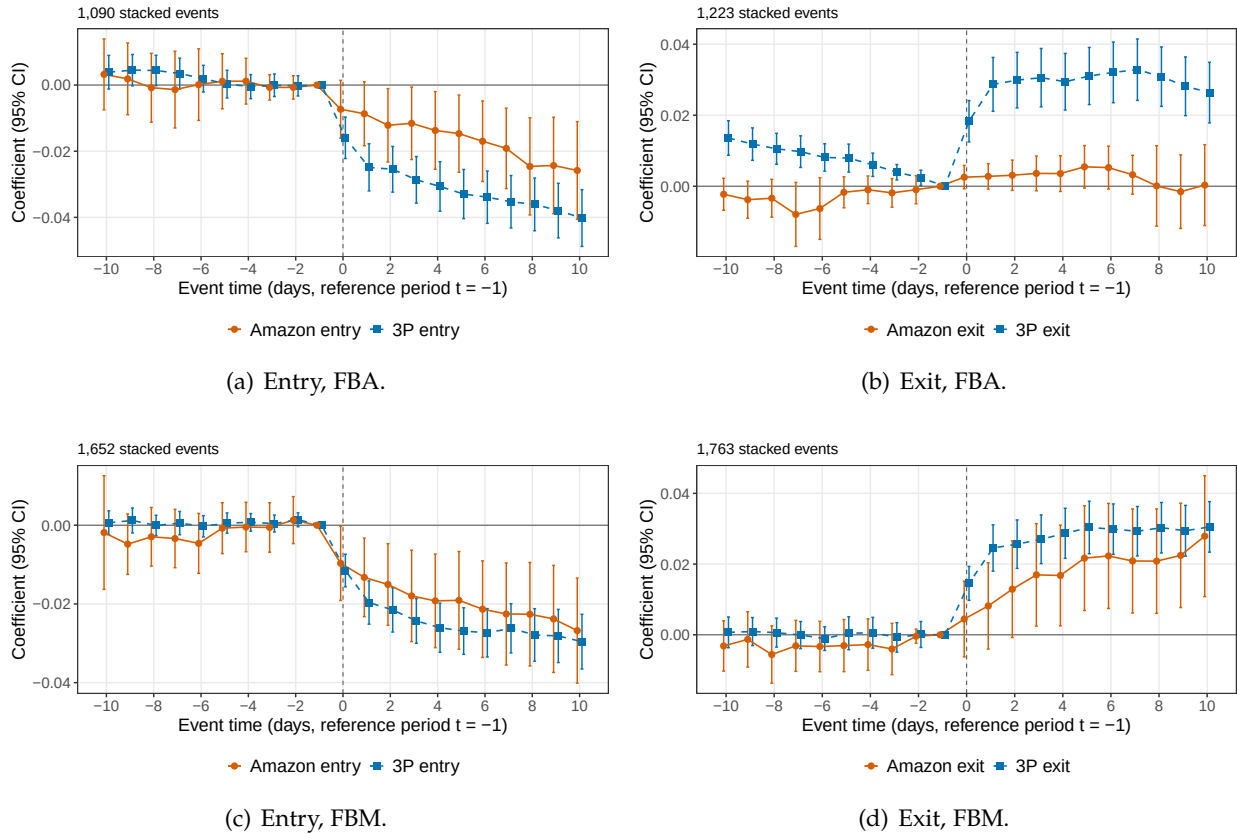
²⁰Panel (a) in Figure D.2 shows that analysis using FBM price exhibits qualitatively similar pattern to estimation using FBA prices. Further, Figure D.3 conducts further subsample analysis by showing the analogue of Figure 5 (b) for exit, as well as an entry analysis split by seller count.

results show that sales rank stays roughly flat during the pre-period (see panel (c) of Figure D.2 in the Appendix), providing little evidence of systematic pre-existing demand trends around the qualifying transitions.

To summarize, our findings show that intermittent changes in Amazon’s retail availability are associated with sizable and reversible movements in the lowest 3P price, with stronger responses when Amazon enters at a relatively competitive price. This heterogeneity is consistent with a price-mediated competitive channel.

4.3 Differential Effects of Amazon and 3P Entry and Exit

Figure 6: Stacked event study of entry and exit by Amazon and 3P seller; outcome variables: $\log(p^{3P,FBA})$ and $\log(p^{3P,FBM})$



Notes: Figures show coefficient estimates stemming from Equation 4 with the lowest 3P sellers’ price as an outcome variable, as well as 95% confidence intervals computed using two-way clustered standard errors (ASIN and calendar-date). All regressions are estimated using “qualifying” events in the daily coffee data, as defined in the text.

Our objective is to compare 3P sellers’ pricing response to Amazon entry with their response to 3P entry, which we can do cleanly by estimating equation 4.²¹ Figure 6 plots both outcomes,

²¹As mentioned above, we additionally require that the marketplace seller count is constant during the event window except for $t = -1$, which means that for Amazon, we now use only a subset of the events studied before.

Table 2: Average post-event effects in stacked Amazon-vs-3P event studies

Event	3P price	Window	3P event	Amzn event	$ \text{Amzn} - \text{3P} $	p-value: $ \text{Amzn} = \text{3P} $	Events: Amzn / 3P
Entry	FBA	0–10	-3.11*** (0.35)	-1.63*** (0.56)	-1.48** (0.65)	0.023	147 / 943
Entry	FBA	3–10	-3.45*** (0.38)	-1.89*** (0.61)	-1.56** (0.71)	0.028	147 / 943
Entry	FBM	0–10	-2.45*** (0.29)	-1.92*** (0.57)	-0.52 (0.61)	0.387	207 / 1445
Entry	FBM	3–10	-2.70*** (0.31)	-2.17*** (0.62)	-0.54 (0.67)	0.420	207 / 1445
Exit	FBA	0–10	2.89*** (0.38)	0.26 (0.26)	-2.63*** (0.46)	0.000	222 / 1001
Exit	FBA	3–10	3.02*** (0.41)	0.25 (0.32)	-2.77*** (0.51)	0.000	222 / 1001
Exit	FBM	0–10	2.73*** (0.33)	1.78*** (0.68)	-0.95 (0.76)	0.209	232 / 1531
Exit	FBM	3–10	2.94*** (0.35)	2.12*** (0.74)	-0.82 (0.82)	0.317	232 / 1531

Notes: Estimates are average event-time coefficients over the indicated post-event window, relative to $t = -1$. Estimates and standard errors are reported as log points multiplied by 100. 3P denotes the 3P-seller-event path; Amzn denotes the implied Amazon-event path. $|\text{Amzn}| - |\text{3P}|$ compares the absolute magnitudes of the implied Amazon-event and 3P-event paths; negative values indicate that the Amazon-event response is smaller in absolute value. Event counts are unique event IDs in the estimation sample. Clustered (ASIN $\times$ calendar date) standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

while Table 2 reports average post-event effects and contrasts the coefficients of different event types. Across outcomes, the point estimates suggest smaller 3P price movements after Amazon entry/exit than after 3P seller entry/exit. The contrast is strongest when considering prices under FBA: over days 0–10, the lowest 3P price falls by around 3.1% after 3P entry and 1.6% after Amazon entry. It rises again by nearly 3% after 3P exit, but stays essentially unchanged after Amazon exit. Given the apparent pre-trends for the FBA specification, we prefer the FBM estimates which point in the same direction while having flat pre-event paths; however, here, the Amazon–3P differences are not statistically significant. Overall, we read the stacked evidence as consistent with weaker Amazon-induced price responses. The patterns therefore provide suggestive evidence for the theoretical possibility that platform entry may exert less downward pressure on 3P prices than entry by an otherwise identical 3P seller.

4.4 Robustness and Additional Results

Alternative qualifying-spell requirements. Our baseline stacked event studies set $K = 10$, requiring the relevant seller compositions to persist for at least 10 consecutive days around each qualifying transition. We assess the sensitivity of the results to this requirement by re-estimating the event studies using $K = 5$ and $K = 15$. Since K enters the qualifying-event definition itself, varying K also changes the set of transitions included in the estimation, rather than merely altering the displayed event-time window. Figure D.1 in the Appendix reports the corresponding estimates. The baseline patterns are generally more closely reproduced under the more restrictive definition $K = 15$ than under $K = 5$. For FBA prices, the $K = 15$ estimates preserve the baseline ordering for both entry and exit. The FBM entry estimates also remain relatively close across

Amazon and 3P events, as in the baseline specification, although the FBM exit differential is less stable and is estimated imprecisely, which is consistent with the decreased number of events in this case. By contrast, when admitting shorter-lived regimes under $K = 5$, the FBA entry differential reverses sign, while the difference between Amazon and 3P entry under FBM becomes substantially larger than in the baseline. These results suggest that the estimated entry differential is sensitive to the inclusion of relatively short-lived qualifying-spell, whereas transitions between more persistent regimes produce patterns closer to the baseline estimates.

Additional market outcomes. Panels (c) and (d) of Figure D.2 examine changes in product-level sales rank and Buy Box assignment around Amazon entry and exit. As stated in previous paragraph, a lower sales rank indicates stronger sales performance, the decline in log sales rank following Amazon entry, together with the corresponding increase following exit, is consistent with Amazon availability being associated with improved sales performance or visibility for the ASIN. However, these estimates do not reveal how sales are allocated between Amazon and 3P sellers or distinguish between demand expansion and substitution across offers. Amazon entry also raises the probability that Amazon occupies the Buy Box by approximately 30 percentage points, while Amazon exit produces a corresponding decline. It shows that Amazon’s participation may change which offer is prominently displayed to consumers, but Amazon does not automatically win the Buy Box whenever it enters.

Alternative specifications. For the effects pertaining to Amazon’s presence, we assess whether a similar association holds in other product categories and outside our somewhat selective qualifying events. A two-way fixed effects specification (Tables D.4 and D.5) yields uniformly negative and statistically significant associations of Amazon presence with FBA prices, with magnitudes ranging from 2.8% for TVs to 10.8% for pasta. While this specification alone cannot support the causal interpretation of the event studies, the broader pattern is consistent with our event-study evidence in the coffee category.

5 Empirical Evidence: Asymmetric Pass-Through of Cost Shocks

5.1 Empirical Approach

To examine the model’s prediction of asymmetric common-shock pass-through on a hybrid platform, we exploit upstream cost shocks that are common to Amazon and 3P sellers and compare how the two seller types adjust their posted prices. Our reduced-form design therefore traces the dynamic response of posted retail prices to commodity-cost shocks that are plausibly exogenous to individual ASINs and common across seller types.

We focus on variation in coffee-bean and cocoa-bean prices for three reasons. First, both are major internationally traded commodities with transparent and frequently observed market prices. Second, each has a direct downstream counterpart in our data, namely ground coffee and baking cocoa, for which the raw commodity is a salient and economically important input. Third, both commodity prices exhibit substantial variation over our sample period, providing meaningful time-series variation for estimating pass-through. Taken together, these features make coffee- and cocoa-bean prices well suited as common upstream cost shifters for comparing the pricing responses of Amazon and 3P sellers.²²

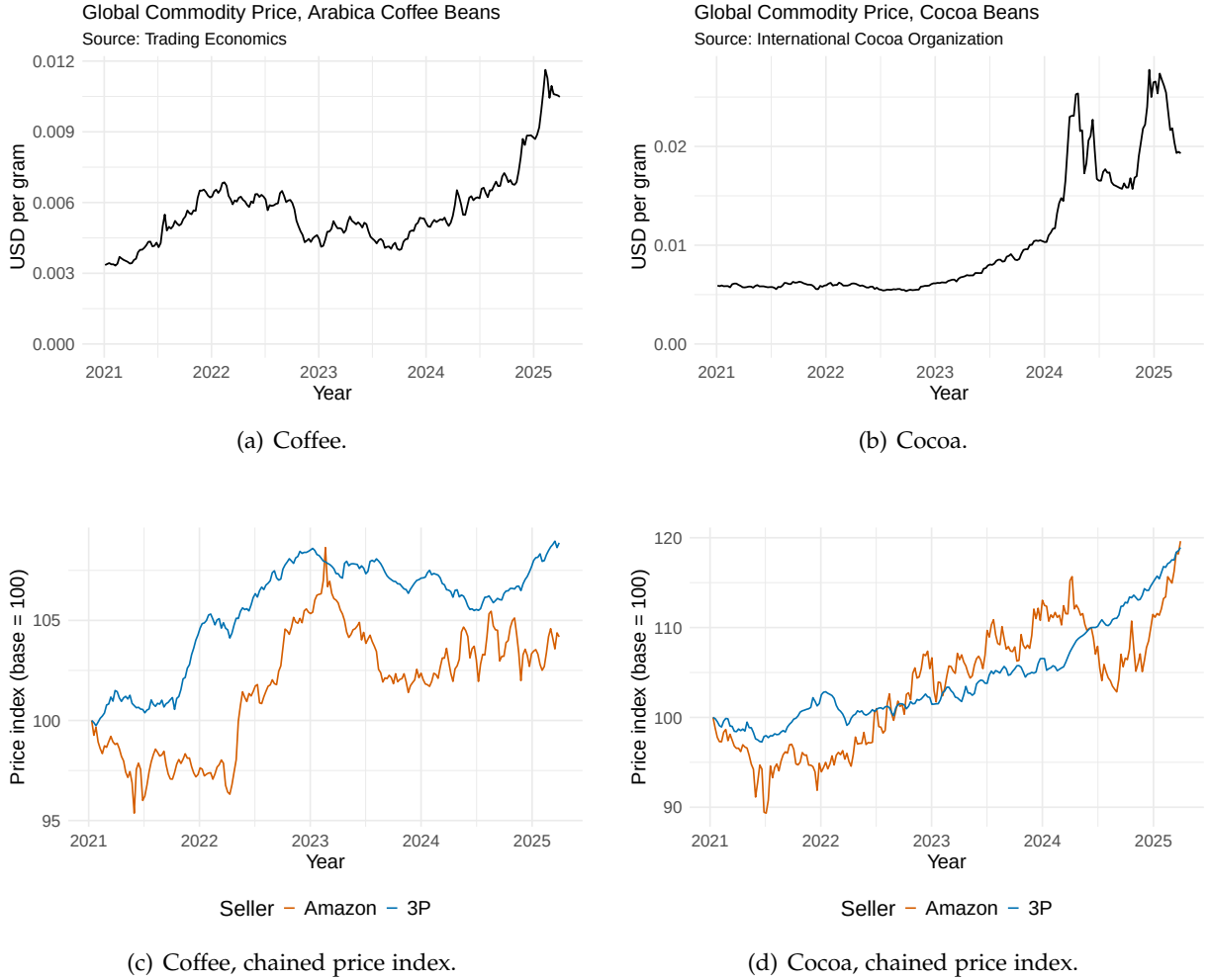
We obtain daily commodity-price data from Trading Economics (tradingeconomics.com) and the International Cocoa Organization (www.icco.org; see Appendix B.2). To express commodity and retail prices in comparable units, we use conversion factors that account for roasting losses in coffee and bean-to-powder processing yields in cocoa, thereby translating raw-bean prices into implied commodity input costs per gram of the corresponding downstream product. This approach follows the related pass-through literature (e.g., Nakamura and Zerom, 2010; Sangani, 2026; Bonnet et al., 2013). The adopted conversion factors and their sources are documented in Appendix Table B.1. Panels (a) and (b) of Figure 7 plot the resulting commodity-cost series. During our January 2021-March 2025 sample period, the major increases in coffee and cocoa prices were widely attributed to adverse supply conditions, including poor weather in Brazil for coffee and in Côte d’Ivoire and Ghana for cocoa.²³ This supply-side origin supports the interpretation of the commodity price changes as common upstream cost shifters.

We next provide a descriptive summary of posted-price changes. Cross-sectional means or medians would combine within-ASIN repricing with changes in the mix of ASINs offered by Amazon and 3P sellers. Panels (c) and (d) of Figure 7 therefore report chained Jevons indices for Amazon’s posted price and the lowest observed 3P offer, which are less sensitive to changes in product composition. For each price series and pair of adjacent calendar weeks, we compare an ASIN’s price with its own price in the preceding week, conditional on that price series being observed for the ASIN in both weeks. We then take the equal-weighted geometric mean of these within-ASIN price relatives and chain the resulting weekly links. Both product categories exhibit an upward trend in within-ASIN posted prices against the backdrop of large increases in

²²Prior work has also used coffee-bean price shocks to study pass-through (Bettendorf and Verboven, 2000; Nakamura and Zerom, 2010; Bonnet, Dubois, Villas-Boas and Klapper, 2013; Sangani, 2026). We do not study other products affected by cocoa-bean prices, such as chocolate bars, because these products often involve a large share of so-called “local costs” (Nakamura and Zerom, 2010), such as advertising expenditures. In addition, chocolate manufacturers have reportedly adjusted product recipes in response to input-price shocks (see <https://www.ft.com/content/169c9fca-1a7a-4c0f-84e9-398777335629>, last accessed February 26, 2025), which further limits the suitability of these products for studying price pass-through.

²³See <https://www.ft.com/content/4ea3116d-6a55-4d14-8a47-8988209dc1e5> and <https://www.ft.com/content/8a917bea-8ce3-11e3-ad57-00144feab7de> (last accessed October 10, 2025). We verified that no material changes in U.S. tariffs on the relevant coffee and cocoa inputs occurred during the sample period.

Figure 7: Commodity prices and price indices of products sold on Amazon



Notes: Panels (a) and (b) plot weekly average prices for Arabica coffee beans and cocoa beans, respectively, adjusted by the conversion factors discussed in Appendix B.2. Panels (c) and (d) report seller-type-specific chained matched-ASIN Jevons price indices. For seller type s and adjacent calendar weeks $t - 1$ and t , let $\mathcal{M}_{s,t}$ denote the set of ASINs for which the relevant weekly average posted price is observed in both weeks, and let $N_{s,t} = |\mathcal{M}_{s,t}|$. The bilateral link is $J_{s,t} = \exp \left\{ \frac{1}{N_{s,t}} \sum_{i \in \mathcal{M}_{s,t}} [\log(p_{i,s,t}) - \log(p_{i,s,t-1})] \right\}$. Thus, each ASIN for which the relevant price series is observed in both adjacent weeks receives equal weight in that weekly link. We chain the adjacent-week links and normalize each series to 100 in the first full sample week of 2021. An ASIN observed in only one of two adjacent weeks does not enter that bilateral link. The set $\mathcal{M}_{s,t}$ may change across links, so the indices are rolling adjacent-week indices rather than fixed-basket indices over the full sample period.

commodity prices. In coffee, the index for the lowest 3P offer rises more than the Amazon index. In cocoa, both indices rise substantially and finish at broadly similar levels, although their paths differ. Taken together, these descriptive patterns point to a certain degree of heterogeneity in both the magnitude and timing of price adjustment across seller types: the divergence is particularly visible in coffee, while in cocoa similar cumulative changes mask different adjustment paths. Because the seller-specific indices may rely on different matched-ASIN sets, we next turn to the

regression analysis for formal evidence on differential pass-through.²⁴

We treat coffee as our primary empirical setting, as its larger sample allows for more precise estimates, and as the coffee commodity price exhibits more variation across the whole sample period. Cocoa provides a second application based on a second, independent commodity-cost shock.

Econometric specification. We estimate pass-through using a standard distributed-lag specification, which is widely used in the pass-through literature (e.g., [Besanko, Dubé and Gupta \(2005\)](#); [Nakamura and Zerom \(2010\)](#); [Conlon and Rao \(2020\)](#); [Alvarez, Cavallo, MacKay and Mengano \(2024\)](#); [Sangani \(2026\)](#)). This specification provides a flexible reduced-form characterization of the dynamic response of retail prices to commodity-cost shocks. In particular, as our goal is to compare pass-through across seller types while holding the product-week support fixed, we stack Amazon’s posted price and the lowest 3P seller price into a seller-type panel. The stacked specification provides an apples-to-apples comparison across seller types by estimating both price responses on the same set of observed ASIN-week cells.

Let $s \in \{3P, A\}$ index seller type, where A denotes Amazon, and let $\mathbf{1}\{s = A\}$ indicate that the seller type is Amazon. In our baseline specification, we retain only ASIN-week cells for which the weekly price change is observed for *both* seller types – i.e., where they have common support over ASIN-weeks – so that each retained ASIN-week contributes one observation for Amazon and one for the 3P offer. For each product category and each window length $K \in \{6, 12, 24, 48\}$, we estimate

$$\Delta p_{ist} = \sum_{k=0}^{K-1} \beta_k \Delta c_{t-k} + \sum_{k=0}^{K-1} \delta_k (\mathbf{1}\{s = A\} \times \Delta c_{t-k}) + \mu_{is} + \varepsilon_{ist}, \quad (5)$$

where Δp_{ist} denotes the change in the posted retail price per gram for ASIN i and seller type s between weeks $t = 1$ and t . For $s = 3P$, the price indicates is the lowest observed 3P price, while for $s = A$, the price corresponds to Amazon’s posted price. The variable Δc_t is the weekly change in the relevant raw-commodity price in dollars per gram after using the abovementioned the conversion factors ([Appendix B.2](#)). The fixed effect μ_{is} is specific to an ASIN-seller-type price series and allows Amazon’s and the 3P price series for the same ASIN to have different average weekly price changes.

For 3P sellers, $\mathbf{1}\{s = A\} = 0$, so β_k is the pass-through of the commodity-cost change at lag k . For Amazon, the corresponding pass-through is $\beta_k + \delta_k$. The cumulative pass-through

²⁴A common observation is that rising prices can affect package sizes (“shrinkflation”); however, we do not find any evidence of a change in the prevailing package size distribution over the sample period.

measures are:

$$\hat{\rho}_{3P}(K) \equiv \sum_{k=0}^{K-1} \hat{\beta}_k, \quad (6)$$

$$\hat{\rho}_A(K) \equiv \sum_{k=0}^{K-1} (\hat{\beta}_k + \hat{\delta}_k), \quad (7)$$

$$\hat{\rho}_A(K) - \hat{\rho}_{3P}(K) = \sum_{k=0}^{K-1} \hat{\delta}_k. \quad (8)$$

Thus, expression 8 characterizes the cumulative Amazon-minus-3P pass-through differential, and a negative value indicates that Amazon passes through less of the commodity-cost shock than 3P sellers. Given that downstream prices and upstream commodity costs are measured in the same units in our data, the specification has a direct economic interpretation: the sum of the coefficients measures the cumulative pass-through of a one-unit increase in commodity cost per gram into the product price per gram.

In light of [Sangani \(2026\)](#), who shows that for common commodity shocks, pass-through may appear incomplete in logs even when it is close to complete in levels, we take the levels specification as our baseline. We confirm that first differences in commodity prices are stationary for both cocoa and coffee using an augmented Dickey-Fuller test.

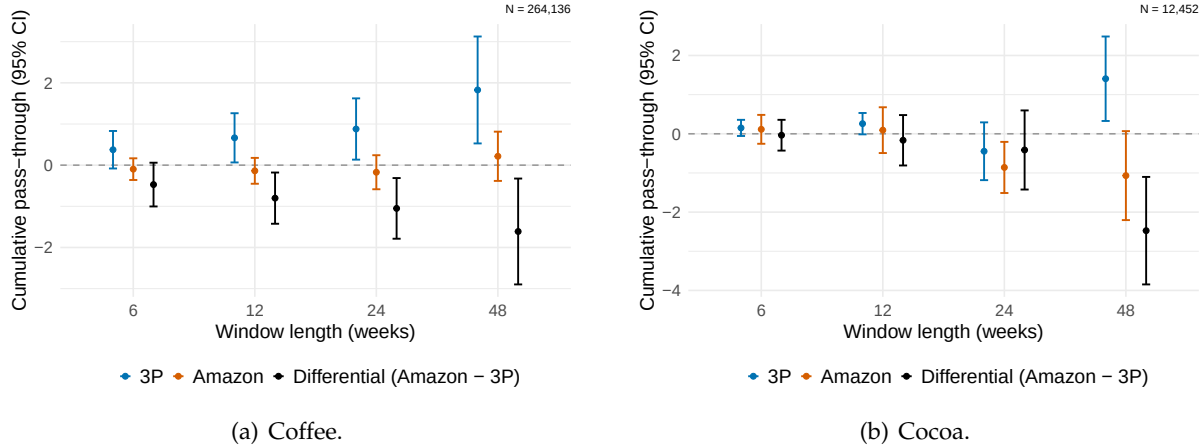
5.2 Differential Cost Pass-Through by Seller Type

Figure 8 and Table 3 report the cumulative pass-through estimates from the baseline specification. The central result is a negative Amazon-minus-3P pass-through differential: Conditional on the same set of ASIN-week observations, Amazon transmits less of the common commodity-cost shock into posted retail prices than the lowest 3P offer does.

The evidence is strongest for coffee. The estimated Amazon-minus-3P differential is negative at every horizon and is statistically significant at the 12-, 24-, and 48-week horizons. The gap persists as the horizon lengthens, suggesting that it is not eliminated by allowing additional time for Amazon’s price adjustment. At the 48-week horizon, the estimated cumulative pass-through is 1.827 for 3P sellers and 0.215 for Amazon, corresponding to an Amazon-minus-3P differential of -1.612. In line with [\(Sangani, 2026\)](#), long-run pass-through in levels is statistically indistinguishable from one for 3P sellers, but not for Amazon.

The estimates for cocoa point in the same qualitative direction but are, given the much smaller sample size, considerably less precise; the Amazon-minus-3P differential is statistically distinguishable from zero only at the 48-week horizon.

Figure 8: Cumulative commodity-cost pass-through by seller type



Notes: Cumulative pass-through estimates from distributed-lag regressions of weekly changes in posted retail prices on weekly changes in converted commodity costs. At window length K , the contemporaneous coefficient and first $K - 1$ lag coefficients are summed. Both variables are measured in dollars per gram, so an estimate of one denotes full pass-through. Regressions use a matched sample of ASIN-week cells with observed weekly price changes for both Amazon's posted price and the lowest 3P price, and include $\text{ASIN} \times \text{seller-type}$ fixed effects. "Differential (Amazon - 3P)" is the difference between the two estimates; negative values indicate lower pass-through by Amazon. Vertical bars show 95% confidence intervals based on standard errors clustered by ASIN and calendar week.

Table 3: Cumulative commodity-cost pass-through by seller type

	$K = 6$	$K = 12$	$K = 24$	$K = 48$
Panel A. Coffee				
3P	0.375 (0.232)	0.664** (0.305)	0.879** (0.380)	1.827*** (0.663)
Amazon	-0.097 (0.134)	-0.138 (0.160)	-0.172 (0.211)	0.216 (0.305)
Differential (Amazon - 3P)	-0.472* (0.271)	-0.801** (0.318)	-1.050*** (0.376)	-1.612** (0.656)
Observations	264,136	264,136	264,136	264,136
Panel B. Cocoa				
3P	0.150 (0.105)	0.258* (0.139)	-0.446 (0.376)	1.406** (0.550)
Amazon	0.114 (0.188)	0.093 (0.298)	-0.859*** (0.333)	-1.068* (0.580)
Differential (Amazon - 3P)	-0.036 (0.200)	-0.165 (0.328)	-0.413 (0.515)	-2.474*** (0.700)
Observations	12,452	12,452	12,452	12,452

Notes: Distributed-lag regressions of weekly changes in posted retail prices on weekly changes in converted commodity costs; each column is a separate regression at window length K . 3P reports $\sum_{k=0}^{K-1} \hat{\beta}_k$, Amazon $\sum_{k=0}^{K-1} (\hat{\beta}_k + \hat{\delta}_k)$, and the Differential $\sum_{k=0}^{K-1} \hat{\delta}_k$; negative values indicate lower pass-through by Amazon. Prices and commodity costs are in dollars per gram (commodity prices converted at 1.235 for coffee and 2.44 for cocoa), so one denotes full pass-through. The sample is ASIN-week cells with both seller types' price changes observed, stacked so each cell contributes one Amazon and one 3P observation; all regressions include $\text{ASIN} \times \text{seller-type}$ fixed effects. Standard errors clustered by ASIN and calendar week in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

5.3 Robustness and Alternative Outcome Variables

We assess the sensitivity of the baseline seller-type differential to using different subsamples, functional forms, and the relaxing common-support restriction. We then consider Buy Box-winning prices as an alternative, consumer-facing price measure.

Subsample analyses. The divergence between 3P and Amazon pass-through is not driven by particularly high-priced products; in fact, it tends to become stronger when excluding products that are “consistently” high-priced, defined by products whose median prices are strictly higher than the 80th percentile of all prices (Figure E.1). Given that 3P and Amazon presence within a given ASIN is often temporary, we also estimate our regressions on a subsample where restrict to ASINs with at least 48 consecutive weeks of joint Amazon and 3P availability. The Amazon-minus-3P differential remains negative (Figure E.2): at the 48-week horizon it is nearly unchanged in both categories. Intermediate-horizon estimates become less precise for coffee, whereas the 3P vs. Amazon difference becomes more pronounced for cocoa.

Alternative specifications. The pass-through difference remains substantial even when we relax the requirement that Amazon and 3P prices be jointly observed and instead estimate the pass-through regressions separately for Amazon and the lowest 3P prices (see Figure E.3). In the online appendix, we report the analogous log specification. The results is consistent with the findings of Sangani (2026): for 3P sellers, pass-through tend to be complete in levels, but less so in logs.

Buy Box-winning price. Lastly, we also estimate pass-through on the Buy Box price conditionally on the seller being Amazon or a 3P, respectively (imposing that 3P and Amazon prices are jointly observable in the Buy Box is not possible, as only one of the parties can win the Buy Box at any time). While this price arguably captures the price that most consumers will in the end pay when making purchasing decision, the Buy Box price captures not only pricing decision, but also the algorithm’s ranking. We observe generally lower pass-through for 3P sellers. For coffee, where we obtain mostly insignificant results in the case of 3P sellers, and a positive coefficient for a lag of 48 weeks in the case of Amazon; see Figure E.4. This is consistent with evidence showing that the Buy Box algorithm is very price elastic (Chen et al., 2016; Lee and Musolf, 2025; Breslin, 2025). In particular, the evidence supports the idea that Amazon may be more likely to win a Buy Box previously held by a 3P seller after that seller raises their price.

6 Conclusion

Our paper highlights the distinct pricing incentives faced by a platform that resells on its own marketplace relative to 3P sellers. In a theoretical model, these incentives generate two key implications: platform entry and exit generically induce different incumbent-3P price responses than comparable 3P entry and exit, and common cost shocks induce larger price responses by 3P sellers than by Amazon. Our model links both patterns to the platform’s dual role as retailer and commission-earning intermediary. Using Amazon U.S. data, we document reduced-form evidence consistent with both implications. Incumbent 3P prices tend to fall less following Amazon entry than following comparable 3P entry, and to rise less following Amazon exit than following comparable 3P exit. In response to common sector-level input-cost shocks, 3P prices exhibit significantly greater pass-through than Amazon prices, with Amazon’s estimated response much smaller and statistically indistinguishable from zero at the relevant horizons.

These findings point to a pricing channel distinct from the platform-controlled prominence and recommendation mechanisms studied in the steering literature. The pass-through asymmetry, in particular, implies that common cost shocks need not translate into uniform price changes across sellers on the same marketplace. Instead, they can change the relative prices of 1P and 3P offers, and hence their competitive positions, even when sellers are exposed to the same underlying cost shock. In our model, this asymmetry arises from the commission wedge, implying that the platform’s fee structure affects not only the pricing incentives of 3P sellers in normal times, but also how sector-wide cost pressure is transmitted across seller types. Common cost shocks can therefore amplify the competitive consequences of the platform’s dual role rather than simply shifting all marketplace prices in parallel.

Our results also speak to ongoing policy debates over price competition on hybrid platforms. These issues are particularly salient in light of recent antitrust cases in Germany and the United States concerning Amazon’s pricing practices, the level and structure of fees charged to 3P sellers, and restraints on 3P seller pricing ([Federal Trade Commission, 2024](#); [Bundeskartellamt, 2025](#)). This paper contributes to this conversation by showing that the platform’s fee structure can shape not only the 3P pricing incentives, but also the relative price responses of 1P and 3P sellers and the transmission of common cost shocks across the marketplace. A natural direction for future work is to quantify these effects in a structural framework and evaluate how alternative commission regimes or other platform regulations affect pass-through, competition, and consumer welfare.

References

- Alvarez, Santiago E., Alberto Cavallo, Alexander MacKay, and Paolo Mengano (2024) "Markups and Cost Pass-through Along the Supply Chain," *Working Paper*, <https://alexandermackay.org/files/Markups%20and%20Cost%20Pass-through%20Along%20the%20Supply%20Chain.pdf>.
- Anderson, Simon and Özlem Bedre Defolie (2024) "Hybrid platform model: monopolistic competition and a dominant firm," *The RAND Journal of Economics*, 55 (4), 684–718, <https://doi.org/10.1111/1756-2171.12478>.
- Arcidiacono, Peter, Paul B. Ellickson, Carl F. Mela, and John D. Singleton (2020) "The Competitive Effects of Entry: Evidence from Supercenter Expansion," *American Economic Journal: Applied Economics*, 12 (3), 175–206, [10.1257/app.20180047](https://doi.org/10.1257/app.20180047).
- Bedre Defolie, Özlem and Senay Sokullu (2026) "Estimating the Implications of Hybrid Platform Model: Amazon," SSRN Working Paper, April, [10.2139/ssrn.6540918](https://ssrn.com/abstract=6540918), Posted April 20, 2026.
- Bennati, Luca (2026) "Vertical Integration and E-commerce Competition: Evidence from Amazon Marketplace," *Working Paper*, https://drive.google.com/file/d/1GCzljX1haX5pLL_IXGlQ4r7fWgCJI3ee/view.
- Besanko, David, Jean-Pierre Dubé, and Sachin Gupta (2005) "Own-Brand and Cross-Brand Retail Pass-Through," *Marketing Science*, 24 (1), 123–137.
- Bettendorf, L and F Verboven (2000) "Incomplete transmission of coffee bean prices: evidence from The Netherlands," *European Review of Agricultural Economics*, 27 (1), 1–16, [10.1093/erae/27.1.1](https://doi.org/10.1093/erae/27.1.1).
- Bonnet, Céline, Pierre Dubois, Sofia B. Villas-Boas, and Daniel Klapper (2013) "Empirical Evidence on the Role of Nonlinear Wholesale Pricing and Vertical Restraints on Cost Pass-Through," *The Review of Economics and Statistics*, 95 (2), 500–515, [10.1162/REST_a_00267](https://doi.org/10.1162/REST_a_00267).
- Breslin, Seamus (2025) "How to Win the Amazon Buy Box," February, <https://www.repricerexpress.com/win-amazon-buy-box/>, Accessed: 2026-02-12.
- Bresnahan, Timothy F. and Peter C. Reiss (1991) "Entry and Competition in Concentrated Markets," *Journal of Political Economy*, 99 (5), 977–1009, [10.1086/261786](https://doi.org/10.1086/261786).
- Bundeskartellamt (2025) "Bundeskartellamt has concerns about Amazon's use of so-called price control mechanisms," https://www.bundeskartellamt.de/SharedDocs/Meldung/EN/Pressemitteilungen/2025/2025_06_02_Amazon.html, June, Press release.

- Cabral, Luís and Lei Xu (2021) "Seller reputation and price gouging: Evidence from the COVID-19 pandemic," *Economic Inquiry*, 59 (3), 867–879, <https://doi.org/10.1111/ecin.12993>.
- Cachon, Gérard P and Martin A Lariviere (2005) "Supply chain coordination with revenue-sharing contracts: strengths and limitations," *Management science*, 51 (1), 30–44.
- Chen, Le, Alan Mislove, and Christo Wilson (2016) "An Empirical Analysis of Algorithmic Pricing on Amazon Marketplace," *Proceedings of the 25th international conference on World Wide Web*, 1339–1349.
- Chen, Nan and Hsin-Tien Tsai (2024) "Steering via algorithmic recommendations," *The RAND Journal of Economics*, 55 (4), 501–518, <https://doi.org/10.1111/1756-2171.12481>.
- Choe, Chongwoo, Antoine Dubus, Noriaki Matsushima, and Shiva Shekhar (2025) "Data Regulation, Platform Encroachment, and the Matthew Effect," *Platform Encroachment, and the Matthew Effect* (December 15, 2025).
- Conlon, Christopher T. and Nirupama L. Rao (2020) "Discrete Prices and the Incidence and Efficiency of Excise Taxes," *American Economic Journal: Economic Policy*, 12 (4), 111–143, [10.1257/pol.20180658](https://doi.org/10.1257/pol.20180658).
- Crawford, Gregory S, Matteo Courthoud, Regina Seibel, and Simon Zuzek (2022) "Amazon entry on Amazon marketplace," *CEPR Discussion Paper No. 17531*.
- Dendorfer, Florian and Regina Seibel (2026) "What's In the Box? The Effect Of Self-Preferencing On Amazon Marketplace," Technical report, https://reginaseibel.github.io/publication/amazon_box/amazon_box.pdf, Mimeo.
- Etro, Federico (2021) "Product selection in online marketplaces," *Journal of Economics & Management Strategy*, 30, 614–637, [10.1111/jems.12428](https://doi.org/10.1111/jems.12428).
- (2023) "Hybrid Marketplaces with Free Entry of Sellers," *Review of Industrial Organization*, 62, 119–148.
- Fairtrade International (2026) "Fairtrade Standard for Cocoa," https://www.fairtrade.net/content/dam/fairtrade/fairtrade-international/standards-spo-cocoa/Cocoa_SPO_EN.pdf, Version 27.09.2023_v2.14, updated March 24, 2026; accessed July 22, 2026.
- Farronato, Chiara, Andrey Fradkin, and Alexander MacKay (2023) "Self-Preferencing at Amazon: Evidence from Search Rankings," *AEA Papers and Proceedings*, 113, 239–43, [10.1257/pandp.20231068](https://doi.org/10.1257/pandp.20231068).

- (2025) “Vertical Integration and Consumer Choice: Evidence from a Field Experiment,” NBER Working Paper 34135, National Bureau of Economic Research, [10.3386/w34135](https://www.nber.org/papers/w34135).
- Federal Trade Commission (2024) “Amended Complaint: Amazon.com, Inc. (Amazon eCommerce),” https://www.ftc.gov/system/files/ftc_gov/pdf/01712024.03.14RedactedAmendedComplaint%20%28002%29.pdf.
- Fischer, Kai, Simon Martin, and Philipp Schmidt-Dengler (2025) “The Heterogeneous Effects of Entry on Prices,” Technical Report 10566, https://drive.google.com/file/d/1kMBYUHED-sU9xMH76LaQerI-Mi5V_Uc/view, Forthcoming in *Journal of the European Economic Association*.
- Foros, Øystein, Hans Jarle Kind, and Greg Shaffer (2017) “Apple’s agency model and the role of most-favored-nation clauses,” *The RAND Journal of Economics*, 48 (3), 673–703.
- Gautier, Pieter, Bo Hu, and Makoto Watanabe (2023) “Marketmaking middlemen,” *The Rand journal of economics*, 54 (1), 83–103.
- Gutiérrez, Germán (2022) “The Welfare Consequences of Regulating Amazon,” http://germangutierrezg.com/Gutierrez2021_AMZ_welfare.pdf.
- Hagiu, Andrei, Tat-How Teh, and Julian Wright (2022) “Should platforms be allowed to sell on their own marketplaces?” *The RAND Journal of Economics*, 53 (2), 297–327.
- Hagiu, Andrei and Julian Wright (2015) “Marketplace or Reseller?” *Management Science*, 61 (1), 184–203, <http://www.jstor.org/stable/24551078>.
- He, Leshui, Imke Reimers, and Benjamin Shiller (2022) “Does Amazon Exercise Its Market Power? Evidence from Toys “R” Us,” *The Journal of Law and Economics*, 65 (4), 665–685.
- He, Sherry, Brett Hollenbeck, and Davide Proserpio (2022) “The Market for Fake Reviews,” *Marketing Science*, 41 (5), 896–921.
- Hervas-Drane, Andres and Sandro Shelegia (2022) “Price competition with a stake in your rival,” *International Journal of Industrial Organization*, 84, 102862.
- (2025) “Retailer-led marketplaces,” *Management Science*, 71 (11), 9650–9669.
- Hu, Honggang, Quan Zheng, and Xiajun Amy Pan (2022) “Agency or wholesale? The role of retail pass-through,” *Management Science*, 68 (10), 7538–7554.
- Johnson, Justin P (2017) “The agency model and MFN clauses,” *The Review of Economic Studies*, 84 (3), 1151–1185.

- Kleven, Henrik, Camille Landais, and Jakob Egholt Sogaard (2019) "Children and Gender Inequality: Evidence from Denmark," *American Economic Journal: Applied Economics*, 11 (4), 181–209, [10.1257/app.20180010](https://doi.org/10.1257/app.20180010).
- Korganbekova, Malika and Alibek Korganbekov (2026) "Private Labels and Product Imitation: Evidence from Amazon," Mimeo, July, July 22, 2026.
- Lee, Kwok Hao and Leon Musolff (2025) "Two-Sided Markets Shaped by Platform-Guided Search," *Working Paper*, <https://lmusolff.com/papers/TwoSidedMarketsPlatformGuidedSearch.pdf>.
- Madsen, Erik and Nikhil Vellodi (2025) "Insider imitation," *Journal of Political Economy*, 133 (2), 652–709.
- Miklos-Thal, Jeanine and Greg Shaffer (2021) "Pass-through as an economic tool: on exogenous competition, social incidence, and price discrimination," *Journal of Political Economy*, 129 (1), 323–335.
- Musolff, Leon (2026) "Algorithmic Pricing, Price Wars, and Tacit Collusion: Evidence from E-Commerce," *Management Science*, o (o), Published online ahead of print.
- Nakamura, Emi and Dawit Zerom (2010) "Accounting for Incomplete Pass-Through," *The Review of Economic Studies*, 77 (3), 1192–1230, <https://doi.org/10.1111/j.1467-937X.2009.589.x>.
- Rainforest Alliance (2025) "Traceability Annex," <https://knowledge.rainforest-alliance.org/docs/traceability-annex-v14>, Document A-05-SRCL-B-CH, Version 1.4, published September 8, 2025, effective March 1, 2026; accessed July 22, 2026.
- Sangani, Kunal (2026) "Pass-Through in Levels and the Unequal Incidence of Commodity Shocks," *forthcoming in the Quarterly Journal of Economics*, https://kunalsangani.com/files/complete_passthrough_live.pdf.
- Varian, Hal R. (1980) "A Model of Sales," *The American Economic Review*, 70 (4), 651–659, <http://www.jstor.org/stable/1803562>.
- Weyl, E. Glen and Michal Fabinger (2013) "Pass-Through as an Economic Tool: Principles of Incidence under Imperfect Competition," *Journal of Political Economy*, 121 (3), 528–583.
- Zhu, Feng and Qihong Liu (2018) "Competing with complementors: An empirical look at Amazon.com," *Strategic Management Journal*, 39 (10), 2618–2642, <https://doi.org/10.1002/smj.2932>.

Appendix

A Supplementary Material for the Theoretical Analysis

A.1 Omitted Proofs in Section 3

Proof of Proposition 1

Proof. Under the pure marketplace mode with n retailers, the symmetric equilibrium $\mathbf{p}_M^* = (p_M^*(n), \dots, p_M^*(n))$ is characterized by

$$D_i(\mathbf{p}_M^*) + \left(p_M^*(n) - \frac{c}{1-\tau} \right) \frac{\partial D_i(\mathbf{p}_M^*)}{\partial p_i} = 0.$$

Substituting the linear demand function in Assumption ?? yields

$$p_M^*(n) = \frac{(1-\gamma)(1-\tau) + c(1 + (n-2)\gamma)}{(1-\tau)(2 + (n-3)\gamma)}.$$

Denote the semi-symmetric equilibrium under the hybrid mode by $\hat{\mathbf{p}}^* = (\hat{p}_P^*(n), \hat{p}_S^*(n), \dots, \hat{p}_S^*(n))$. Let $D_P(\hat{\mathbf{p}}^*) = D_1(\hat{\mathbf{p}}^*)$ denote equilibrium demand for the platform's own product and let $D_S(\hat{\mathbf{p}}^*) = D_i(\hat{\mathbf{p}}^*)$ for all $i \in \{2, \dots, n\}$ denote equilibrium demand for each 3P seller. The equilibrium is characterized by

$$\begin{cases} D_P(\hat{\mathbf{p}}^*) + (\hat{p}_P^*(n) - c) \frac{\partial D_P(\hat{\mathbf{p}}^*)}{\partial p_P} + \tau(n-1)\hat{p}_S^*(n) \frac{\partial D_S(\hat{\mathbf{p}}^*)}{\partial p_P} = 0, \\ D_S(\hat{\mathbf{p}}^*) + \left(\hat{p}_S^*(n) - \frac{c}{1-\tau} \right) \frac{\partial D_S(\hat{\mathbf{p}}^*)}{\partial p_S} = 0. \end{cases}$$

The solution is:

$$\begin{cases} \hat{p}_P^*(n) = \frac{c(1+(n-2)\gamma)[2(1-\tau)+\gamma(2n-3+\tau)]+(1-\gamma)(1-\tau)[2+\gamma(2n-3+(n-1)\tau)]}{(1-\tau)[4+6(n-2)\gamma+\gamma^2(2n^2-9n+9-(n-1)\tau)]}, \\ \hat{p}_S^*(n) = \frac{c(1+(n-2)\gamma)[2+\gamma(2n-3-\tau)]+(1-\gamma)(1-\tau)[2+(2n-3)\gamma]}{(1-\tau)[4+6(n-2)\gamma+\gamma^2(2n^2-9n+9-(n-1)\tau)]}. \end{cases}$$

Direct comparison yields

$$\hat{p}_P^*(n) \geq \hat{p}_S^*(n) \iff \hat{p}_S^*(n) \geq p_M^*(n) \iff \tau \leq 1 - \frac{2c(1 + (n-2)\gamma)}{(n-1)\gamma}.$$

Finally, we explain why the two price comparisons share the same threshold. Evaluating the platform's first-order condition at the symmetric pure-marketplace price profile $\mathbf{p}_M^*(n)$ and

setting the resulting marginal-profit equal to zero, we obtain

$$D_P(\mathbf{p}_M^*) + (p_M^*(n) - c) \frac{\partial D_P(\mathbf{p}_M^*)}{\partial p_P} + \tau(n-1)p_M^*(n) \frac{\partial D_S(\mathbf{p}_M^*)}{\partial p_P} = 0$$

$$\Leftrightarrow \tau = 1 - \frac{2c(1 + (n-2)\gamma)}{(n-1)\gamma}.$$

This implies that, when

$$\tau = 1 - \frac{2c(1 + (n-2)\gamma)}{(n-1)\gamma},$$

the revenue-sharing effect exactly offsets the effect associated with eliminating the double-markup distortion. The platform therefore finds it optimal to charge the pure-marketplace equilibrium price, and

$$\hat{p}_P^*(n) = \hat{p}_S^*(n) = p_M^*(n).$$

When

$$\tau < 1 - \frac{2c(1 + (n-2)\gamma)}{(n-1)\gamma},$$

the revenue-sharing effect dominates the effect associated with eliminating the double-markup distortion, so the platform finds it profitable to raise its price above $p_M^*(n)$. Because prices are strategic complements under the linear demand system, 3P sellers respond by raising their prices as well. Consequently,

$$\hat{p}_P^*(n) > \hat{p}_S^*(n) > p_M^*(n).$$

When

$$\tau > 1 - \frac{2c(1 + (n-2)\gamma)}{(n-1)\gamma},$$

the effect associated with eliminating the double-markup distortion dominates, so the platform finds it profitable to set a price below $p_M^*(n)$. The 3P sellers respond by lowering their prices. Consequently, the hybrid-mode equilibrium features lower prices for all retailers than the pure-marketplace equilibrium:

$$\hat{p}_P^*(n) < \hat{p}_S^*(n) < p_M^*(n).$$

□

Proof of Proposition 2

Proof. First, we derive the conditions under which the equilibria in both modes are interior.

Under the pure marketplace mode, the equilibrium demand of each 3P seller is

$$D_M^*(n) = \frac{[1 + (n-2)\gamma] (1 - \tau - c)}{(1 - \tau) [2 + (n-3)\gamma] [1 + (n-1)\gamma]}.$$

Therefore, this demand is strictly positive if and only if $c < 1 - \tau$.

Under the hybrid mode, the equilibrium demands of the platform and each 3P sellers are, respectively:

$$\begin{aligned}\hat{D}_P^*(n) &= \frac{(1-c)AB - \tau(n-1)\gamma Q}{Q\Delta}, \\ \hat{D}_S^*(n) &= \frac{A[(1-\tau)B - cH]}{(1-\tau)Q\Delta},\end{aligned}$$

where $A \equiv 1 + (n-2)\gamma$, $B \equiv 2 + (2n-3)\gamma$, $H \equiv 2 + (2n-3+\tau)\gamma = B + \tau\gamma$, $Q \equiv 1 + (n-1)\gamma$, and $\Delta \equiv 4 + 6(n-2)\gamma + \gamma^2 [2n^2 - 9n + 9 - (n-1)\tau]$. All of these auxiliary quantities are strictly positive under the maintained parameter restrictions $n \geq 2$, $0 < \gamma < 1$, and $0 < \tau < 1$.

It follows that

$$\begin{aligned}\hat{D}_P^*(n) > 0 &\iff c < \hat{c}_P \equiv 1 - \frac{\tau(n-1)\gamma [1 + (n-1)\gamma]}{[1 + (n-2)\gamma] [2 + (2n-3)\gamma]}, \\ \hat{D}_S^*(n) > 0 &\iff c < \hat{c}_S \equiv \frac{(1-\tau) [2 + (2n-3)\gamma]}{2 + (2n-3+\tau)\gamma} = \frac{(1-\tau)B}{H}.\end{aligned}$$

Because $\hat{c}_P - \hat{c}_S = \frac{\tau Q \Delta}{A B H} > 0$, we have $\hat{c}_S < \hat{c}_P$. Hence, the equilibrium under the hybrid mode is interior if and only if $c < \hat{c}_S$.

Comparing $p_S^*(n)$ and $p_M^*(n-1)$ yields

$$\hat{p}_S^*(n) - p_M^*(n-1) = \frac{\gamma(1-\gamma)K}{(1-\tau) [2 + (n-4)\gamma] \Delta} (c - c_E),$$

where $K \equiv 2(1-\tau) + \gamma [2n(1-\tau) + 5\tau - 3] > 0$ and $c_E \equiv \frac{(1-\tau)[2 + \gamma(2n-3-(n-1)\tau)]}{K}$. Therefore, $\text{sign} [\hat{p}_S^*(n) - p_M^*(n-1)] = \text{sign}(c - c_E)$.

A direct comparison of $\hat{c}_S$, c_E and $1 - \tau$ yields

$$\begin{aligned}\hat{c}_S &< 1 - \tau, \\ c_E - (1 - \tau) &= \frac{(1-\tau)\tau [2 + (n-4)\gamma]}{K} > 0,\end{aligned}$$

which implies that $\hat{c}_S < 1 - \tau < c_E$. Thus, the condition $c < \hat{c}_S$ guarantees that the equilibria in both modes are interior and that $c < c_E$.

Equivalently, the interiority condition can be written as

$$0 < \tau < \hat{\tau} \equiv \frac{(1-c)[2+(2n-3)\gamma]}{2+(2n-3+c)\gamma}.$$

Therefore, over the relevant range of commission rates, $\tau < \hat{\tau}$, we have

$$\hat{p}_S^*(n) < p_M^*(n-1).$$

Hence, the platform's entry lowers the equilibrium prices charged by 3P sellers. $\square$

Proof of Proposition 3

Proof. Substituting the linear demand system into the equilibrium conditions—Equation (1) for 3P sellers, $i = 2, \dots, n$, and Equation (2) for the platform—yields the following result.

$$2\hat{p}_P^* - \delta(1 + \tau_R) \sum_{i=2}^n \hat{p}_i^* = \frac{1 - \gamma}{1 + (n-2)\gamma} + c_P,$$

and

$$2\hat{p}_i^* - \delta \sum_{j \neq i}^n \hat{p}_j^* = \frac{1 - \gamma}{1 + (n-2)\gamma} + \frac{c_i}{1 - \tau_D}, \quad i \in \{2, \dots, n\}.$$

Stacking the equilibrium pricing conditions yields

$$M(\tau_R) \hat{\mathbf{p}}^* = \frac{1 - \gamma}{1 + (n-2)\gamma} \mathbf{1}_n + C(\tau_D) \mathbf{c},$$

where

$$M(\tau_R) \equiv \begin{bmatrix} 2 & -\delta(1 + \tau_R) \mathbf{1}_{n-1}^\top \\ -\delta \mathbf{1}_{n-1} & (2 + \delta) I_{n-1} - \delta J_{n-1} \end{bmatrix}, \text{ and } C(\tau_D) \equiv \begin{bmatrix} 1 & \mathbf{0}_{n-1}^\top \\ \mathbf{0}_{n-1} & \frac{1}{1 - \tau_D} I_{n-1} \end{bmatrix}.$$

Differentiating this equation with respect to the marginal-cost vector $\mathbf{c}$ yields

$$M(\tau_R) \rho(\tau_R, \tau_D) = C(\tau_D),$$

and hence

$$\rho \equiv \frac{\partial \hat{\mathbf{p}}^*}{\partial \mathbf{c}^\top} = M(\tau_R)^{-1} C(\tau_D).$$

$\square$

Proof of Proposition 4

Proof. Let $\kappa_D \equiv \frac{1}{1-\tau_D}$, $\ell \equiv 2 - (n-2)\delta$, and $\Omega_R \equiv 2\ell - (n-1)\delta^2(1+\tau_R)$. The pass-through matrix from the platform's and 3P sellers' individual cost shocks to their equilibrium prices is given by

$$\rho = M(\tau_R)^{-1}C(\tau_D) = \begin{bmatrix} \frac{\ell}{\Omega_R} & \frac{\delta(1+\tau_R)\kappa_D}{\Omega_R} \mathbf{1}_{n-1}^\top \\ \frac{\delta}{\Omega_R} \mathbf{1}_{n-1} & \frac{\kappa_D}{2+\delta} I_{n-1} + \frac{\kappa_D}{n-1} \left[\frac{2}{\Omega_R} - \frac{1}{2+\delta} \right] J_{n-1} \end{bmatrix},$$

where $\tau_R = \tau_D = \tau$.

Therefore, The four blocks of the pass-through matrix contain five distinct types of pass-through rates. The upper-left block of the pass-through matrix $\rho_{PP} \equiv \frac{\partial \hat{p}_P^*}{\partial c_P} = \frac{\ell}{\Omega_R}$ denotes the platform's own-cost pass-through rate. For any $S \in \{2, \dots, n\}$, the upper-right part contains $\rho_{PS} \equiv \frac{\partial \hat{p}_P^*}{\partial c_S} = \frac{\delta(1+\tau_R)\kappa_D}{\Omega_R}$, denotes the pass-through of a 3P seller's cost shock to the platform's price. The lower-left block, by contrast, contains $\rho_{SP} \equiv \frac{\partial \hat{p}_S^*}{\partial c_P} = \frac{\delta}{\Omega_R}$ denotes the pass-through of the platform's cost shock to a 3P seller's price. The lower-right block concerns only the 3P sellers. A 3P seller's own-cost pass-through rate is $\rho_{SS} \equiv \frac{\partial \hat{p}_S^*}{\partial c_S} = \frac{\kappa_D}{2+\delta} + \frac{\kappa_D}{n-1} \left[\frac{2}{\Omega_R} - \frac{1}{2+\delta} \right]$. For any distinct $S, S' \in \{2, \dots, n\}$, $\rho_{SS'} \equiv \frac{\partial \hat{p}_S^*}{\partial c_{S'}} = \frac{\kappa_D}{n-1} \left[\frac{2}{\Omega_R} - \frac{1}{2+\delta} \right]$ denotes the pass-through of one 3P seller's cost shock to another 3P seller's price.

Thus, the pass-through rates of the common cost shock to the platform's price and to a 3P seller's price are, respectively,

$$\rho_P = \rho_{PP} + \sum_{S=2}^n \rho_{PS} = \frac{\ell + (n-1)\delta(1+\tau_R)\kappa_D}{\Omega_R},$$

and

$$\rho_S = \rho_{SS} + \rho_{SP} + \sum_{\substack{S'=2 \\ S' \neq S}}^n \rho_{SS'} = \frac{2\kappa_D + \delta}{\Omega_R}, \quad S \in \{2, \dots, n\}.$$

We now impose $\tau_R = \tau_D = \tau \in (0, 1)$, so that $\kappa_D = 1/(1-\tau)$. Under the maintained assumptions on the demand system, $0 < \delta < 1/(n-1)$, and hence $\Omega_R > 0$.

For own-cost pass-through, direct comparison gives

$$\rho_{SS} - \rho_{PP} = \frac{2\tau [2 - (n-3)\delta - (n-2)\delta^2]}{(1-\tau)(2+\delta)\Omega_R} > 0.$$

The inequality follows from $\tau > 0$, $\Omega_R > 0$, and $0 < \delta < 1/(n-1)$, which also implies $2 - (n-3)\delta - (n-2)\delta^2 > 0$. Therefore, $\rho_{PP} < \rho_{SS}$, establishing part (1).

For cross-cost pass-through, when $n \geq 3$, there exist two distinct 3P sellers S and S' . We have

$$\rho_{SS'} - \rho_{SP} = \frac{2\delta\tau(1+\delta)}{(1-\tau)(2+\delta)\Omega_R} > 0$$

and

$$\rho_{PS} - \rho_{SS'} = \frac{2\delta\tau}{(1-\tau)(2+\delta)\Omega_R} > 0.$$

Hence, $\rho_{SP} < \rho_{SS'} < \rho_{PS}$, which establishes part (2).

Finally, comparing the pass-through rates of an industry-wide cost shock gives

$$\rho_S - \rho_P = \frac{2\tau[1 - (n-1)\delta]}{(1-\tau)\Omega_R} > 0,$$

where the inequality follows from $0 < \delta < 1/(n-1)$. Thus, $\rho_P < \rho_S$, establishing part (3). $\square$

A.2 Numerical Examples for Proposition 1

As stated in Proposition 1, the comparison between $\hat{p}_P^*(n)$ and $\hat{p}_S^*(n)$, i.e., the two equilibrium prices on the hybrid platform, is equivalent to the comparison between the platform's entry effect, $\hat{p}_S^*(n) - p_M^*(n-1)$, and a 3P seller's entry effect, $p_M^*(n) - p_M^*(n-1)$. The numerical examples shown in Figure A.1 therefore focus on the comparison between $\hat{p}_P^*(n)$ and $\hat{p}_S^*(n)$ to illustrate how the equilibrium price ranking varies across different parameter configurations.

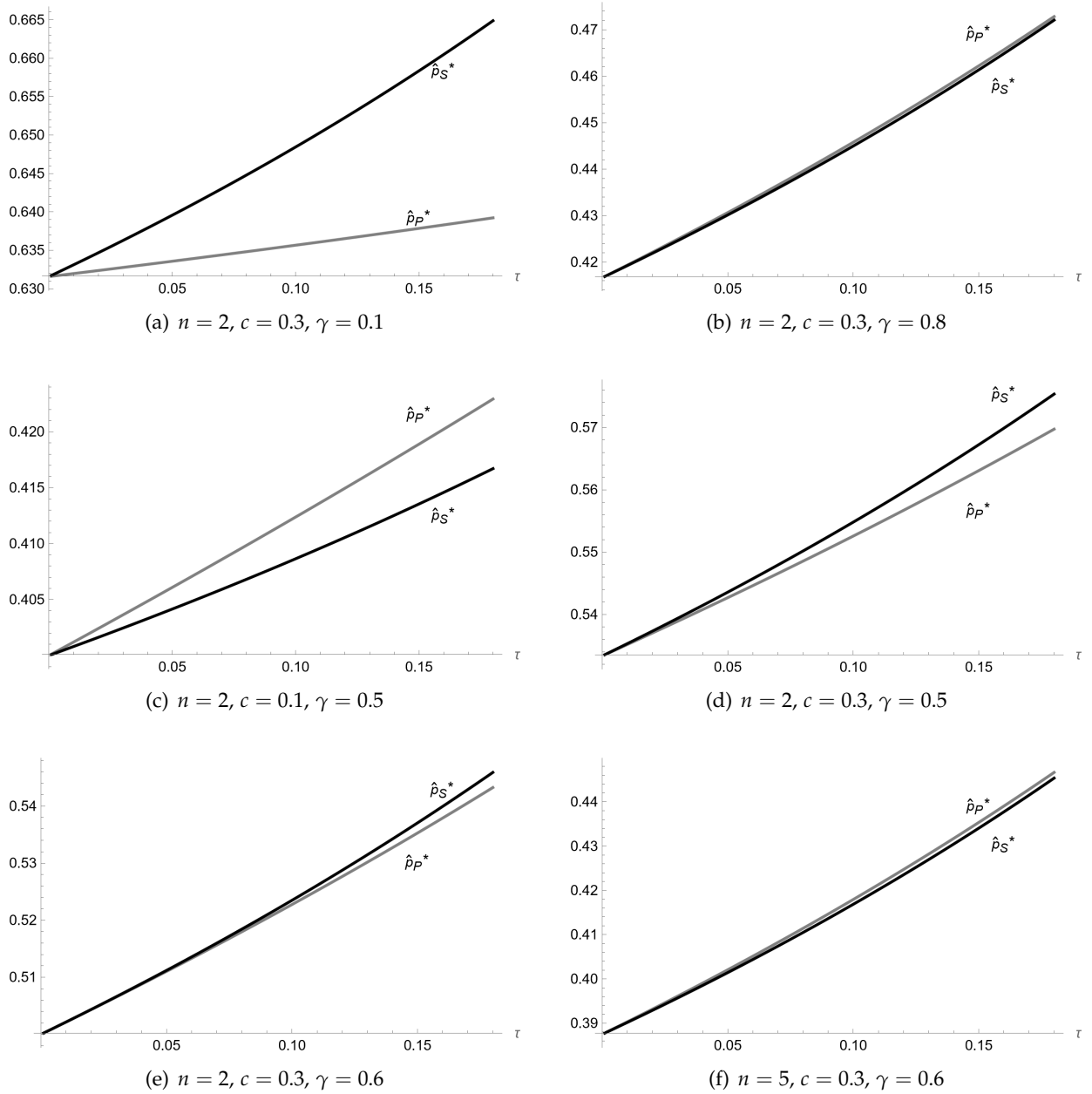
In panels (a) and (b) of Figure A.1, n and c are held fixed, while γ increases from 0.1 to 0.8. When γ is relatively low, as in panel (a), the effect associated with eliminating the double-markup distortion dominates, so that the platform charges a lower price than 3P sellers. When γ is sufficiently high, as in panel (b), the revenue-sharing effect dominates, so that the platform charges a higher price than 3P sellers. Intuitively, a larger γ makes the platform's product and the 3P products closer substitutes. As a result, an increase in the platform's own price shifts more demand toward 3P sellers. Because the platform earns commission revenue from those sales, it internalizes this demand diversion and therefore has a stronger incentive to raise its own price. This can be seen directly from the platform's first-order condition:

$$D_P + (p_P - c) \frac{\partial D_P}{\partial p_P} + \tau \sum_{i=2}^n p_i \frac{\partial D_i}{\partial p_P} = 0.$$

The last term captures the revenue-sharing effect. Under the linear demand system, the aggregate diversion ratio is

$$D \equiv \left| \frac{\sum_{i=2}^n \frac{\partial D_i}{\partial p_P}}{\frac{\partial D_P}{\partial p_P}} \right| = \frac{(n-1)\gamma}{1 + (n-2)\gamma},$$

Figure A.1: Comparison of semi-symmetric equilibrium prices under hybrid mode across parameter configurations



Notes: The figure plots the semi-symmetric equilibrium prices under hybrid mode as functions of the commission rate τ across different parameter configurations. The gray line represents the platform's equilibrium price, $\hat{p}_P^*(n)$, and the black line represents the equilibrium price of 3P sellers, $\hat{p}_S^*(n)$. The values of n , c , and γ are reported in the title of each panel. When $\hat{p}_P^*(n) > \hat{p}_S^*(n)$, the revenue-sharing effect dominates the effect associated with eliminating the double-markup distortion; when $\hat{p}_P^*(n) < \hat{p}_S^*(n)$, the latter dominates.

which is increasing in γ . Thus, as products become more substitutable, a larger share of the demand lost by the platform when it raises its price is captured by 3P sellers. This makes the associated increase in commission revenue more important relative to the platform's loss of retail

profit.

Then, in panels (c) and (d), n and γ are held fixed, while c increases from 0.1 to 0.3. When c is relatively low, as in panel (c), the revenue-sharing effect dominates, so that the platform charges a higher price than 3P sellers. When c is sufficiently high, as in panel (d), the effect associated with eliminating the double-markup distortion dominates, so that the platform charges a lower price than 3P sellers. Intuitively, as c increases, the double-markup distortion faced by 3P sellers becomes more pronounced, making its elimination relatively more important for the platform’s pricing decision.

Finally, in panels (e) and (f), c and γ are held fixed, while n increases from 2 to 5. When n is relatively small, as in panel (e), the effect associated with eliminating the double-markup distortion dominates, so that the platform charges a lower price than 3P sellers. When n is sufficiently large, as in panel (f), the revenue-sharing effect dominates, so that the platform charges a higher price than 3P sellers. Intuitively, as the number of 3P sellers increases, the platform internalizes the effect of its own price on a larger set of commission-generating sales. When the platform raises its price, demand is diverted toward the $n - 1$ 3P sellers, increasing the commission revenue it earns from their sales. Consistent with this intuition, the platform’s aggregate diversion ratio D is increasing in n . Thus, with more competitors, a larger share of the demand lost by the platform following a price increase is captured by commission-paying 3P sellers, making the revenue-sharing effect relatively more important.

B Data

B.1 Amazon Data

Sales rank. Given that a product’s sales rank is not always recorded on a daily basis, we forward-fill this variable, so that a product’s sales rank on a given day reflects the latest sales rank that was recorded on Keepa.

After creating a daily panel of products’ prices (recorded at midnight of a given date), we further exclude products if their average 3P sellers’ (Amazon’s) price (standardized by weight when considering grocery products) across all observed days lies above the 98th, or below the 2nd percentile of all 3P sellers’ (Amazon’s) daily prices. Very often these products in fact do not belong to the category considered and hence were misclassified, and should hence not be part of our sample.

Defining Amazon “entry” and “exit”. We define Amazon entry to be the first day on which Amazon’s price variable becomes available, following a day at which it was not available.

Likewise, we define exit as the first day on which Amazon’s price variable becomes unavailable, following a day of Amazon’s price being available.

Identifying Amazon private label products. While not core to our analysis, we report the percentage of Amazon private label products in Table 1. We identify these private label products by tagging products whose manufacturer is listed as “365 by Whole Foods Market,” “Amazon.com Services, Inc.,” “Amazon.com Services LLC.,” or “Amazon,” as well as products whose brand is listed as “Amazon Fresh” or “Amazon Basics.”

Building a weekly panel. From a daily panel of products with prices recorded at midnight, we build a weekly panel by taking the average across all observed prices and sales rank. If a price variable is not available on some days of a given week, these take the average across the remaining days on which it is observed. While our analysis covers January 2021 until March 2025, note that, in order to generate the pass-through results, we take lagged variables going back into 2020.

Identifying higher-turnover products. We use the sales rank variable to identify relatively high-turnover products: we call a product “popular” if it is consistently highly-ranked (which is defined as its sales rank being below the median of the overall sales rank distribution in at least 50% of its observations). We also consider a product’s percentile in the sales-rank distribution.

B.2 Commodity Price Data

Cocoa beans. The cocoa bean pricing data we use stems from the International Cocoa Organization (ICCO) and is available via www.icco.org/statistics. We use the ICCO daily price for cocoa beans in US\$, defined as “the average of the quotations of the nearest three active futures trading months on ICE Futures Europe (London) and ICE Futures US (New York) at the time of London close”. We refer the interested reader to the website for further information.

Arabica coffee beans. Daily pricing data on Arabica coffee beans are obtained from Trading Economics, available on tradingeconomics.com/commodity/coffee. The data corresponds to Arabica Coffee C Futures contracts in US\$ that trade on the Intercontinental Exchange (see <https://www.ice.com/products/15/Coffee-C-Futures>), and is the global benchmark for Arabica Coffee. Because Arabica and Robusta coffee-bean prices tend to co-move, and because Arabica accounts for roughly 90% of U.S. coffee imports,²⁵ we use the Arabica coffee price as a proxy for the commodity input cost of ground coffee sold in the United States.

²⁵See <https://www.ers.usda.gov/data-products/charts-of-note/110079?> (accessed 13/07/2026).

For both of these commodity price series, we create weekly average prices by averaging across all observed prices in a given week. Figure 7 shows the weekly time series of both prices.

Commodity-Price Conversion Factors. For the levels regressions, we need to express each raw-commodity price in downstream-product-equivalent units (Sangani, 2026). Let c_{jt}^{raw} denote the raw-commodity price per gram in category j , and let m_j denote the adopted raw-commodity-equivalent factor for one gram of the downstream product. We define $c_{jt} = m_j c_{jt}^{\text{raw}}$ and use Δc_{jt} in the distributed-lag regressions. Thus, $\sum_{k=0}^K \beta_k$ measures the cumulative retail-price response to a change in the converted commodity input-price measure, with both variables expressed per gram of downstream product. The factors and their sources are reported in Table B.1. (Note that the conversion rescales level coefficients and standard errors proportionally and therefore leaves t -statistics and the within-category Amazon-3P comparison unchanged.) The coffee factor is a processing-yield adjustment (weight is lost during roasting). For cocoa, the factor is conceptually different: it measures the beans processed per unit of powder, but the same beans also produce cocoa butter.

Table B.1: Commodity-price conversion factors

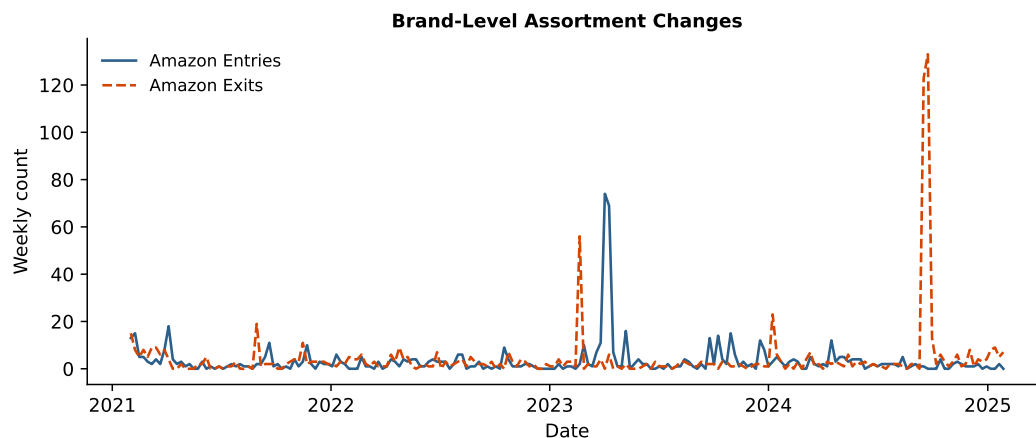
Downstream product category	Conversion factor m_j
Ground roasted coffee	1.235
Baking cocoa (cocoa powder)	2.44

Notes: Raw-commodity prices are multiplied by m_j . For coffee, $m_{\text{coffee}} = 1/(1 - 0.19) = 1.235$, using the 19 percent roasting-weight loss in Nakamura and Zerom (2010) and Sangani (2026). For cocoa, $m_{\text{cocoa}} = 1/(0.82 \times 0.50) = 2.44$, using the mass-balance conversion ratios reported by Fairtrade International (2026, Requirement 2.1.5). The same ratios are also used by Rainforest Alliance (2025, Section 5).

C Facts

C.1 Further Evidence on Fact 2

Figure C.1: Amazon assortment change on the brand level: case of “The Coffee Fool”



Notes: To illustrate that Amazon sometimes enters or exits systematically many ASINs of a particular brand, the Figure shows the weekly number of (re-)entries/exits of Amazon over time in the coffee category, focusing on the brand “The Coffee Fool” only. We only count a re-entry (re-exit) as such if they are followed by a spell of presence (absence) of 5 days or more.

Table C.1: Predictors of Amazon entry using Logit models (3P-only days at risk, using FBA prices)

Dependent variable	Outcome: qualifying Amazon entry ($t + 1$) event (continuous 3P availability under FBA)			
	(1) Baseline	(2) + 3P price	(3) + Sales rank	(4) + lagged changes
Num. 3P sellers ($t - 1$)	0.0400*** (0.0140)	0.0417*** (0.0143)	0.0412*** (0.0144)	0.0348** (0.0161)
weekdayTue	0.4278** (0.1845)			
weekdayWed	0.7976*** (0.1777)			
weekdayThu	0.6070*** (0.1821)			
weekdayFri	0.5886*** (0.1830)			
weekdaySat	0.4593** (0.1864)			
weekdaySun	0.0822 (0.1976)			
FBA price ($t - 1$)		0.0090 (0.0111)	0.0107 (0.0107)	0.0228*** (0.0083)
log Sales rank ($t - 1$)			-0.0730* (0.0396)	-0.0961** (0.0444)
Δ_3 log Sales rank ($t - 1$)				-0.0508 (0.0701)
Δ Num. 3P sellers ($t - 1$)				0.1874*** (0.0537)
Δ FBA price ($t - 1$)				-0.0062 (0.0167)
Δ FBM price ($t - 1$)				0.0163 (0.0139)
ASIN	✓	✓	✓	✓
Week	✓	✓	✓	✓
Observations	227,755	227,755	224,960	151,766

Notes: The estimation sample consists of all ASIN-days at risk of entry, that is, every 3P-only day (Amazon absent) for switcher ASINs (ASINs that ever switch between a “dual” and a 3P-only regime). The dependent variable equals one if a baseline qualifying Amazon entry – requiring continuous 3P (under FBA) availability throughout the event window – occurs on the following day, and zero otherwise. Models are estimated by discrete-time logit. Δ_3 log Sales rank ($t - 1$) is the three-day change in the natural log of the Amazon sales rank measured at $t - 1$, i.e. $\log(\text{rank}_{t-1}) - \log(\text{rank}_{t-4})$; it approximates the three-day percentage change in sales rank. Because sales rank is inversely related to units sold, a positive value indicates a deterioration (decline) in the product’s relative sales performance. Clustered (ASIN) standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Table C.2: Predictors of Amazon exit using Logit models (both-regime days at risk, using FBA prices)

Dependent variable	Outcome: qualifying Amazon exit ($t + 1$) event (continuous 3P availability under FBA)			
	(1) Baseline	(2) + Prices	(3) + Sales & buy box	(4) + Amazon higher + lagged changes
Num. 3P sellers ($t - 1$)	0.0455** (0.0194)	0.0454** (0.0193)	0.0464** (0.0188)	0.0417** (0.0205)
weekdayTue	0.1438 (0.1663)			
weekdayWed	0.2188 (0.1729)			
weekdayThu	-0.0301 (0.1750)			
weekdayFri	0.1057 (0.1742)			
weekdaySat	0.2361 (0.1659)			
weekdaySun	0.0168 (0.1863)			
Amazon price ($t - 1$)		-0.0271 (0.0188)	-0.0227 (0.0196)	-0.0152 (0.0218)
FBA price ($t - 1$)		0.0011 (0.0104)	-0.0045 (0.0119)	-0.0031 (0.0120)
log Sales rank ($t - 1$)			0.1307** (0.0578)	0.1331** (0.0646)
Amazon wins buy box ($t - 1$)			0.2608 (0.1766)	0.4202* (0.2358)
Δ_3 log Sales rank ($t - 1$)				0.0384 (0.1449)
Amazon > 3P price ($t - 1$)				0.1232 (0.2079)
Δ Num. 3P sellers ($t - 1$)				0.1239 (0.0857)
Δ FBA price ($t - 1$)				-0.0021 (0.0278)
Δ FBM price ($t - 1$)				0.0024 (0.0062)
Δ Amazon price ($t - 1$)				-0.0080* (0.0046)
ASIN	✓	✓	✓	✓
Week	✓	✓	✓	✓
Observations	73,220	73,220	72,387	46,037

Notes: The estimation sample consists of all ASIN-days at risk of exit, that is, every day with Amazon present for switcher ASINs (ASINs that ever switch between a “dual” and a 3P-only regime). The dependent variable equals one if a baseline qualifying Amazon exit – requiring continuous 3P (under FBA) availability throughout the event window – occurs on the following day, and zero otherwise. Models are estimated by discrete-time logit. Δ_3 log Sales rank ($t - 1$) is the three-day change in the natural log of the Amazon sales rank measured at $t - 1$, i.e. $\log(\text{rank}_{t-1}) - \log(\text{rank}_{t-4})$; it approximates the three-day percentage change in sales rank. Because sales rank is inversely related to units sold, a positive value indicates a deterioration (decline) in the product’s relative sales performance. Clustered (ASIN) standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

C.2 Further Evidence on Fact 3

Differences in price levels. To make a precise comparison of 3P sellers’ and Amazon’s prices in product-weeks in which both are selling, Tables C.3 to C.6 display the results from a regression

of logged prices on an indicator for Amazon, with ASIN and year-week fixed effects. It becomes apparent that, if anything, Amazon tends to set a lower price than 3P sellers, especially in the coffee category.

Table C.3: Price differences for identical products: Coffee

Dependent variable Compared to Model:	All products			Popular products		
	Lowest 3P (1)	FBA (2)	FBM (3)	Lowest 3P (4)	FBA (5)	FBM (6)
1{Amazon}	-0.2652*** (0.0072)	-0.1860*** (0.0082)	-0.4245*** (0.0069)	-0.2609*** (0.0078)	-0.1910*** (0.0088)	-0.4266*** (0.0075)
ASIN FE	✓	✓	✓	✓	✓	✓
Week FE	✓	✓	✓	✓	✓	✓
Observations	286,808	202,772	273,438	247,398	177,311	236,672
Adjusted R ²	0.88493	0.92332	0.91610	0.88489	0.92270	0.91726

Notes: The unit of observation is an ASIN-week in the Coffee category. The dependent variable is the log of the posted price. For each column, we restrict the sample to ASIN-week observations in which both Amazon and at least one third-party seller of the relevant type (any 3P, lowest FBA, or FBM) are active. The coefficient on 1{Amazon} measures the average log price difference between Amazon and the corresponding third-party offer for the same product in the same week. “Popular products” are defined above. Standard errors, clustered at the ASIN level, are reported in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table C.4: Price differences for identical products: Cocoa

Dependent variable Compared to Model:	All products			Popular products		
	Lowest 3P (1)	FBA (2)	FBM (3)	Lowest 3P (4)	FBA (5)	FBM (6)
1{Amazon}	-0.0395 (0.0469)	-0.0568* (0.0306)	-0.2676*** (0.0351)	-0.0339 (0.0529)	-0.0449 (0.0319)	-0.2808*** (0.0382)
ASIN FE	✓	✓	✓	✓	✓	✓
Week FE	✓	✓	✓	✓	✓	✓
Observations	13,504	10,371	12,917	11,730	9,195	11,231
Adjusted R ²	0.90274	0.93924	0.93063	0.87096	0.91966	0.90680

Notes: The unit of observation is an ASIN-week in the Cocoa category. The dependent variable is the log of the posted price. For each column, we restrict the sample to ASIN-week observations in which both Amazon and at least one third-party seller of the relevant type (any 3P, lowest FBA, or FBM) are active. The coefficient on 1{Amazon} measures the average log price difference between Amazon and the corresponding third-party offer for the same product in the same week. “Popular products” are defined above. Standard errors, clustered at the ASIN level, are reported in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table C.5: Price differences for identical products: SD Cards

Dependent variable Compared to Model:	All products			Popular products		
	Lowest 3P (1)	FBA (2)	FBM (3)	Lowest 3P (4)	FBA (5)	FBM (6)
1{Amazon}	-0.0309 (0.0192)	-0.0006 (0.0243)	-0.1016*** (0.0226)	-0.0150 (0.0298)	0.0578 (0.0376)	-0.0895** (0.0360)
ASIN FE	✓	✓	✓	✓	✓	✓
Week FE	✓	✓	✓	✓	✓	✓
Observations	20,218	15,719	19,126	10,620	7,963	9,968
Adjusted R ²	0.91216	0.92688	0.91114	0.89789	0.91437	0.89695

Notes: The unit of observation is an ASIN-week in the SD Cards category. The dependent variable is the log of the posted price. For each column, we restrict the sample to ASIN-week observations in which both Amazon and at least one third-party seller of the relevant type (any 3P, lowest FBA, or FBM) are active. The coefficient on 1{Amazon} measures the average log price difference between Amazon and the corresponding third-party offer for the same product in the same week. “Popular products” are defined above. Standard errors, clustered at the ASIN level, are reported in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table C.6: Price differences for identical products: USB Hubs

Dependent variable Compared to Model:	All products			Popular products		
	Lowest 3P (1)	FBA (2)	FBM (3)	Lowest 3P (4)	FBA (5)	FBM (6)
1{Amazon}	0.0028 (0.0170)	0.0289 (0.0211)	-0.0914*** (0.0146)	0.0046 (0.0214)	0.0542 (0.0336)	-0.0967*** (0.0178)
ASIN FE	✓	✓	✓	✓	✓	✓
Week FE	✓	✓	✓	✓	✓	✓
Observations	36,666	27,215	31,872	19,786	14,702	16,881
Adjusted R ²	0.92494	0.93599	0.93954	0.92581	0.93043	0.94510

Notes: The unit of observation is an ASIN-week in the USB Hubs category. The dependent variable is the log of the posted price. For each column, we restrict the sample to ASIN-week observations in which both Amazon and at least one third-party seller of the relevant type (any 3P, lowest FBA, or FBM) are active. The coefficient on 1{Amazon} measures the average log price difference between Amazon and the corresponding third-party offer for the same product in the same week. “Popular products” are defined above. Standard errors, clustered at the ASIN level, are reported in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

D Empirical Evidence: Price Levels

D.1 Pre-trend Tests

Table D.1: Pre-trend Wald tests: joint significance of pre-event coefficients, Amazon entry/exit event studies

Outcome	Event	Restrictions	F-stat	df1	df2	p-value
log(FBA)	Entry	9	0.887	9	413.0	0.5366
log(FBM)	Entry	9	0.539	9	604.0	0.8467
log(FBA)	Exit	9	1.337	9	353.0	0.2162
log(FBM)	Exit	9	1.457	9	488.0	0.1610

Notes: Ho: all pre-event coefficients (entry: $t = -\text{pre_window}, \dots, -2$; exit: $t = -\text{exit_pre_window}, \dots, -2$) are jointly zero; a rejection would signal pre-existing differential trends. F-tests use `fixest::wald()`, which respects the two-way clustered (ASIN x date) covariance used in each event-study model.

Table D.2: Pre-trend Wald tests: joint significance of pre-event coefficients, stacked event studies (main price outcomes: log(FBA), log(FBM))

Outcome	Event	Path	Restrictions	F-stat	df1	df2	p-value
log(FBA)	Entry	3P	9	1.237	9	502.0	0.2695
log(FBA)	Entry	Amazon - 3P	9	1.606	9	502.0	0.1105
log(FBA)	Entry	Amazon (implied)	9	1.069	9	502.0	0.3844
log(FBA)	Exit	3P	9	4.763	9	560.0	0.0000***
log(FBA)	Exit	Amazon - 3P	9	4.229	9	560.0	0.0000***
log(FBA)	Exit	Amazon (implied)	9	0.565	9	560.0	0.8258
log(FBM)	Entry	3P	9	1.478	9	683.0	0.1520
log(FBM)	Entry	Amazon - 3P	9	0.868	9	683.0	0.5534
log(FBM)	Entry	Amazon (implied)	9	0.507	9	683.0	0.8699
log(FBM)	Exit	3P	9	0.611	9	703.0	0.7885
log(FBM)	Exit	Amazon - 3P	9	0.365	9	703.0	0.9514
log(FBM)	Exit	Amazon (implied)	9	0.518	9	703.0	0.8624

Notes: Ho: all pre-event coefficients ($t = -10, \dots, -2$, relative to the omitted $t = -1$) are jointly zero for the indicated path. 3P = third-party event-time path; Amazon - 3P = differential path; Amazon (implied) = 3P + differential. F-tests use `fixest::wald()`, which respects the two-way clustered (ASIN x date) covariance used in each stacked event-study model.

D.2 Estimation Results

Table D.3: Event-studies outcomes: 3P lowest price under FBA or FBM

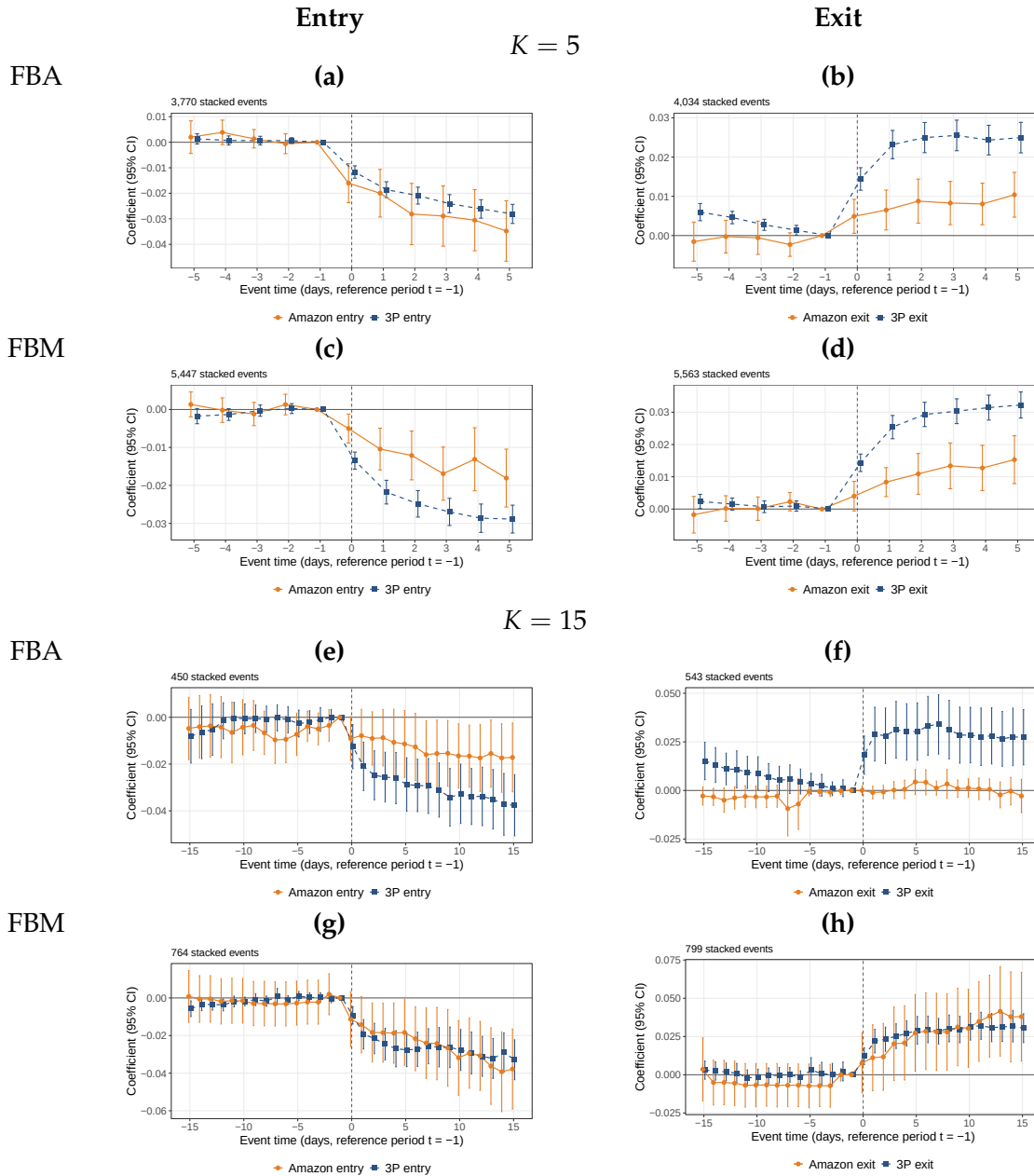
Dependent Variables:	$\log(p^{3P,FBA})$	$\log(p^{3P,FBM})$	$\log(p^{3P,FBA})$	$\log(p^{3P,FBM})$
Event:	Entry		Exit	
Model:	(1)	(2)	(3)	(4)
event_time = -10	0.0001 (0.0048)	0.0038 (0.0037)	0.0021 (0.0040)	-0.0002 (0.0040)
event_time = -9	0.0007 (0.0046)	0.0022 (0.0029)	0.0026 (0.0041)	3.69×10^{-5} (0.0040)
event_time = -8	-0.0006 (0.0045)	0.0033 (0.0027)	0.0010 (0.0045)	-0.0025 (0.0037)
event_time = -7	0.0005 (0.0042)	0.0018 (0.0026)	-0.0003 (0.0043)	1.16×10^{-5} (0.0036)
event_time = -6	7.58×10^{-5} (0.0038)	0.0009 (0.0025)	0.0034 (0.0044)	0.0076 (0.0050)
event_time = -5	0.0025 (0.0032)	7.86×10^{-5} (0.0024)	0.0018 (0.0037)	0.0006 (0.0040)
event_time = -4	0.0028 (0.0029)	-0.0009 (0.0026)	-0.0016 (0.0036)	0.0051 (0.0033)
event_time = -3	0.0015 (0.0027)	-0.0011 (0.0017)	0.0017 (0.0032)	0.0032 (0.0029)
event_time = -2	0.0013 (0.0023)	-0.0009 (0.0017)	-0.0013 (0.0024)	0.0029 (0.0024)
event_time = 0	-0.0186*** (0.0038)	-0.0086*** (0.0022)	0.0106*** (0.0033)	0.0069*** (0.0021)
event_time = 1	-0.0253*** (0.0042)	-0.0153*** (0.0031)	0.0156*** (0.0038)	0.0099*** (0.0032)
event_time = 2	-0.0326*** (0.0045)	-0.0181*** (0.0036)	0.0183*** (0.0041)	0.0192*** (0.0043)
event_time = 3	-0.0379*** (0.0049)	-0.0189*** (0.0041)	0.0242*** (0.0046)	0.0257*** (0.0043)
event_time = 4	-0.0438*** (0.0054)	-0.0204*** (0.0042)	0.0256*** (0.0048)	0.0273*** (0.0048)
event_time = 5	-0.0507*** (0.0058)	-0.0230*** (0.0040)	0.0288*** (0.0050)	0.0320*** (0.0046)
event_time = 6	-0.0536*** (0.0058)	-0.0247*** (0.0044)	0.0286*** (0.0053)	0.0312*** (0.0048)
event_time = 7	-0.0562*** (0.0060)	-0.0260*** (0.0044)	0.0290*** (0.0051)	0.0322*** (0.0050)
event_time = 8	-0.0606*** (0.0063)	-0.0299*** (0.0045)	0.0265*** (0.0061)	0.0330*** (0.0054)
event_time = 9	-0.0635*** (0.0067)	-0.0283*** (0.0048)	0.0267*** (0.0060)	0.0333*** (0.0053)
event_time = 10	-0.0635*** (0.0068)	-0.0295*** (0.0048)	0.0287*** (0.0061)	0.0349*** (0.0056)
Event	✓	✓	✓	✓
Observations	11,403	17,703	10,353	14,301
Number of Events	543	843	493	681
Number of ASINs	414	605	354	489
R ²	0.97950	0.98778	0.98561	0.98530

Notes: Two-way clustered (ASIN $\times$ date) standard errors in parentheses. Columns (1) and (2) report entry event studies; columns (3) and (4) report exit event studies. Dependent variables are $\log(\text{lowest 3P seller price under FBA})$ in columns (1) and (3) and $\log(\text{lowest 3P seller price under FBM})$ in columns (2) and (4). Coefficients are event-time indicators relative to $t = -1$ (omitted period), as specified in equation 3.

D.3 Robustness and Additional Outcomes

D.3.1 Alternative qualifying-spell requirements

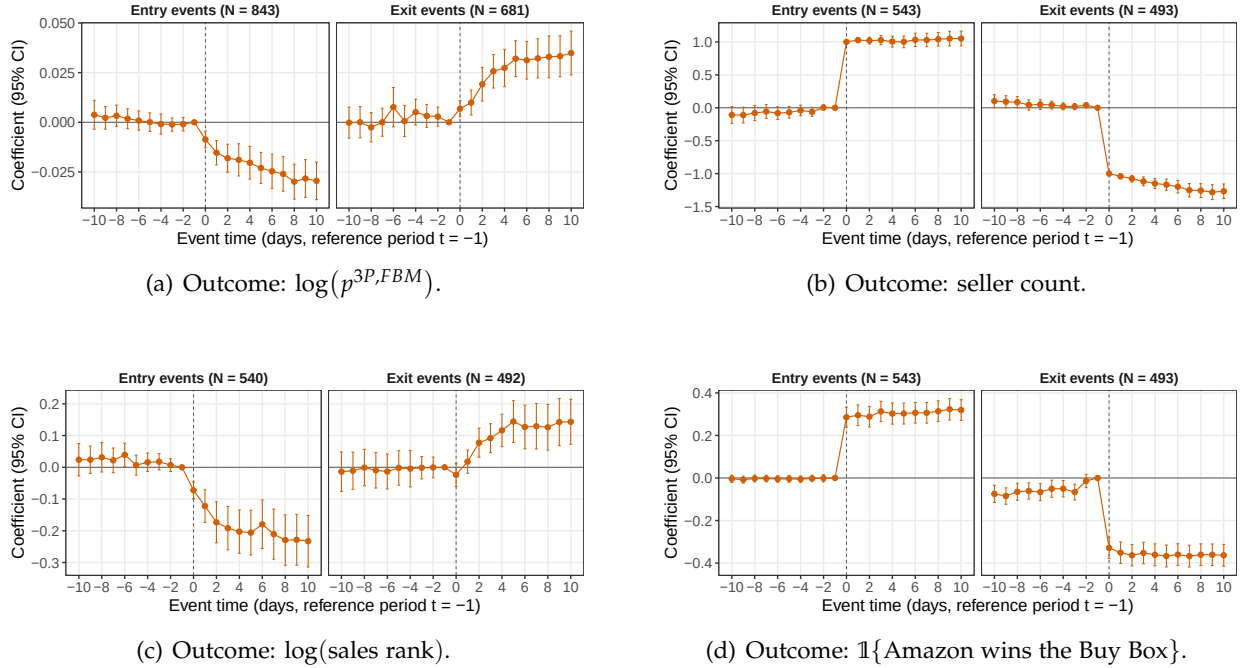
Figure D.1: Stacked event studies under alternative qualifying-event definitions



Notes: The figure reports stacked event-study estimates for Amazon and 3P seller entry and exit under alternative qualifying-event definitions. Panels (a)-(d) use $K = 5$, and panels (e)-(h) $K = 15$. Within each group, the top row outcome is the logarithm of the lowest 3P price under FBA, the bottom under FBM. The left column reports entry events and the right column exit events. Relative to the baseline definition $K = 10$, $K = 5$ requires shorter pre- and post-event periods and admits a larger set of qualifying events, whereas $K = 15$ selects a smaller set of events with more persistent seller-presence regimes. The omitted event day is $r = -1$. Vertical bars report 95% confidence intervals based on standard errors two-way clustered by ASIN and calendar date. All regressions use the daily coffee data and otherwise follow Equation 4.

D.3.2 Additional market outcomes

Figure D.2: Amazon entry and exit: additional event-study outcomes



Notes: The figure reports event-study estimates from Equation 3 using outcomes other than the baseline lowest FBA price. Panel (a) uses the logarithm of the lowest FBM price, panel (b) uses the total number of marketplace sellers, panel (c) uses the logarithm of sales rank, and panel (d) uses an indicator for whether Amazon occupies the Buy Box. The omitted event day is $r = -1$. Vertical bars report 95% confidence intervals based on standard errors two-way clustered by ASIN and calendar date. All regressions use clean Amazon entry and exit events in the daily coffee sample, as defined in the text.

D.3.3 Alternative specifications

Tables D.4 and D.5 show the results from a more "naive", but more standard, two-way fixed effects (TWFE) specification using data at the daily level. We run separate regressions for each product category. For product i observed on day t , we estimate:

$$\log(p_{it}^{3P,m}) = \beta \cdot \mathbb{I}(\text{Amazon}_{it}) + \mu_i + \lambda_t + \varepsilon_{it} \quad (9)$$

where, as in the main text, $p_{it}^{3P,m}$ denotes the lowest price offered by 3P sellers under $m \in \{FBA, FBM\}$ for ASIN i on day t . $\mathbb{I}(\text{Amazon}_{it})$ is an indicator taking the value 1 if Amazon is active as a seller for product i at time t . The inclusion of product fixed effects (μ_i) and time fixed effects (λ_t) absorbs time-invariant product characteristics (e.g., quality) and common macro-shocks, respectively.

Table D.4: Amazon's presence and 3P lowest price (FBA), by product category

Product Dependent Variable: Model:	Coffee (1)	Cocoa (2)	Pasta $\log(p^{3P,FBA})$ (3)	Shirts (4)	TVs (5)	USBhubs (6)
1(Amazon present)	-0.092*** (0.006)	-0.073*** (0.020)	-0.108*** (0.011)	-0.070*** (0.010)	-0.028*** (0.008)	-0.076*** (0.020)
asin	✓	✓	✓	✓	✓	✓
yearmonthday	✓	✓	✓	✓	✓	✓
Observations	929,797	66,371	394,602	96,467	116,668	124,066
N ASINs	1,411	82	590	206	386	172
R ²	0.90951	0.94421	0.92521	0.77564	0.98214	0.94066

Notes: Standard errors clustered by ASIN in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$. All columns cover the sample period January 2021 to March 2025. Every column is further restricted to ASINs with ≥ 10 consecutive days with Amazon and 3P both priced, and ≥ 10 consecutive days with 3P priced while Amazon is absent.

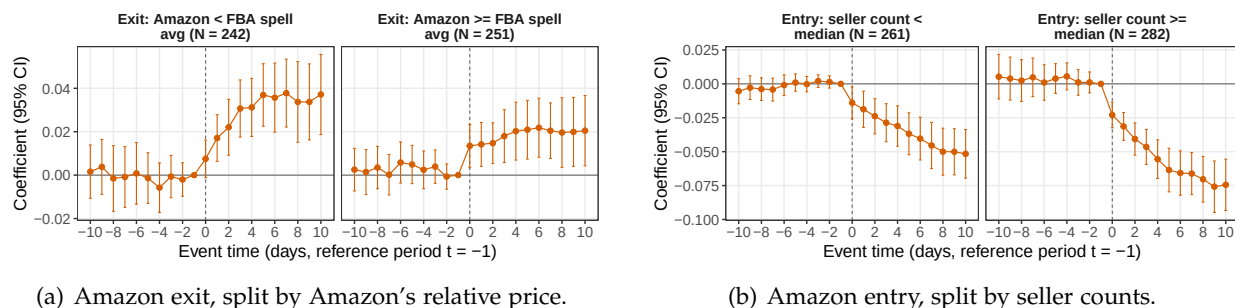
Table D.5: Amazon's presence and 3P lowest price (FBM), by product category

Product Dependent Variable: Model:	Coffee (1)	Cocoa (2)	Pasta $\log(p^{3P,FBM})$ (3)	Shirts (4)	TVs (5)	USBhubs (6)
1(Amazon present)	-0.030*** (0.006)	-0.054*** (0.018)	-0.048*** (0.008)	-0.020** (0.009)	-0.048*** (0.007)	-0.067*** (0.016)
asin	✓	✓	✓	✓	✓	✓
yearmonthday	✓	✓	✓	✓	✓	✓
Observations	1,487,611	87,099	641,785	182,797	367,899	111,807
N ASINs	1,601	89	746	243	497	131
R ²	0.91264	0.95081	0.91872	0.80469	0.96714	0.91209

Notes: Standard errors clustered by ASIN in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$. All columns cover the sample period January 2021 to March 2025. Every column is further restricted to ASINs with ≥ 10 consecutive days with Amazon and 3P both priced, and ≥ 10 consecutive days with 3P priced while Amazon is absent.

D.3.4 Sample splits and response heterogeneity

Figure D.3: Heterogeneity in 3P price responses



Notes: Panel (a) reports Amazon exit event-study estimates for subsamples defined by whether Amazon's price at entry is smaller, or at least as great as the pre-period lowest 3P price under FBA. The outcome is the logarithm of the lowest FBA price. The omitted event day is $t = -1$. Vertical bars report 95% confidence intervals based on standard errors two-way clustered by ASIN and calendar date. See the text for the precise definition of the subsamples. Panel (b) reports event-study estimates from Equation 3 for the logarithm of the lowest 3P FBA price, separately by whether the pre-period total seller count is below, or greater or equal than the median seller count among the products considered. The omitted event day is $t = -1$. Vertical bars report 95% confidence intervals based on standard errors two-way clustered by ASIN and calendar date. All regressions use clean Amazon entry events in the daily coffee sample, as defined in the text.

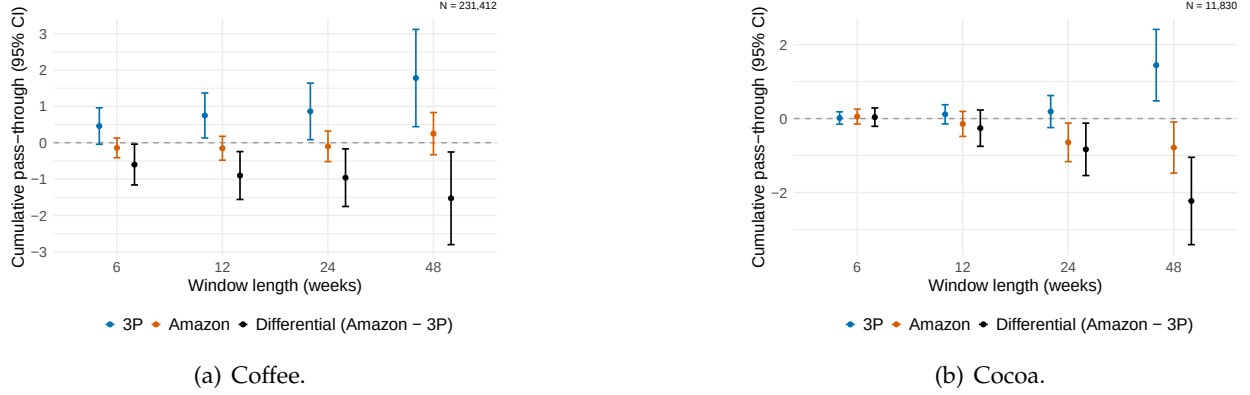
E Empirical Evidence: Asymmetric Pass-Through of Cost Shocks

E.1 Subsample analyses

E.2 Alternative specification

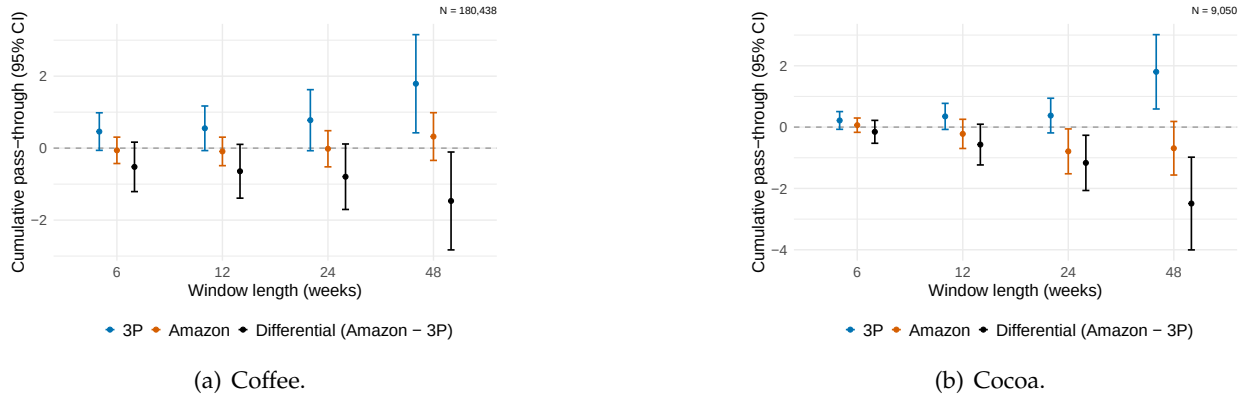
E.3 Buy Box-winning price

Figure E.1: Cumulative commodity-cost pass-through by seller type: matched ASIN-week sample excluding high-priced ASINs



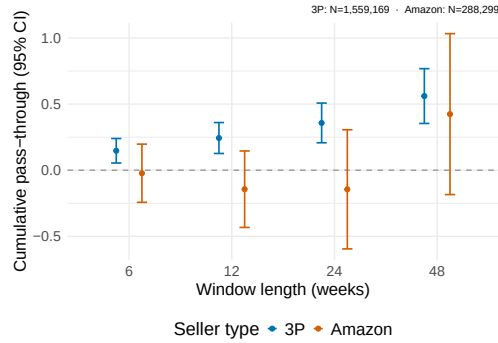
Notes: The figure reports cumulative pass-through estimates from distributed-lag regressions of weekly changes in posted retail prices on weekly changes in converted commodity costs. At window length K , the contemporaneous coefficient and first $K - 1$ lag coefficients are summed. Both variables are measured in dollars per gram, so an estimate of one denotes full pass-through. Regressions use a matched sample of ASIN-week cells with observed weekly price changes for both Amazon's posted price and the lowest 3P price, and include $\text{ASIN} \times \text{seller-type}$ fixed effects. The sample additionally excludes ASINs whose median price exceeds the 80th percentile of prices in the category. "Differential (Amazon - 3P)" is the difference between the two estimates; negative values indicate lower pass-through by Amazon. Vertical bars show 95% confidence intervals based on standard errors two-way clustered by ASIN and calendar week.

Figure E.2: Cumulative commodity-cost pass-through by seller type: matched ASIN-week sample excluding ASINs without ≥ 48 week consecutive presence by both seller types

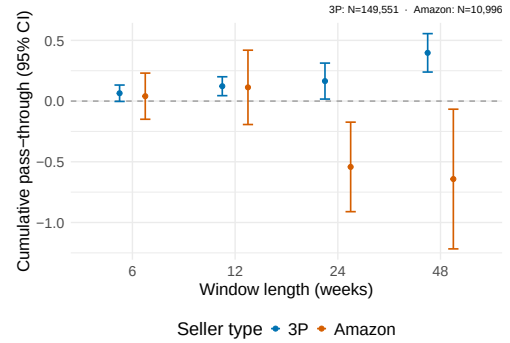


Notes: The figure reports cumulative pass-through estimates from distributed-lag regressions of weekly changes in posted retail prices on weekly changes in converted commodity costs. At window length K , the contemporaneous coefficient and first $K - 1$ lag coefficients are summed. Both variables are measured in dollars per gram, so an estimate of one denotes full pass-through. Regressions use a matched sample of ASIN-week cells with observed weekly price changes for both Amazon's posted price and the lowest 3P price, and include $\text{ASIN} \times \text{seller-type}$ fixed effects. The sample excludes ASINs that do not have at least 48 weeks of consecutive presence by both seller types. "Differential (Amazon - 3P)" is the difference between the two estimates; negative values indicate lower pass-through by Amazon. Vertical bars show 95% confidence intervals based on standard errors two-way clustered by ASIN and calendar week.

Figure E.3: Cumulative commodity-cost pass-through by seller type: seller-specific ASIN-week samples



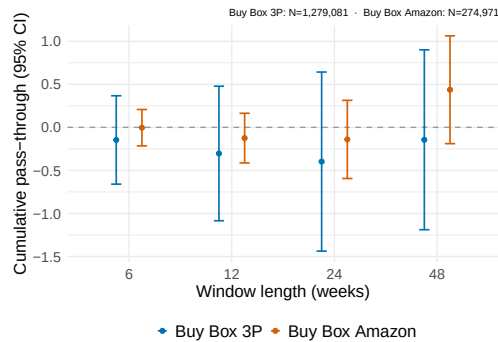
(a) Coffee.



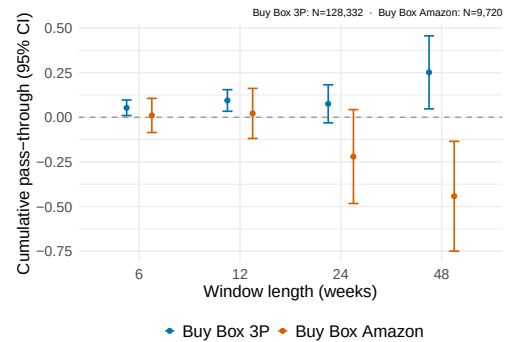
(b) Cocoa.

Notes: The figure reports cumulative pass-through estimates from distributed-lag regressions of weekly changes in posted retail prices on weekly changes in converted commodity costs. At window length K , the contemporaneous coefficient and first $K - 1$ lag coefficients are summed. Both variables are measured in dollars per gram, so an estimate of one denotes full pass-through. Regressions are estimated separately for Amazon's posted price and the lowest 3P price, using all ASIN-week cells with an observed weekly price change for the relevant seller type and including ASIN fixed effects. Vertical bars show 95% confidence intervals based on standard errors two-way clustered by ASIN and calendar week.

Figure E.4: Cumulative commodity-cost pass-through using Buy Box prices: seller-specific ASIN-week samples



(a) Coffee.



(b) Cocoa.

Notes: The figure reports cumulative pass-through estimates from distributed-lag regressions of weekly changes in posted retail prices on weekly changes in converted commodity costs. At window length K , the contemporaneous coefficient and first $K - 1$ lag coefficients are summed. Both variables are measured in dollars per gram, so an estimate of one denotes full pass-through. Regressions are estimated separately by seller type using Buy Box prices in ASIN-weeks in which that seller type is displayed in the Buy Box, and include ASIN fixed effects. Vertical bars show 95% confidence intervals based on standard errors two-way clustered by ASIN and calendar week.

Online Appendix of “Pricing Patterns on Dual-Mode E-Commerce Platforms”

Luise Eisfeld Jun Yan Li Yu

August 13, 2026

F Data: Additional Information

F.1 Data Collection from Keepa

As a first step, we use Keepa’s “Product Finder” tool to arrive at a list of ASINs related to a given product category (e.g., Coffee or Men’s T-Shirts). Table 1 shows the query parameters used for each of the products that were queried. Note that a product (ASIN) on Amazon can belong to one *Root Category*, and to multiple *Browse Nodes*¹. The latter are called *Subcategories* on the Keepa website. We configure the Keepa Product Finder filters so that only subcategories directly related to our target product (e.g., “Ground Coffee”) are included, while certain adjacent but irrelevant subcategories (e.g., “Instant Coffee”) are explicitly excluded. This enables us to focus on truly similar products.

For product categories with over 10,000 products listed on amazon.us, we only collect data on those 10,000 products with the lowest current sales rank or the lowest 180-day average sales rank at the time of scraping, thereby focusing our attention on products that actually generate sales and are (normally) in stock. After obtaining a list of ASINs for a given product category, we query these ASINs one by one using “Product Object” queries² through Keepa’s Application Programming Interface (API), and thereby obtain product and price information. We query all categories at multiple instances in time in the period of July 2024 through March 2026.

After collecting the list of ASINs, we exclude products which (either based on manual inspection or based on the variable “type”) seem to belong to a different category and are thus misclassified. For products in the Pasta category, we exclude products marked with terms like “buckwheat”, “rice” or “gluten-free”, so as to focus on products made from wheat; for products in the coffee category, we exclude chicory or instant coffee sticks, for instance.

¹See <https://webservices.amazon.com/paapi5/documentation/use-cases/organization-of-items-on-amazon/browse-nodes/browse-nodes-and-items.html> (accessed on 25/11/2024). The Browse Node IDs of the subcategories listed in Table 1 are not displayed for readability reasons, but can be provided upon request.

²See <https://keepa.com/#!discuss/t/product-object/116> (last accessed 18/02/2026).

Table 1: Query parameters used in initial product data collection process on Keepa

Root Category and Product Category	Included and excluded subcategories
Root Category: Grocery & Gourmet Food	
Coffee	· include: Ground Coffee · exclude: Tea; Instant Coffee
Cocoa	· include: Cocoa · exclude: Chocolate Drink Mixes; Hot Chocolate & Malted Drinks; Thanksgiving, Tea & Cocoa
Pasta	· include: Pasta & Noodles · exclude: Fresh Pasta & Sauces; Gnocchi
Root Category: Electronics	
TVs	· include: LED & LCD TVs · exclude: Remote Controls; TV Mounts, Stands & Turntables; Satellite Equipment
USB Hubs	· include: USB Hubs
Micro SD Cards	· include: Micro SD Cards
Root Category: Appliances	
Washing Machines	· include: Washers · exclude: Countertop Dishwashers; Built-in Dishwashers; Portable Dishwashers; Washer Parts & Accessories; Portable Washers
Root Category: Clothing, Shoes & Jewelry	
Men's T-Shirts	· include: Men's T-Shirts · exclude: Tights, Pants & Shorts; Leggings

Within a given category, we only use data on products that are ever on sale on [amazon.com](https://www.amazon.com) and thus have a price indicated at any point since 2021.

The original “Product Object” data downloaded through the Keepa API essentially consists of separate datasets for the lowest “marketplace” offer (which can be Amazon or a 3P seller’s price, called NEW in Keepa); the lowest price set by any 3P seller price operating under FBA; the lowest price set by any 3P seller price operating under FBM; Amazon’s price; and the price of the seller whose offer is displayed in the Buybox. (It is worth noting that, prior to 2026, Keepa updated the lowest prices under FBA/FBM at slightly lower frequencies than the other prices.) One row in the original data corresponds to a given price change at a given moment in time (a price becoming unavailable or available corresponds to a “change” as well). We define the lowest 3P price (which we call NEW_third) as the marketplace price (NEW) whenever it differs from the Amazon price, and as the minimum of the lowest FBA and FBM prices otherwise.

Adjustment for occasionally occurring delayed Amazon price recording. The product-

level price data from Keepa occasionally exhibits a timing discrepancy in which Amazon’s offer is detected with a short lag. Specifically, when a previously out-of-stock Amazon offer returns to the marketplace, Keepa’s system may register the resulting change in the marketplace price (NEW) and seller count (COUNT_NEW) before identifying the Amazon offer itself. During this interim period – lasting only a few days – the AMAZON price field remains missing even though the marketplace price already reflects Amazon’s presence, which would cause the lowest 3P price (NEW_third) to be incorrectly attributed. We confirmed this data-collection artifact directly with the Keepa team³. To address it, we apply a conservative correction procedure that detects instances in which (i) the seller count increases by exactly one, (ii) the marketplace price simultaneously decreases, (iii) the Amazon price is not yet recorded, (iv) no FBA or FBM price equals the marketplace price on the affected days (ruling out the possibility that the price drop is attributable to a 3P seller), and (v) within seven days the Amazon price appears while the seller count remains *unchanged*. When all four conditions are met, we backfill the Amazon price to the date of the initial seller-count increase. This adjustment affects a small number of product-days (0.001% in the case of Cocoa and 0.002% in the case of coffee, for instance) and ensures that Amazon’s entry date is correctly recorded.

Dealing with a small number of apparent “pricing errors”. We noticed a certain number of apparent “pricing errors” in the original data, i.e., instances where a product’s price jumps up by several multiples of the original price for a period of time. For instance, a package of coffee bearing the ASIN B09B8TP9RQ, typically priced at around \$24.99, is suddenly offered at \$5,889.34 on February 20th to 26th, 2025 – an almost 200-fold increase in its price – before coming back down. These can be due to, e.g., data entry mistakes, or algorithmic malfunctions. We never observe such patterns in the case of Amazon’s offers, only for 3P sellers’ prices. To detect these upward price deviations in a conservative, but sensible way and exclude them from our dataset upfront, we employ a multi-stage procedure to identify these errors at both the observation and product levels. For each ASIN, we calculate the interquartile range ($IQR = Q_{75} - Q_{25}$) of observed prices. An individual price observation is flagged as an outlier if it satisfies all three conditions: (1) $Price > Q_{75} + 30 \times IQR$; (2) $Price - Q_{75} > 2 \times median_all$, where *median_all* is the median price across all products; and (3) $Price > 2 \times median_product$. These criteria jointly ensure that flagged observations represent extreme deviations in both product-specific terms (IQR criterion), absolute terms (preventing

³See <https://keepa.com/#!discuss/t/price-update-frequency-for-amazon/18645> (last accessed 18/02/2026).

spurious flags in low-variance products), and relative terms (substantial deviation from the product’s typical price level). While we allow for the exclusion of entire *products* from the sample when outlier prevalence suggests systematic data quality issues⁴, in the case of coffee or cocoa, there are no such systematic data quality issues for any products in our sample. More generally, only around 0.2% of all ASINs are affected by these pricing errors. Finally, in the coffee category, we exclude one brand (“San Marco Coffee”) which operated under a dropshipping arrangement until 2024. For this brand, the data show an extreme repricing event, followed by a synchronized exit of Amazon offering any products by that brand. The observed patterns are consistent with a termination in the vendor relationship, which was confirmed to us by the brand, and thus is likely independent from any pass-through of cost shocks.

F.2 Extracting Weight Using the OpenAI API

For our analysis, a precise extraction of product package sizes is crucial. However, package sizes from the relevant variables included by Keepa are missing for many of the products. Instead, package sizes are being accurately described in the product title, but due to non-standardized formatting, are difficult to extract using standard string-matching methods. To address this, we employ OpenAI’s gpt-4o-mini model. The prompt instructs the model to identify and compute information on total weight or volume of the content as indicated in the product title, and is displayed below. Note that an ASIN’s weight cannot vary over time; the introduction of a different package size necessarily implies that a new ASIN is created.

You are tasked with extracting the total weight or volume of a grocery product from its description. Follow these specific rules:

1. ****Accurate extraction only****: Use only the information explicitly provided in the product description. Do not infer or assume any details.
2. ****Output format****: Provide the result as ‘number|unit’.
 - If the description includes a total weight or volume (including multipacks), calculate the combined total if necessary.
 - Units like ounces (oz), grams (g), pounds (lb), kilograms (kg), liters (L), or milliliters (ml) should be preserved as stated.

⁴Specifically, we remove all observations for a product if either: (1) more than 30% of observed prices are flagged as outliers, or (2) the product has at least 10 outliers and fewer than 50 total price observations.

```

3. **No weight or volume available**: If no weight or volume
   information is provided, return 'NA|NA'.

**Examples**:
- **Input**: "Starbucks Caffe Verona Coffee, Dark, Ground, 12-Ounce
  Bags (Pack of 3)"
  **Output**: 36|oz
- **Input**: "Four Sigmatic Think Mushroom Coffee | Organic Ground
  Coffee with Lion's Mane Mushroom and Chaga Mushroom | 12oz Bag"
  **Output**: 12|oz
- **Input**: "AmazonFresh Go For The Bold Ground Coffee, Dark Roast"
  **Output**: NA|NA
- **Input**: "AmazonFresh French Vanilla Flavored Coffee, Ground,
  Medium Roast, 500g"
  **Output**: 500|g

Now, extract the total weight or volume from the following
description:

---
{text}
---

```

To guarantee consistency and reproducibility, we set the temperature parameter to 0. We moreover include a system message – "role": "system", "content": "You are an expert at analyzing grocery product descriptions and extracting accurate weight or volume information. You strictly follow instructions, avoid assumptions, and format answers as specified." – which primes the model as a domain expert and ensures it adheres rigorously to the extraction protocol without introducing unwarranted assumptions.

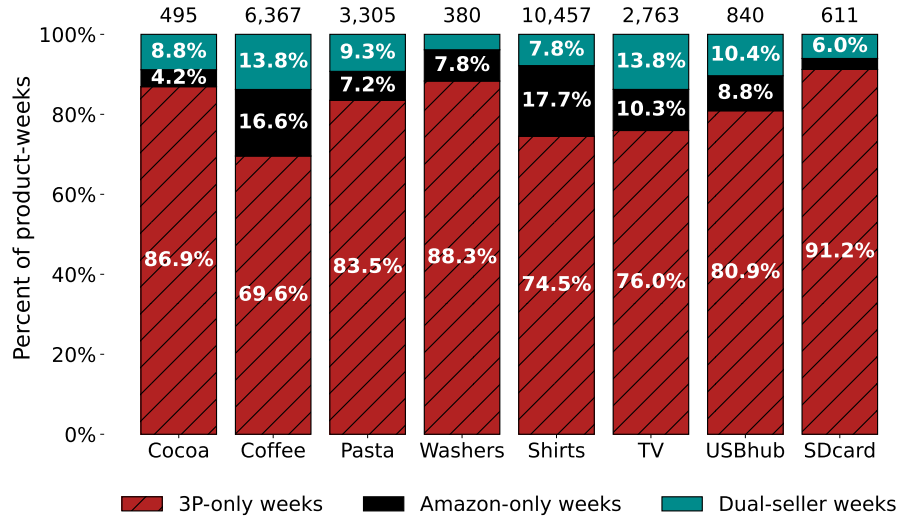
Table 2 displays the resulting distribution of weights across ASINs, showing that we are able to recover the product weight for a vast majority of products, and showing that the median package size is relatively standard.

Table 2: Distribution of product weights across ASINs in grocery categories

Category	ASINs	ASINs w/ weight available	Mean	Median	SD	P10	P90
Cocoa	1,432	1,393	1137.2	453.6	2628.3	170.5	2268.0
Coffee	14,532	14,282	828.1	453.6	894.2	270.0	2041.2
Pasta	7,709	7,477	1861.2	997.9	2258.4	340.2	5000.0

Note: The table reports ASIN-level weight statistics (in grams) for Cocoa, Coffee, and Pasta products.

Figure 1: Extent of dual-mode selling across categories, popular products



Notes: Bars show, for each category, the share of ASIN-week observations in which the product is sold only by 3P sellers, only by Amazon, or by both. The sample contains **only popular** ASINs (as defined in Section ??), observed weekly from January 2021 through December 2024. A product-week is classified as 3P-only (Amazon-only) if at least one 3P (Amazon) offer is recorded and the other side is absent; it is classified as dual-seller if both have offers in the same week.

G Facts

G.1 Fact 1

Survey of Previous Literature On Amazon Entering. Several academic papers shed light on this question by studying the predictors of Amazon’s decision to directly re-sell a product (i.e., to “enter”).

[Hagiu and Wright \(2015\)](#) use data on DVDs that were sold on Amazon in 2014, and find that products offered by Amazon have lower sales rank, i.e., are more popular, than those offered by 3P sellers.

Zhu and Liu (2018) employs data from 2013 and 2014 and finds that Amazon is more likely to begin re-selling products that experience higher sales and have better reviews, and which are not yet using Amazon’s fulfillment services.

A working paper by Crawford, Courthoud, Seibel and Zuzek (2022) uses proprietary data on prices and revenues in Amazon’s Home & Kitchen category. The paper focuses on Amazon’s decision to begin selling products for which only a single 3P merchant supplied that product in the first four weeks of the product’s existence. The authors find that Amazon starts re-selling products that experience high demand growth and where competition is low (i.e., few active sellers are present). This is true both relative to products which Amazon does *not* enter, as well as relative to the entry decisions by big (high-revenue) 3P sellers. Entry by Amazon occurs even if a large or efficient 3P seller is present. Amazon is also more likely to enter when 3P listings exist but generate no sales, suggesting the presence of low-quality or capacity-constrained sellers. The authors argue that these patterns suggest the internalization of platform externalities as a possible motive for Amazon’s entry decisions. Among those externalities are expanding product variety, ensuring product availability, and mitigating seller market power.

Concerning Amazon’s presence as a re-seller more generally, the authors find that nearly half of the revenue from products in which Amazon is a reseller comes from cases of *de novo* entry – products that had not been sold by any 3P seller before Amazon’s entry as a reseller. Amazon’s private label products play a minor role in the Kitchen & Home category, accounting for only 0.3% of total revenues. Amazon is present in products accounting for 39% of revenues in the Home & Kitchen category.

The findings by Bennati (2026), who studies the market for headphones and uses sales estimates stemming from the market intelligence company AmzScout, are very much in line with this. The author’s findings suggest that 3P sellers concentrate on products in the long tail, whereas Amazon re-sells products that are in higher demand and of higher quality.

Correlates of Amazon entry. Taking the example of coffee, we find that, in the cross-section, the brand and the manufacturer are main factors correlated with Amazon’s entry patterns. Controlling for brand and manufacturer, we moreover find that products where Amazon is a reseller at any point during our period of observation are likely to derive more sales, and are likely to feature more other sellers, as can be seen in Table 3. The finding that more competition by 3P sellers is correlated with Amazon re-selling the product could in fact be driven by unobserved (by the econometrician) demand shocks, which might cause entry

by both Amazon and other merchants. Hence, our measure of “high competition” might in fact be a proxy for “high demand”. (Of course, this result can by no means be interpreted causally, and is purely descriptive.)

In the time series, interestingly we find that, across our product categories, the overall number of products re-sold by Amazon has been constant over the years, while the number of products re-sold by 3P sellers has increased over time (not shown in paper).

Table 3: Correlates of whether a product is ever supplied by Amazon, Coffee products only

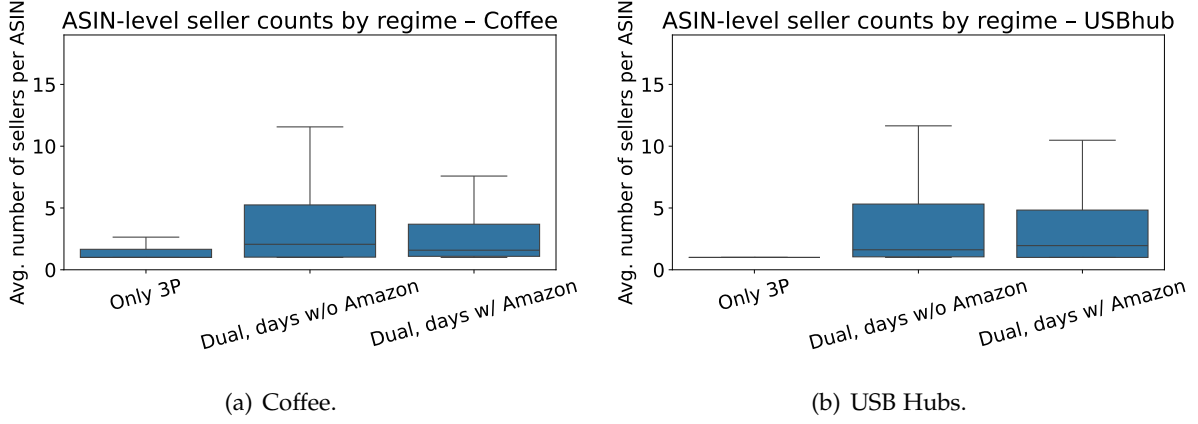
	<i>Dependent variable:</i>	
	1{product ever sold by Amazon}	
1{popular}	0.223** (0.090)	0.162*** (0.047)
1{has any reviews}	0.175*** (0.061)	0.025 (0.033)
avg number of 3P sellers	0.008 (0.013)	0.007*** (0.003)
package size (weight, in kg)	$1.002 e^{-8}$ ($1.913 e^{-8}$)	$8.316 e^{-10}$ ($5.273e^{-9}$)
product was ever stocked out	0.164*** (0.056)	0.014 (0.031)
Manufacturer FE		✓
Brand FE		✓
Observations	12,759	12,664
Adjusted R ²	0.076	0.788

SEs clustered at manufacturer level. *p<0.1; **p<0.05; ***p<0.01

Notes: Cross-sectional dataset was created by beginning with a daily panel ranging from January 1st 2021 to March 31 2025, and collapsing it to the cross-sectional level. 1{popular} is the popularity measure defined in Section ??, indicating the proportion of days on which a product’s sales rank is below the median sales rank across products, and hence varies between 0 and 1. The average number of 3P sellers is defined as an average across all days on which a product is available for sale. The coefficients from a linear regression are shown, and each observation is a product.

As documented in the main text, the presence of Amazon as a seller (in terms of number of products) varies across product categories. Figure 1 mirror the idea that Amazon tends to be active in “popular” products, and Figure 2 analyzes the extent of competition conditional on Amazon presence. Finally, Table 4 presents summary statistics focusing only on products which ever experience Amazon presence in our sample.

Figure 2: Competition by seller regime for exemplary categories



Notes: The figure plots, for the Coffee and USB hub categories, the distribution across ASINs of the average daily number of sellers in three regimes. “Only 3P” includes ASINs that are never supplied by Amazon but have at least one 3P seller; for each such ASIN we average the number of sellers over days on which a 3P seller is present. “Dual, days w/o Amazon” restricts to ASINs that are supplied at some point by both Amazon and 3P sellers, but averages the number of sellers only over days when Amazon is absent and at least one 3P seller is present. “Dual, days w/ Amazon” uses the same set of dual ASINs but averages over all days when Amazon is present, regardless of whether a 3P seller is active. (We restrict to days with at least one seller present.)

G.2 Fact 2

Description of seller regimes.

Table 4: Summary of sample by category, restricted to ASINs ever sold by Amazon (ASIN-week panel; 222 weeks)

Category	Products (ASINs)	Avg. seller count (per ASIN-week)	ASIN-weeks with ≥ 2 sellers	With reviews (across ASINs)	Amazon private label (across ASINs)
Cocoa	160	4.8	62.9%	92.5%	1.9%
Coffee	4,338	3.1	41.0%	97.8%	1.0%
Pasta	1,432	3.7	59.7%	91.3%	2.0%
Washers	93	3.7	45.3%	74.2%	0.0%
TV	1,816	3.5	40.9%	79.2%	0.0%
Shirts	3,365	1.2	7.4%	72.4%	0.0%
USB hubs	879	5.8	36.6%	50.2%	3.3%
SD cards	205	5.3	65.2%	88.3%	2.0%

Notes: Restricted to ASINs where Amazon retail lists the product (AMAZON non-missing) in at least one ASIN-week. We track sampled ASINs daily from January 1, 2021 to March 31, 2025 and collapse the data to the ASIN-week level. The average seller count corresponds to the average number of sellers listing an ASIN in a week, computed over ASIN-weeks with at least one observed seller, and includes Amazon retail when present. “With reviews” and “Amazon private label” are product-level shares. The review indicator is measured from the latest observed product page and is therefore time-invariant within the sample.

Table 5: Descriptives: Seller regimes (FBA as 3P price)

	Obs.	Share	N (ASINs)	
<i>Panel A: ASIN-day regime counts</i>				
3P only	641,316	0.264	1,280	
Amazon only	695,212	0.286	1,420	
Both	287,892	0.119	1,283	
Neither	802,895	0.331	1,559	
Total	2,427,315	1.000	1,565	

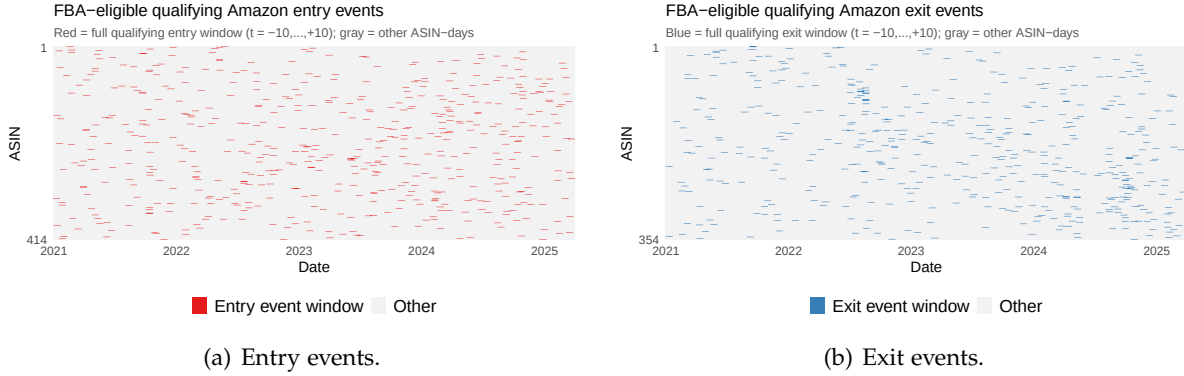
	Mean	Median	P10	P90
<i>Panel B: Within-ASIN share of days in regime</i>				
3P only	0.264	0.145	0.000	0.727
Amazon only	0.286	0.255	0.001	0.681
Both	0.119	0.055	0.000	0.325
Neither	0.331	0.245	0.029	0.756

	N (spells)	Mean duration	Median duration
<i>Panel C: Regime persistence (spell duration in days)</i>			
3P only	15,017	42.71	11.00
Amazon only	12,904	53.88	6.00
Both	12,141	23.71	4.00
Neither	16,146	49.73	6.00
<i>Regime transitions per ASIN</i>			
Mean	34.92		
Median	26.00		

Notes: Coffee category; ASIN-day panel (midnight prices), January 2021–March 2025. ASINs are restricted to those satisfying the event-study screen used in this script: at least 10 consecutive days with both Amazon and the lowest 3P price (NEW_third) non-missing, and at least 10 consecutive days with only the lowest 3P price observed (and Amazon missing).¹⁹ Regimes are defined by price availability of Amazon and FBA. “3P only”: 3P price observed and Amazon missing. “Amazon only”: Amazon observed and 3P missing. “Both”: both observed. “Neither”: neither observed.

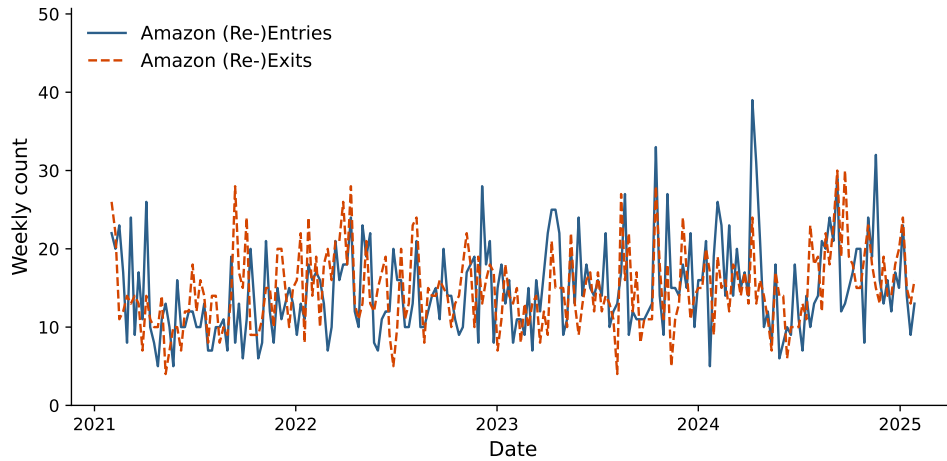
G.2.1 Additional facts on recurring entries/exits

Figure 3: Amazon's presence at product level: Entry and exit events



Notes: The figures plot eligible Amazon entry and exit events over the sample period for ASINs sold by both Amazon and third-party sellers during the sample period. ASINs are sorted vertically by the share of days which Amazon is present in (the ASIN which Amazon is present least often is at the top). Panel (a) shows entry events, with red bars indicating periods in which Amazon enters an ASIN. Panel (b) shows exit events, with blue bars indicating periods in which Amazon exits an ASIN.

Figure 4: Recurring Amazon entries and exits over time: Pasta category



Notes: The Figure shows the analogous to Figure ??, but this time focusing on our data on Pasta. The lines indicate the weekly number of re-exits and re-entries by Amazon over time for the Pasta category. We exclude brands that account for more than 20% of aggregate Amazon entry or exit flows in the ten highest-activity weeks (two brands in the case of pasta), as these spikes reflect systematic brand-level assortment changes rather than product-level entry or exit decisions. We only count a re-entry (re-exit) as such if they are followed by a spell of presence (absence) of 5 days or more (this is robust to other cutoffs).

Table 6: Descriptives on Amazon retail presence spell duration among switcher ASINs (FBA-active days)

	N	Mean	Std. Dev.	Median	P10	P90
<i>Panel A: Spell-level distribution (spell duration in days)</i>						
Amazon present	8,782	31.33	88.39	3.00	1.00	82.00
Amazon absent	9,010	65.90	157.59	9.00	1.00	173.00
<i>Panel B: Per-ASIN number of spells</i>						
Amazon present	1,159	7.58	11.56	3.00	1.00	18.00
Amazon absent	1,159	7.77	11.63	4.00	1.00	18.00
<i>Panel C: Per-ASIN mean spell duration (days)</i>						
Amazon present	1,159	65.17	93.54	34.10	4.16	156.60
Amazon absent	1,159	148.17	204.55	61.50	6.00	427.30

Notes: Coffee category; ASIN-day panel (midnight prices), January 2021–March 2025. ASINs are restricted to those satisfying the event-study screen: at least 10 consecutive days with both Amazon and the lowest 3P price (NEW_third) non-missing, and at least 10 consecutive days with only the lowest 3P price observed (and Amazon missing). We further restrict to switcher ASINs in the 3P-active sample (days with FBA observed): at least one day where Amazon is present and one day where it is absent. A spell is a maximal run in which Amazon’s presence status is unchanged. Panel A reports spell-duration distributions for Amazon-present and Amazon-absent spells. Panel B reports the per-ASIN number of spells (separately for Amazon-present and Amazon-absent spells). Panel C reports each ASIN’s mean spell duration (separately for Amazon-present and Amazon-absent spells).

G.3 Fact 3

Price difference for identical products.

G.4 Further Descriptive Patterns

Price adjustment frequency. Tables 8 and 9 show that, conditional on both parties selling a given ASIN, there are no substantial differences in price adjustment frequency (if anything, Amazon’s price is more likely to display a higher week-to-week price change than the lowest 3P seller’s price).

Table 7: Price differences for identical products: T-Shirts

Dependent variable Compared to Model:	All products			Popular products		
	Lowest 3P (1)	FBA (2)	FBM (3)	Lowest 3P (4)	FBA (5)	FBM (6)
1{Amazon}	0.0002 (0.0014)	0.0027*** (0.0006)	-0.0832*** (0.0114)	-0.0006 (0.0014)	0.0039*** (0.0006)	-0.1621*** (0.0226)
ASIN FE	✓	✓	✓	✓	✓	✓
Week FE	✓	✓	✓	✓	✓	✓
Observations	282,208	270,711	157,649	212,278	208,892	110,913
Adjusted R ²	0.81860	0.86944	0.78345	0.82006	0.86495	0.79109

Notes: The unit of observation is an ASIN-week in the Men's T-Shirts category. The dependent variable is the log of the posted price. For each column, we restrict the sample to ASIN-week observations in which both Amazon and at least one third-party seller of the relevant type (any 3P, lowest FBA, or FBM) are active. The coefficient on 1{Amazon} measures the average log price difference between Amazon and the corresponding third-party offer for the same product in the same week. "Popular products" are defined above. Standard errors, clustered at the ASIN level, are reported in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table 8: Amazon versus 3P weekly price adjustment frequency: Cocoa category

Sample Dependent Variables: Model:	Joint Adj. indicator (>0.5%) (1)	Joint Adj. indicator (>1%) (2)	Broad Adj. indicator (>0.5%) (3)	Broad Adj. indicator (>1%) (4)
Amazon indicator	0.0180 (0.0330)	0.0090 (0.0310)	0.0175 (0.0321)	0.0055 (0.0296)
ASIN	Yes	Yes	Yes	Yes
Week	Yes	Yes	Yes	Yes
Observations	12,470	12,470	160,556	160,556
R ²	0.13828	0.12788	0.25566	0.22198

Notes: Dependent variable equals one if the absolute weekly log price change exceeds the threshold shown in each column (0.5% and 1% respectively), and zero otherwise. Joint sample restricts to ASIN-week observations with non-missing Amazon and third-party prices in both t and $t - 1$; broad sample uses all seller-week observations with non-missing own prices in t and $t - 1$. Weekly data, January 2021 to March 2025. All columns include ASIN and week fixed effects. Clustered (ASIN) standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Table 9: Amazon versus 3P weekly price adjustment frequency: Coffee category

Sample	Joint	Joint	Broad	Broad
Dependent Variables:	Adj. indicator (>0.5%)	Adj. indicator (>1%)	Adj. indicator (>0.5%)	Adj. indicator (>1%)
Model:	(1)	(2)	(3)	(4)
Amazon indicator	0.0203*** (0.0071)	-0.0031 (0.0065)	0.0551*** (0.0064)	0.0254*** (0.0057)
ASIN	Yes	Yes	Yes	Yes
Week	Yes	Yes	Yes	Yes
Observations	263,530	263,530	1,841,761	1,841,761
R ²	0.10842	0.10250	0.26324	0.22855

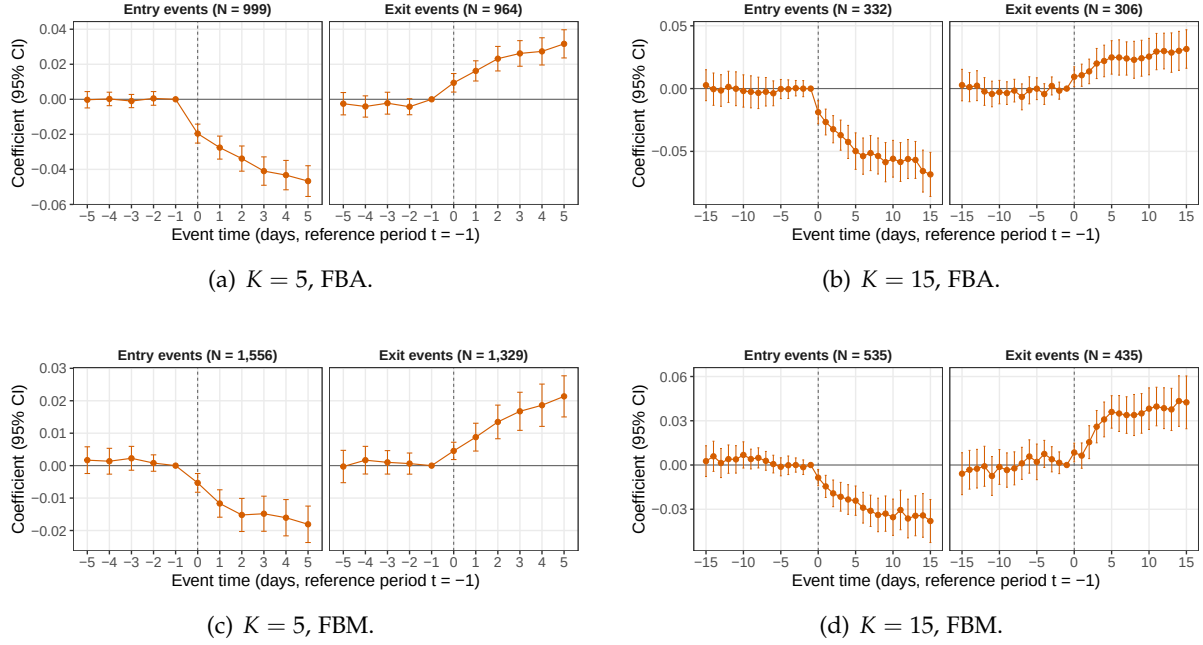
Notes: Dependent variable equals one if the absolute weekly log price change exceeds the threshold shown in each column (0.5% and 1% respectively), and zero otherwise. Joint sample restricts to ASIN-week observations with non-missing Amazon and third-party prices in both t and $t - 1$; broad sample uses all seller-week observations with non-missing own prices in t and $t - 1$. Weekly data, January 2021 to March 2025. All columns include ASIN and week fixed effects. Clustered (ASIN) standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

H Empirical Evidence: Price Levels

H.1 Subsample analyses

Separated.

Figure 5: Amazon entry and exit under alternative qualifying-event definitions

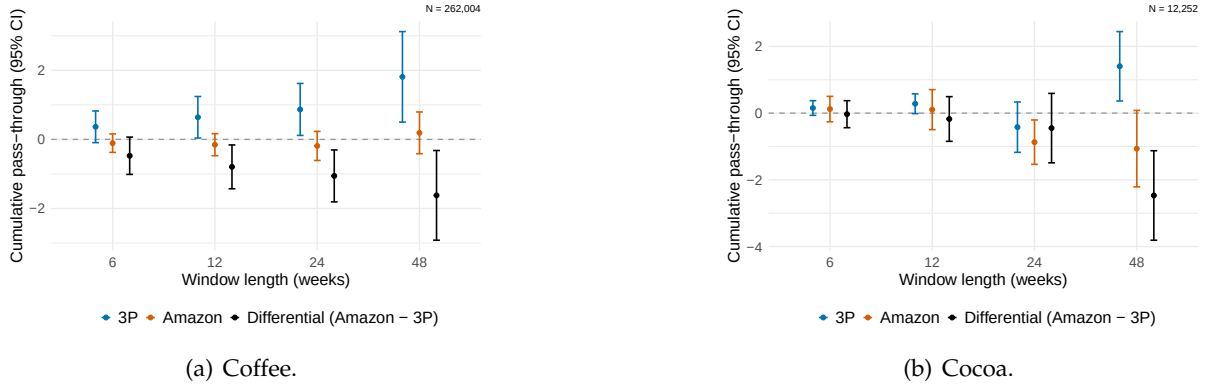


Notes: The figure reports Amazon entry and exit event-study estimates under alternative definitions of qualifying events. Panels (a) and (c) use $K = 5$, which admits events with shorter continuous pre- and post-transition regimes. Panels (b) and (d) use $K = 15$, which imposes a more restrictive event definition requiring longer continuous regimes around the transition. Panels (a) and (b) use the logarithm of the lowest FBA price as the outcome, while panels (c) and (d) use the logarithm of the lowest FBM price. The baseline specification uses $K = 10$. Vertical bars report 95% confidence intervals based on standard errors two-way clustered by ASIN and calendar date.

I Empirical Evidence: Pass-through

I.1 Subsample analysis

Figure 6: Cumulative commodity-cost pass-through by seller type: matched ASIN-week sample excluding ASINs without reviews

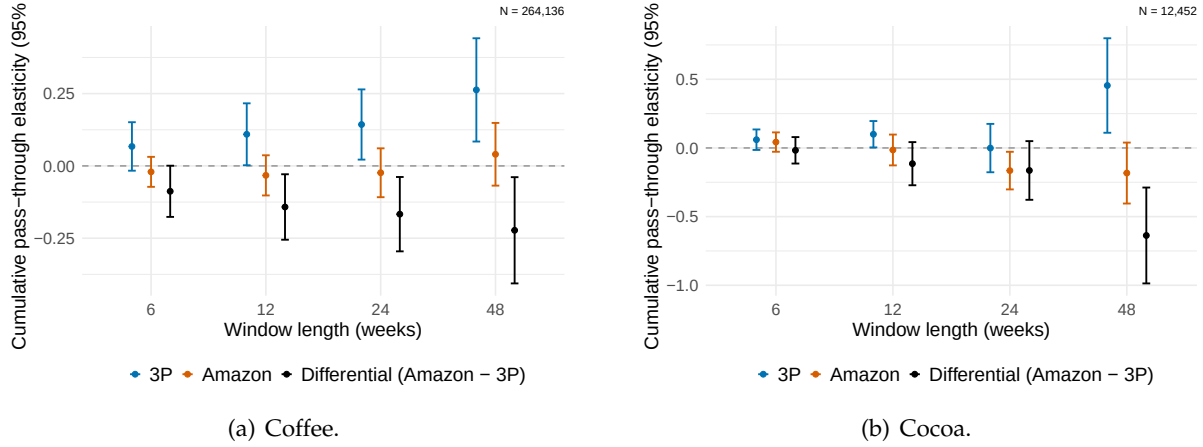


Notes: The sample additionally excludes ASINs who do not have any user review. “Differential (Amazon - 3P)” is the difference between the two estimates; negative values indicate lower pass-through by Amazon. Vertical bars show 95% confidence intervals based on standard errors two-way clustered by ASIN and calendar week.

I.2 Alternative specifications

Log specification.

Figure 7: Commodity-cost pass-through in logarithms

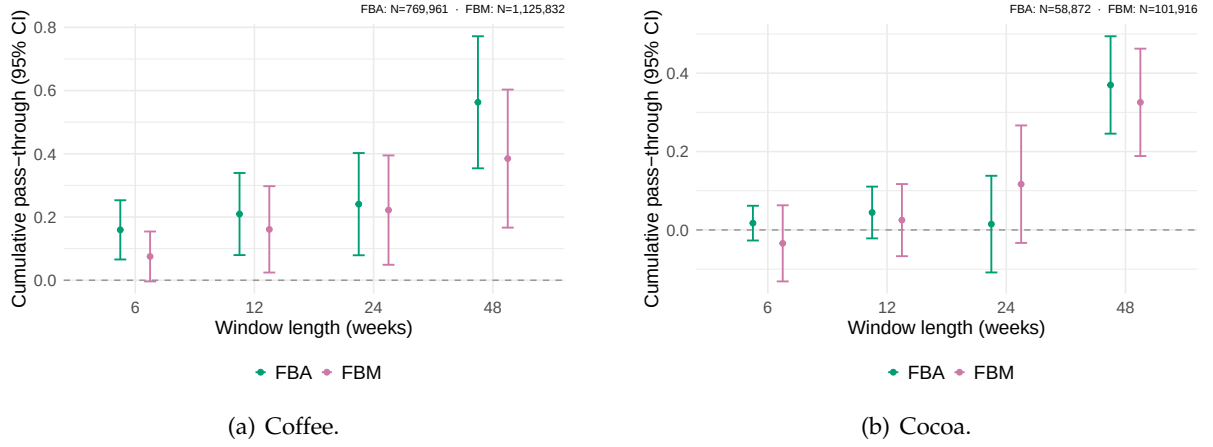


Notes: The figure reports cumulative pass-through estimates from the logarithmic counterpart of the baseline stacked common-support specification. Regressions relate weekly log changes in posted retail prices to weekly log changes in the relevant commodity-cost series. The Amazon and lowest-3P price responses are estimated on the same set of ASIN-week cells. “Differential (Amazon – 3P)” denotes the difference between the cumulative Amazon and 3P estimates. Vertical bars report 95% intervals based on standard errors two-way clustered by ASIN and calendar week.

I.3 Heterogeneity

FBA vs. FBM prices.

Figure 8: Cumulative commodity-cost pass-through for 3P prices by fulfillment mode



Notes: The figure reports cumulative pass-through estimates from distributed-lag regressions of weekly changes in posted retail prices on weekly changes in converted commodity costs. At window length K , the contemporaneous coefficient and first $K - 1$ lag coefficients are summed. Both variables are measured in dollars per gram, so an estimate of one denotes full pass-through. Regressions are estimated separately for the lowest 3P prices under FBA and FBM, using all ASIN-week cells with an observed weekly price change for the relevant fulfillment mode and including ASIN fixed effects. Vertical bars show 95% confidence intervals based on standard errors two-way clustered by ASIN and calendar week.

The pass-through point estimates have a tendency to be slightly higher for 3P sellers operating under FBA, as opposed to sellers operating under FBM (Figure 8).

References

- Bennati, Luca (2026) "Vertical Integration and E-commerce Competition: Evidence from Amazon Marketplace," *Working Paper*, https://drive.google.com/file/d/1GCzljX1haX5pLL_IXG1Q4r7fWgCJI3ee/view.
- Crawford, Gregory S, Matteo Courthoud, Regina Seibel, and Simon Zuzek (2022) "Amazon entry on Amazon marketplace," *CEPR Discussion Paper No. 17531*.
- Hagiu, Andrei and Julian Wright (2015) "Marketplace or Reseller?" *Management Science*, 61 (1), 184–203, <http://www.jstor.org/stable/24551078>.
- Zhu, Feng and Qihong Liu (2018) "Competing with complementors: An empirical look at Amazon.com," *Strategic Management Journal*, 39 (10), 2618–2642, <https://doi.org/10.1002/smj.2932>.